

Canonical W configurations - cluster P3

species: electron
polarization cluster: P3
source cluster configurations: 94
canonical configurations collected: 22

canonical display criterion: exactly 3 retained final-state components
retained-component cutoff: fraction $\geq 2.0\%$
normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

cluster retained-component count distribution:

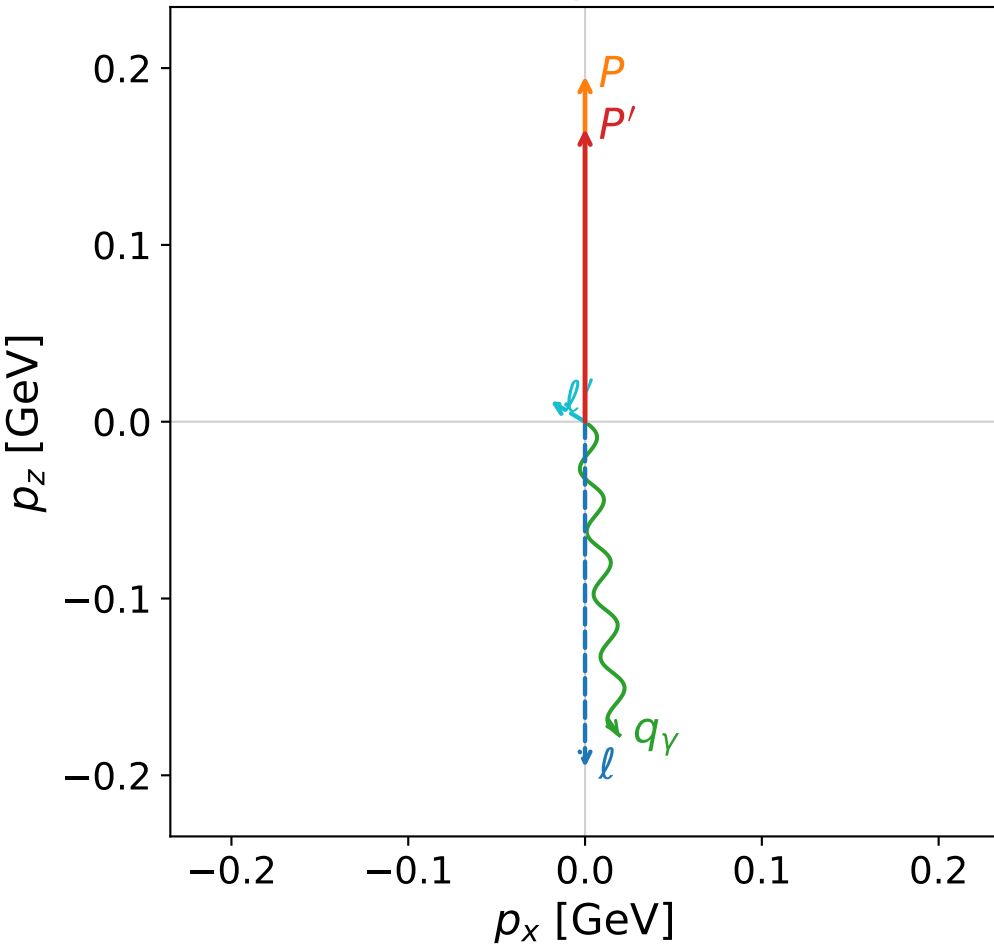
3 components:	22 configurations
4 components:	33 configurations
5 components:	8 configurations
6 components:	31 configurations

CSV index:
Output\GradientPhaseSpaceScan\electron\Data\W\dw\polarization_cluster_03\combined\canonical_dw_polarization_cluster_03_configurations_electron.csv

Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

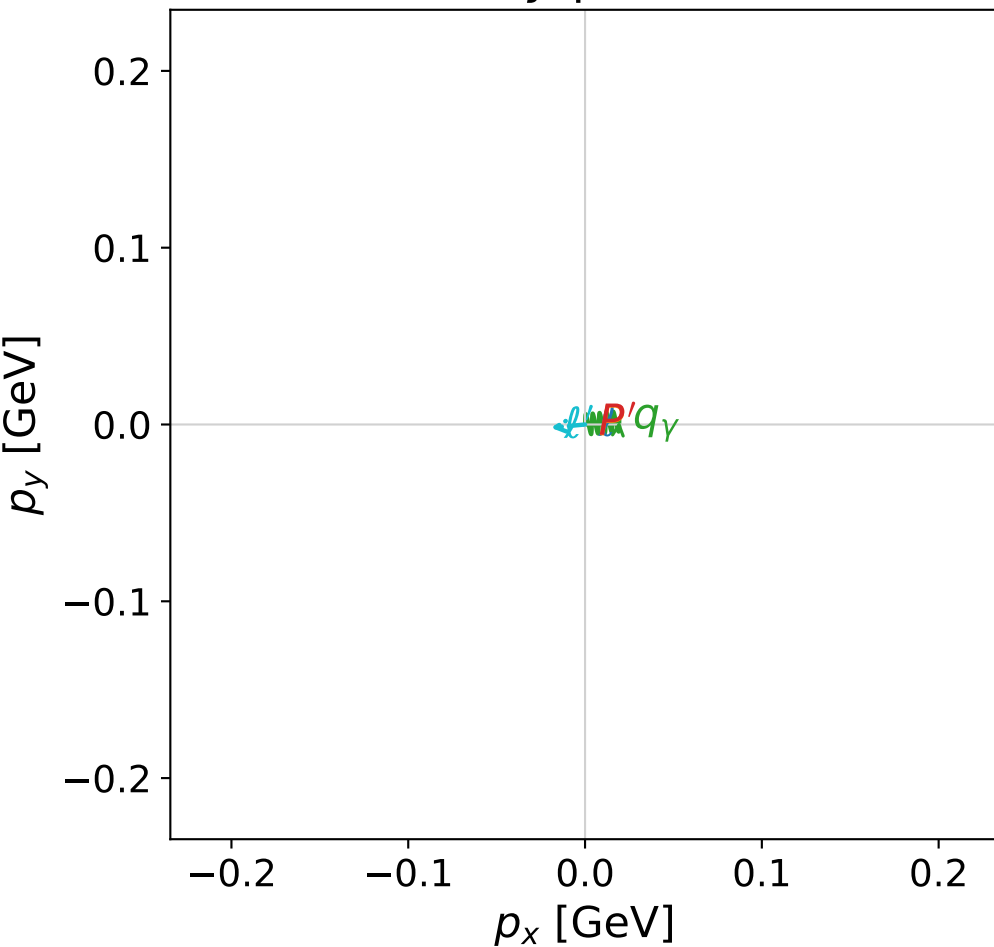


dw_mixing_angles_polarization_03_local_minimum_378
 $D_W=0.0012061$
 region: polarization_cluster_03_local_minimum_378
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0378
 lepton mixing angle: $\alpha_e=0.8336692$ rad
 incoming lepton: $0.672164|+\rangle + 0.740403|-\rangle$
 proton mixing angle: $\alpha_p=3.1197285$ rad
 incoming proton: $-0.999761|+\rangle + 0.0218624|-\rangle$

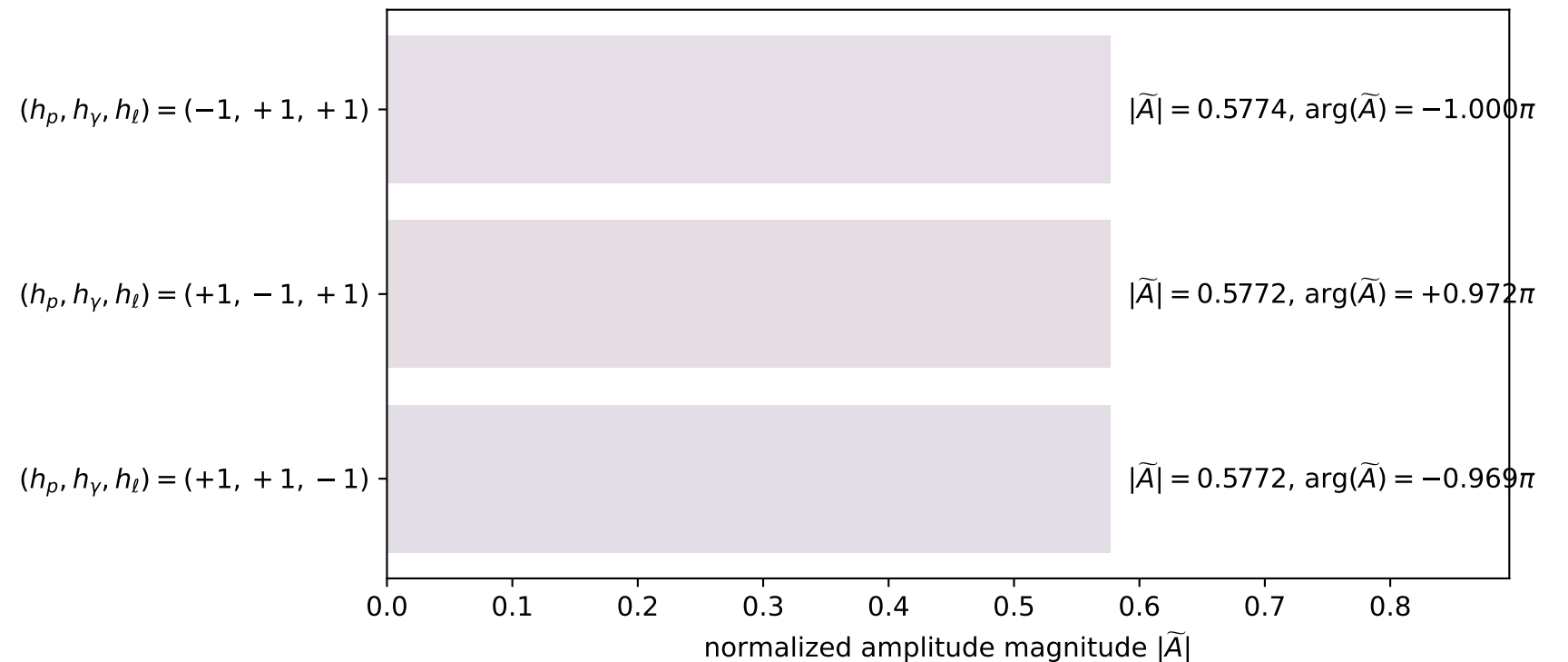
$s=1.33084$, $\sqrt{s}=1.15362$, $|\vec{P}|=0.195471$, $|\vec{P}'|=0.16574$
 $\theta_{P'}=0$, $\theta_Y=3.03202$, $\phi_{P'}=0.000172031$, $\phi_Y=0.0936696$
 $E_Y=0.178397$, $\theta_{\ell'}=1.03482$, $\phi_{\ell'}=3.23526$
 $Q^2=0.0134016$, $x_B=0.0325252$, $t=-0.000852303$, $W^2=1.27848$, $y=0.913613$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1955$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1955$
 P $E=0.9582$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1955$
 ℓ' $E=0.0227$ $p_x=-0.0194$ $p_y=-0.0018$ $p_z=0.0116$
 P' $E=0.9525$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1657$
 q_Y $E=0.1784$ $p_x=0.0194$ $p_y=0.0018$ $p_z=-0.1773$

x--y plane

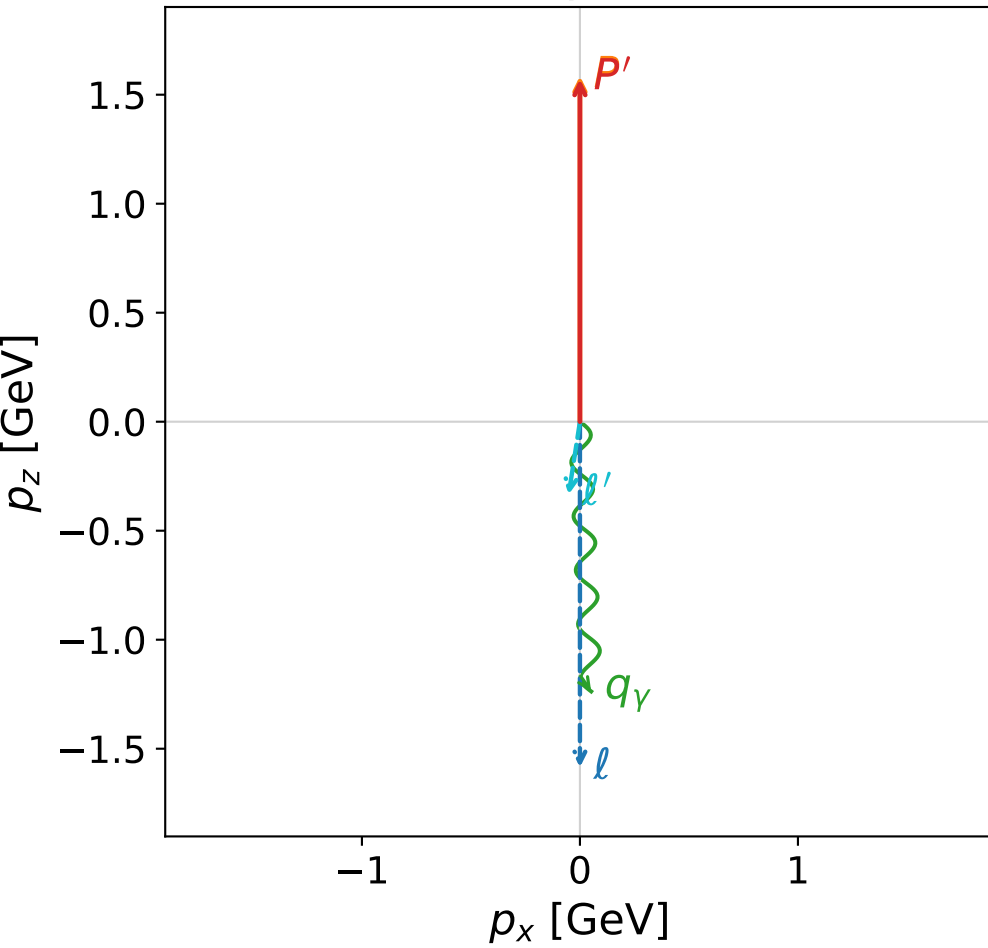


Leading normalized final-state amplitudes
 (N=3, retained=100.0%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



dw_mixing_angles_polarization_03_local_minimum_583

$D_W=0.00377731$

region: polarization_cluster_03_local_minimum_583

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0583

lepton mixing angle: $\alpha_e=0.79847104$ rad

incoming lepton: $0.697803|+\rangle + 0.71629|-\rangle$

proton mixing angle: $\alpha_p=3.0412699$ rad

incoming proton: $-0.994972|+\rangle + 0.100155|-\rangle$

$s=11.746$, $\sqrt{s}=3.42724$, $|\vec{P}|=1.58526$, $|\vec{P}'|=1.57167$

$\theta_{p'}=0$, $\theta_\gamma=3.04754$, $\phi_{p'}=2.35408$, $\phi_\gamma=5.17152$

$E_\gamma=1.24274$, $\theta_{l'}=2.80583$, $\phi_{l'}=2.02993$

$Q^2=0.0627118$, $x_B=0.00737703$, $t=-4.81958e-05$, $W^2=9.31808$, $y=0.782334$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 1.5853$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5853$

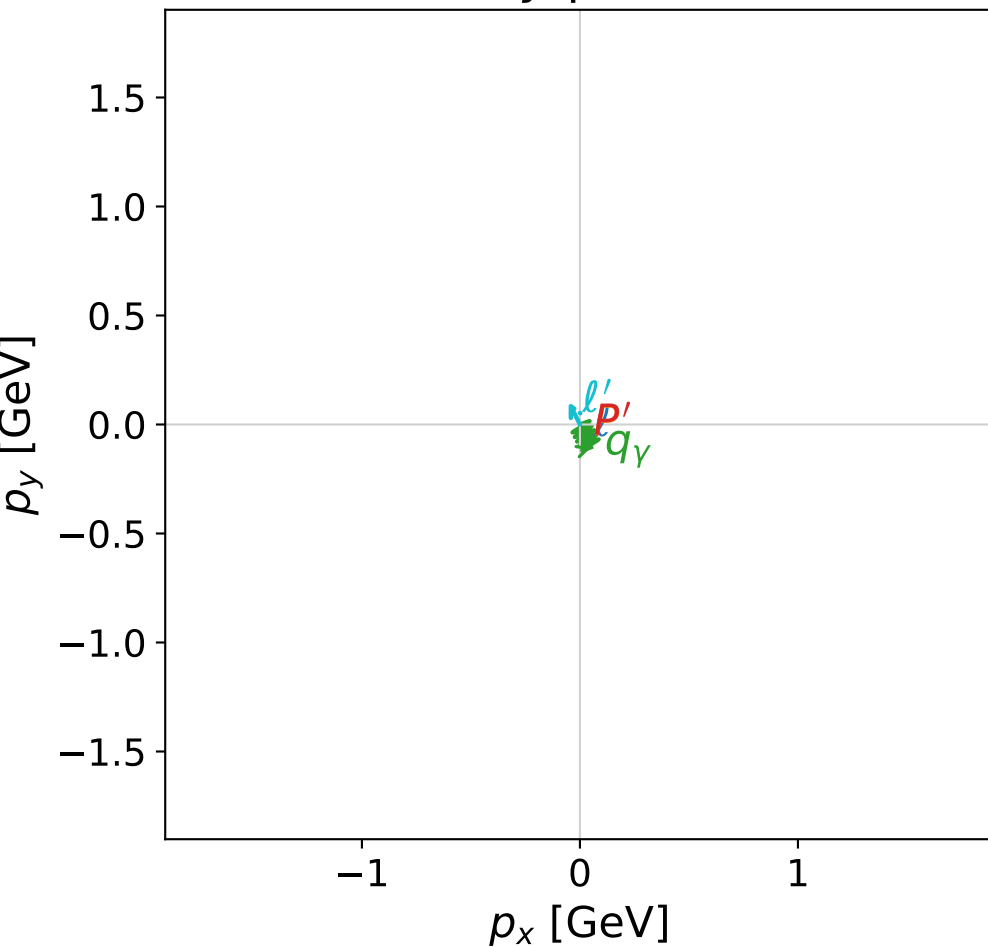
P $E= 1.8420$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5853$

l' $E= 0.3542$ $p_x= -0.0517$ $p_y= 0.1046$ $p_z= -0.3344$

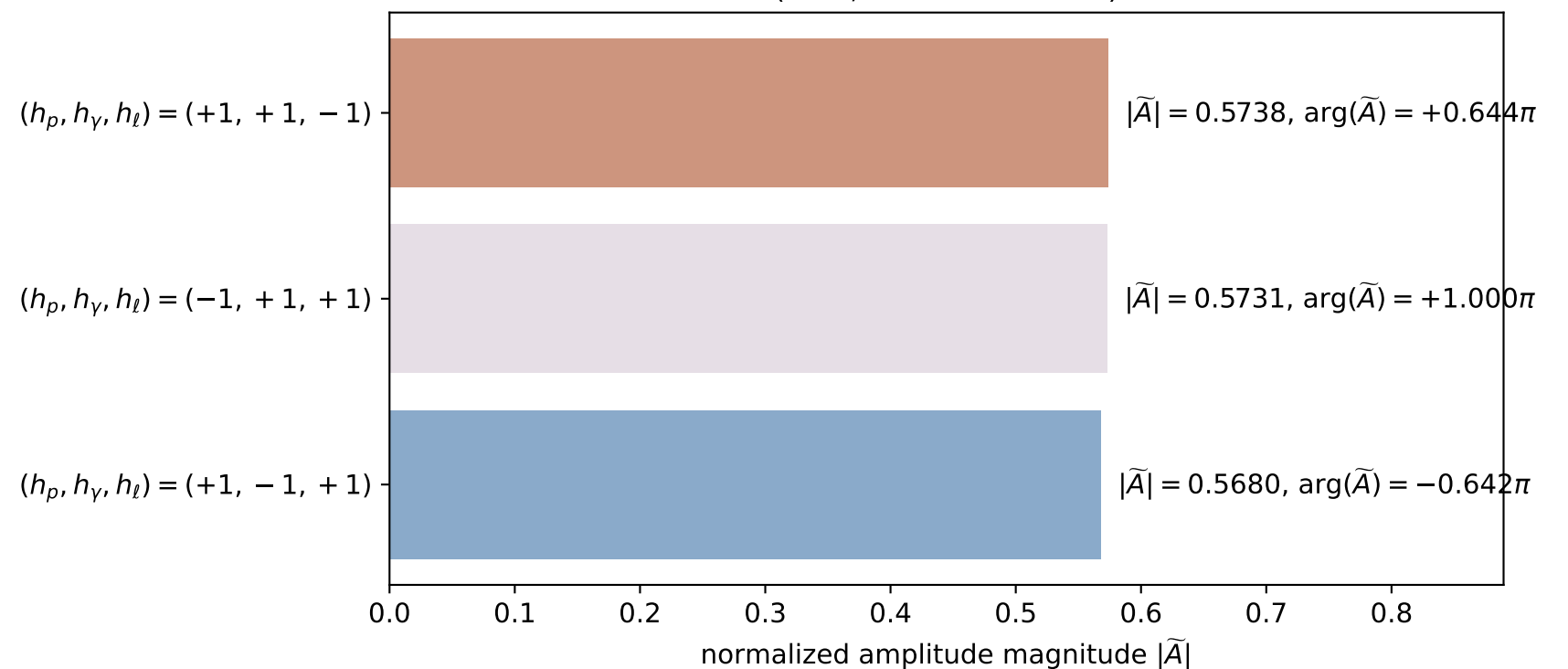
P' $E= 1.8303$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.5717$

q_γ $E= 1.2427$ $p_x= 0.0517$ $p_y= -0.1046$ $p_z= -1.2372$

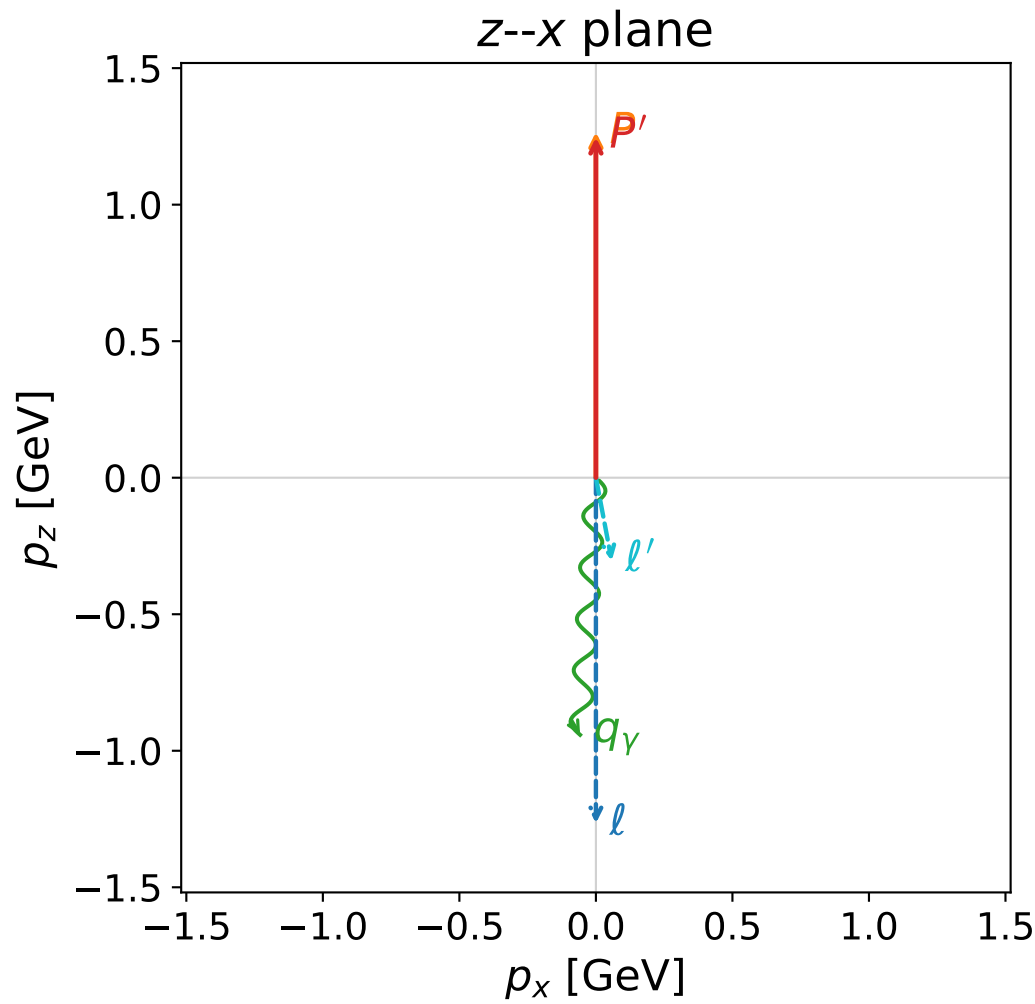
x--y plane



Leading normalized final-state amplitudes
(N=3, retained=98.0%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

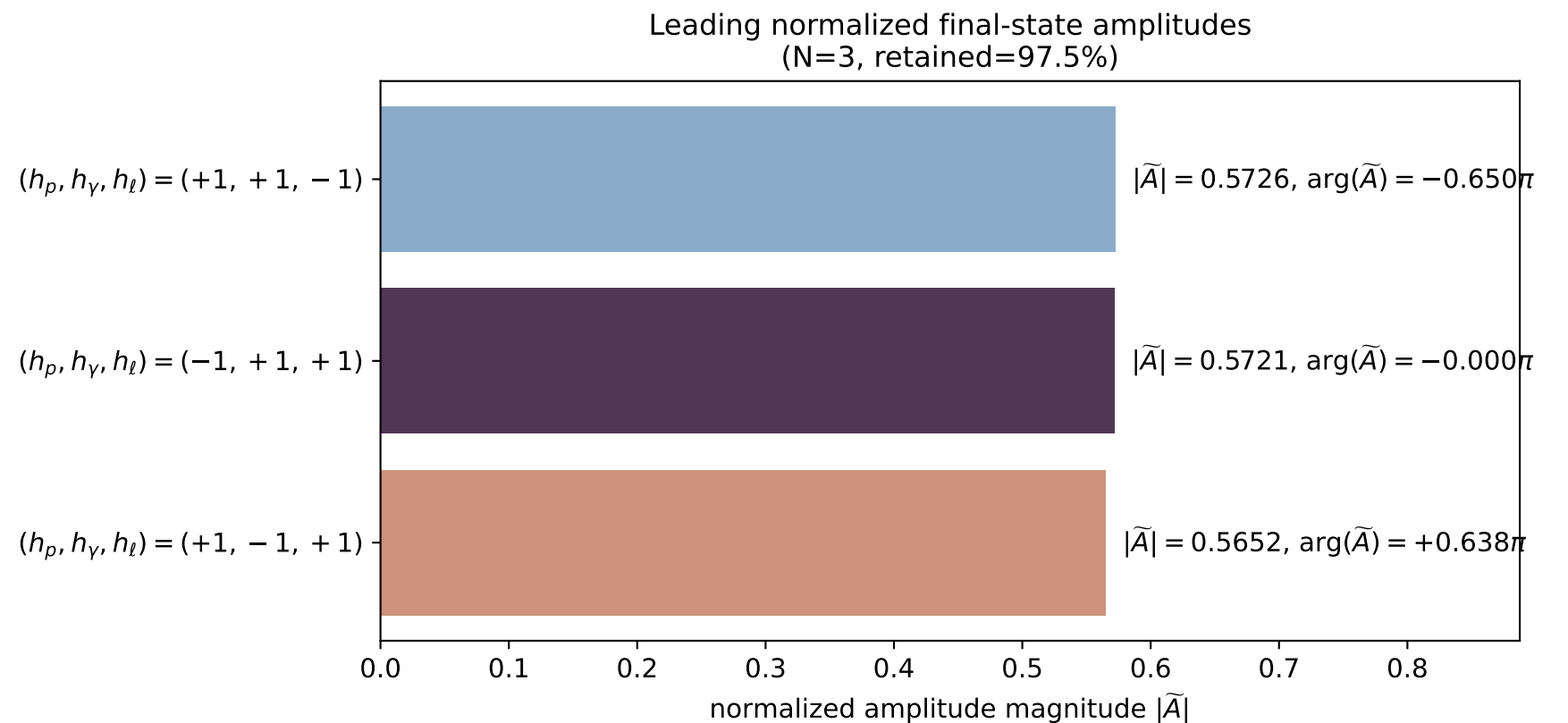
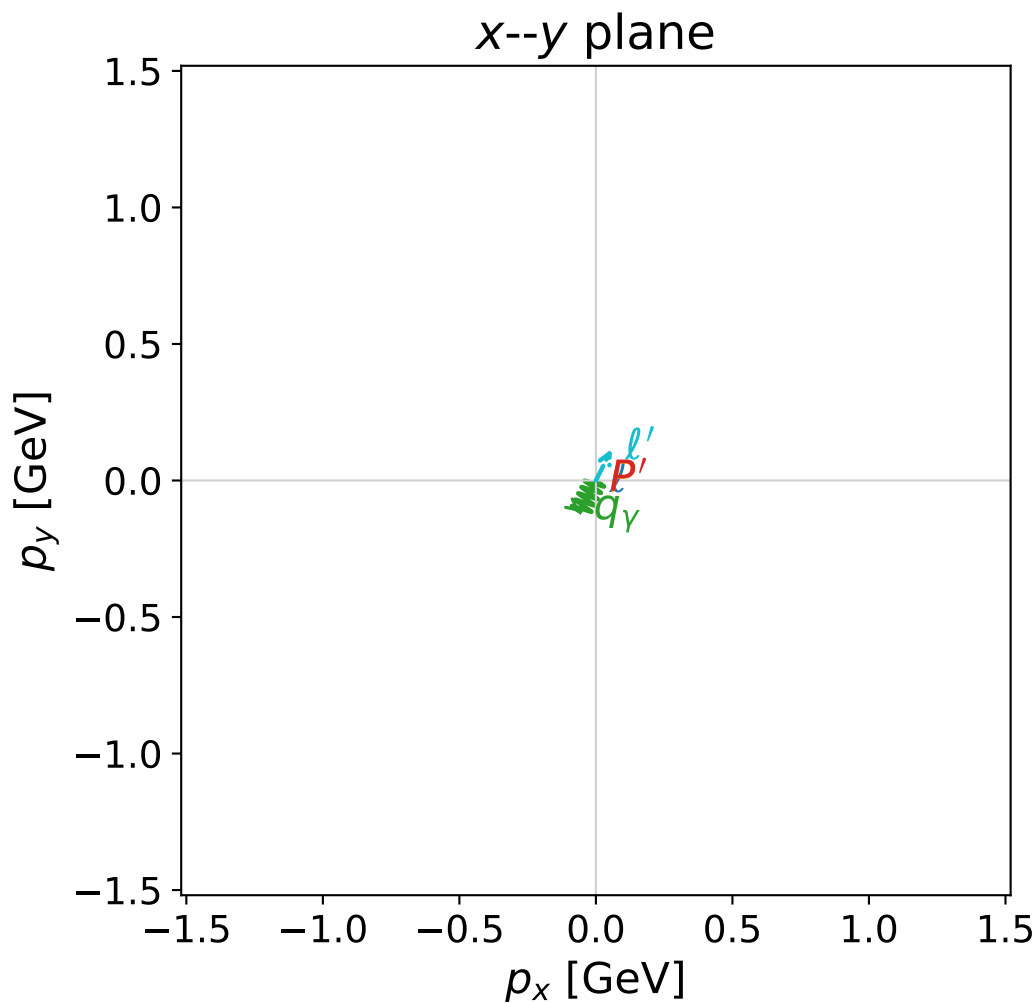


dw_mixing_angles_polarization_03_local_minimum_644
 $D_W=0.00536814$
 region: polarization_cluster_03_local_minimum_644
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0644
 lepton mixing angle: $\alpha_e=0.79589385$ rad
 incoming lepton: $0.699646|+\rangle +0.714489|-\rangle$
 proton mixing angle: $\alpha_p=0.11066879$ rad
 incoming proton: $0.993882|+\rangle +0.110443|-\rangle$

$s=8.07157$, $\sqrt{s}=2.84105$, $|\vec{P}|=1.26568$, $|\vec{P}'|=1.24622$
 $\theta_{p'}=0$, $\theta_\gamma=3.00532$, $\phi_{p'}=4.67144$, $\phi_\gamma=4.24742$
 $E_\gamma=0.95051$, $\theta_{l'}=2.74054$, $\phi_{l'}=1.10583$
 $Q^2=0.066438$, $x_B=0.012352$, $t=-0.000135646$, $W^2=6.19213$, $y=0.747905$

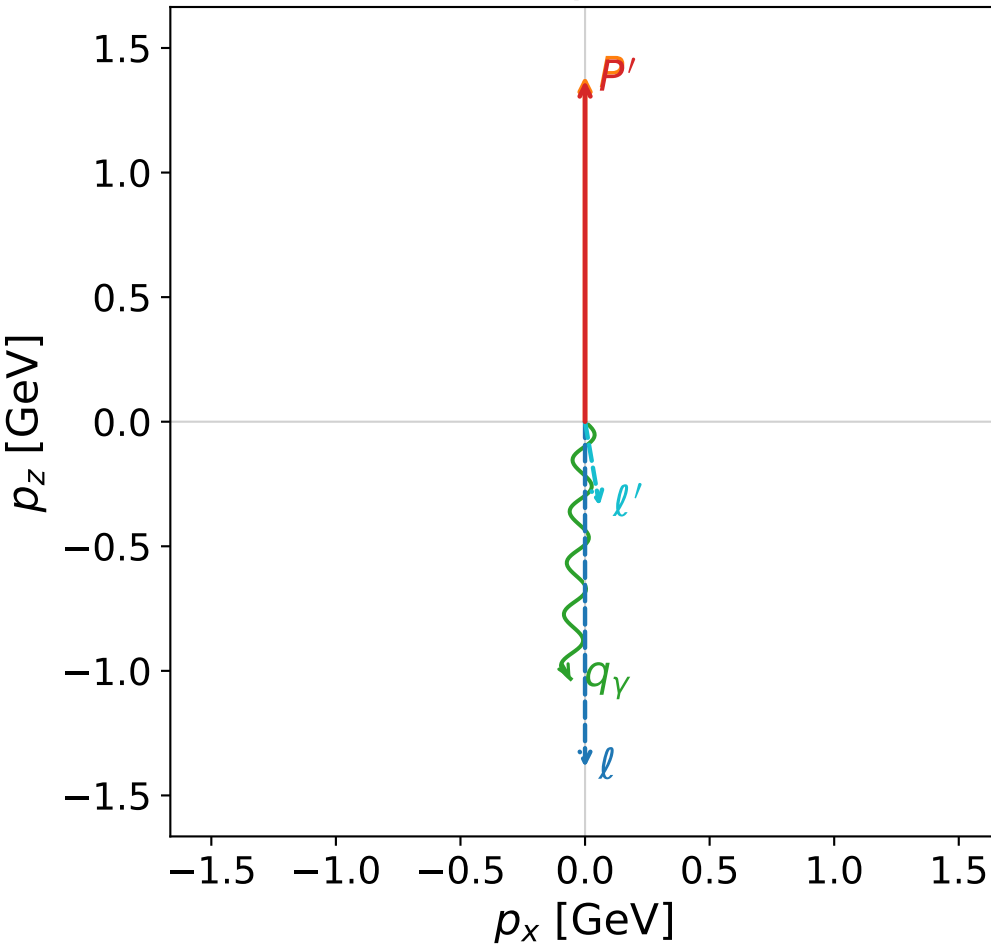
four-momenta $[E, p_x, p_y, p_z]$ GeV:

Particle	E	p_x	p_y	p_z
l	1.2657	0.0000	0.0000	-1.2657
P	1.5754	0.0000	0.0000	1.2657
l'	0.3308	0.0579	0.1154	-0.3045
P'	1.5598	-0.0000	-0.0000	1.2462
q_γ	0.9505	-0.0579	-0.1154	-0.9417



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

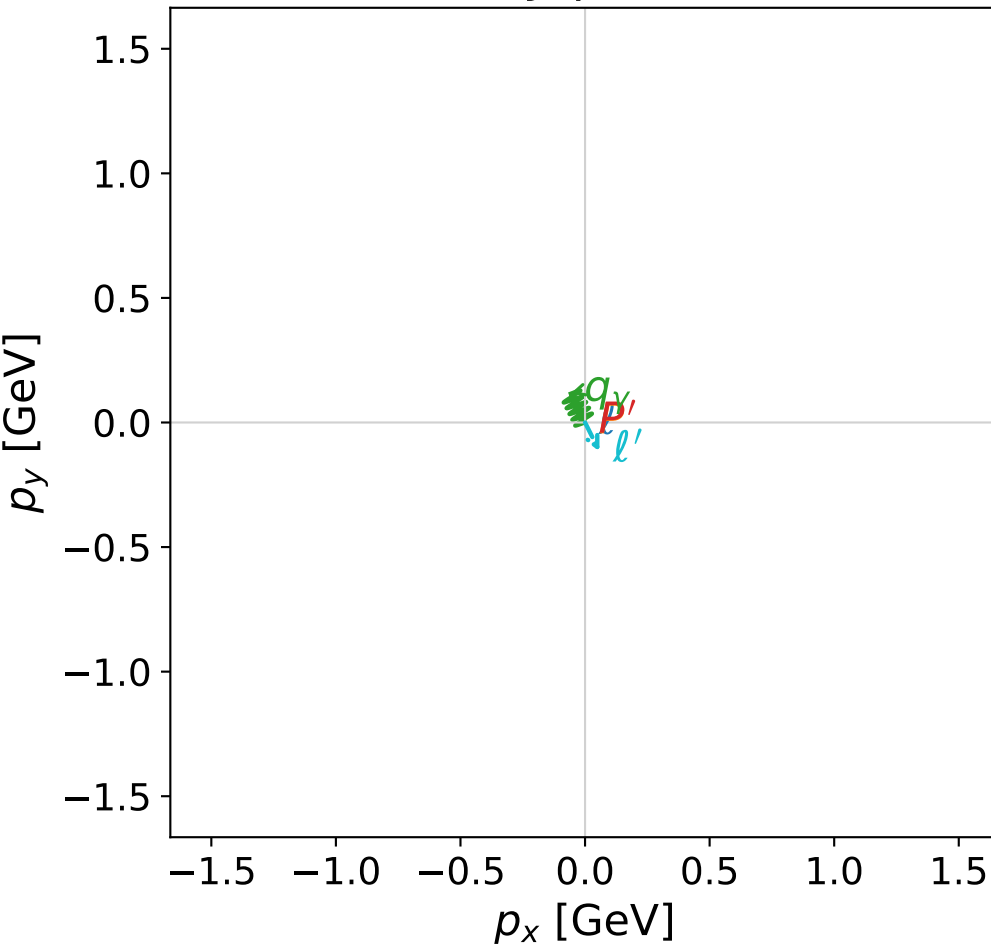


dw_mixing_angles_polarization_03_local_minimum_648
 $D_W=0.0054095$
 region: polarization_cluster_03_local_minimum_648
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0648
 lepton mixing angle: $\alpha_e=0.79612311$ rad
 incoming lepton: $0.699483|+\rangle + 0.71465|-\rangle$
 proton mixing angle: $\alpha_p=0.11130992$ rad
 incoming proton: $0.993811|+\rangle + 0.11108|-\rangle$

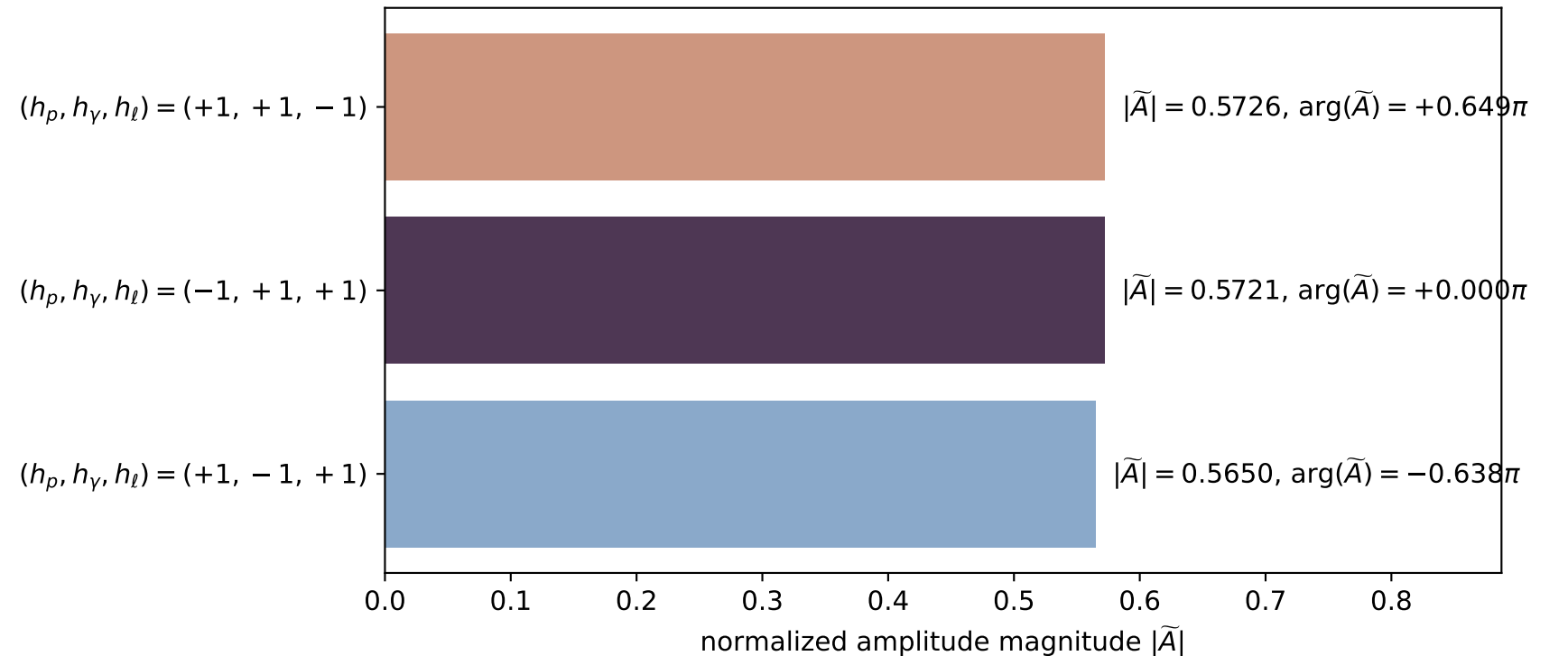
$s=9.37335$, $\sqrt{s}=3.06159$, $|\vec{P}|=1.38711$, $|\vec{P}'|=1.36953$
 $\theta_{P'}=0$, $\theta_\gamma=3.01671$, $\phi_{P'}=0.61743$, $\phi_\gamma=2.03429$
 $E_\gamma=1.04008$, $\theta_{\ell'}=2.77514$, $\phi_{\ell'}=5.17588$
 $Q^2=0.0665991$, $x_B=0.0104943$, $t=-9.77838e-05$, $W^2=7.15947$, $y=0.747186$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.3871$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3871$
 P $E= 1.6745$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3871$
 ℓ' $E= 0.3616$ $p_x= 0.0579$ $p_y= -0.1159$ $p_z= -0.3375$
 P' $E= 1.6600$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3695$
 q_γ $E= 1.0401$ $p_x= -0.0579$ $p_y= 0.1159$ $p_z= -1.0320$

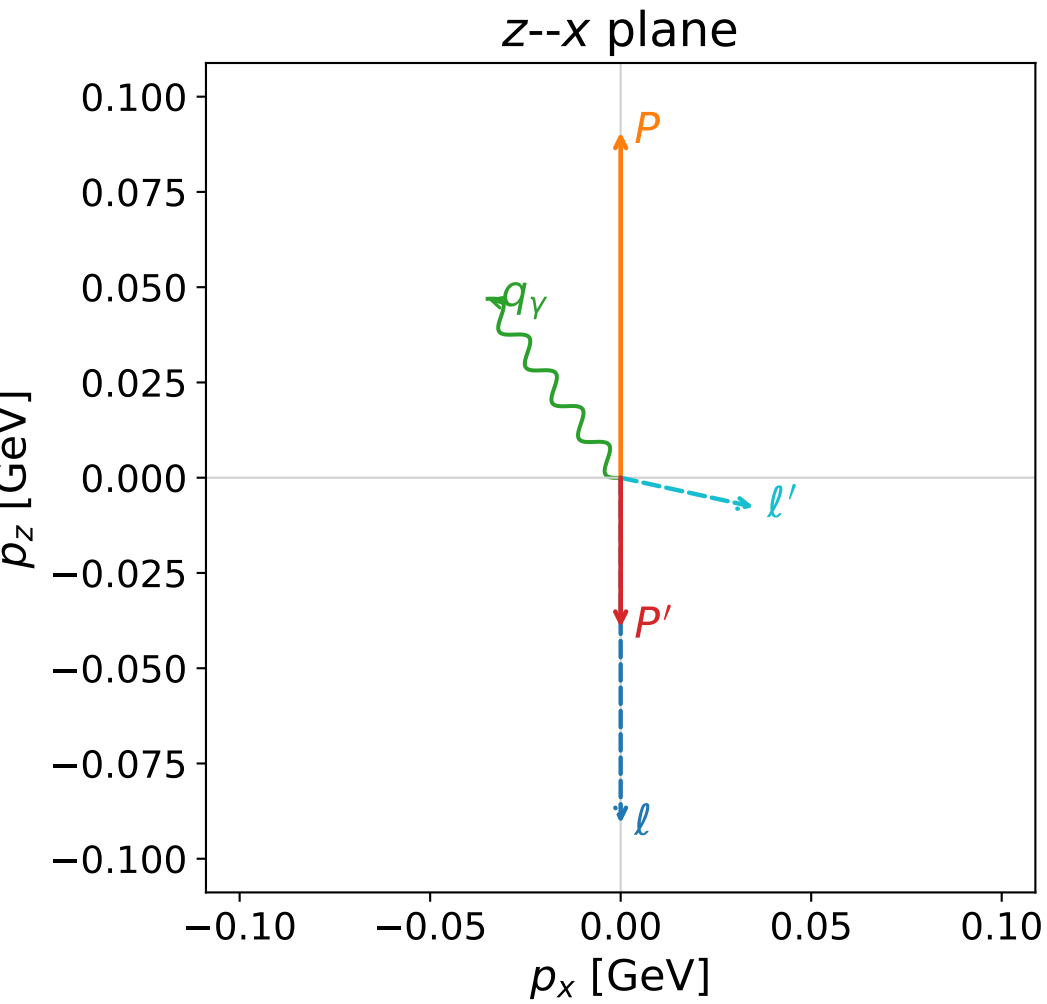
x--y plane



Leading normalized final-state amplitudes
(N=3, retained=97.4%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

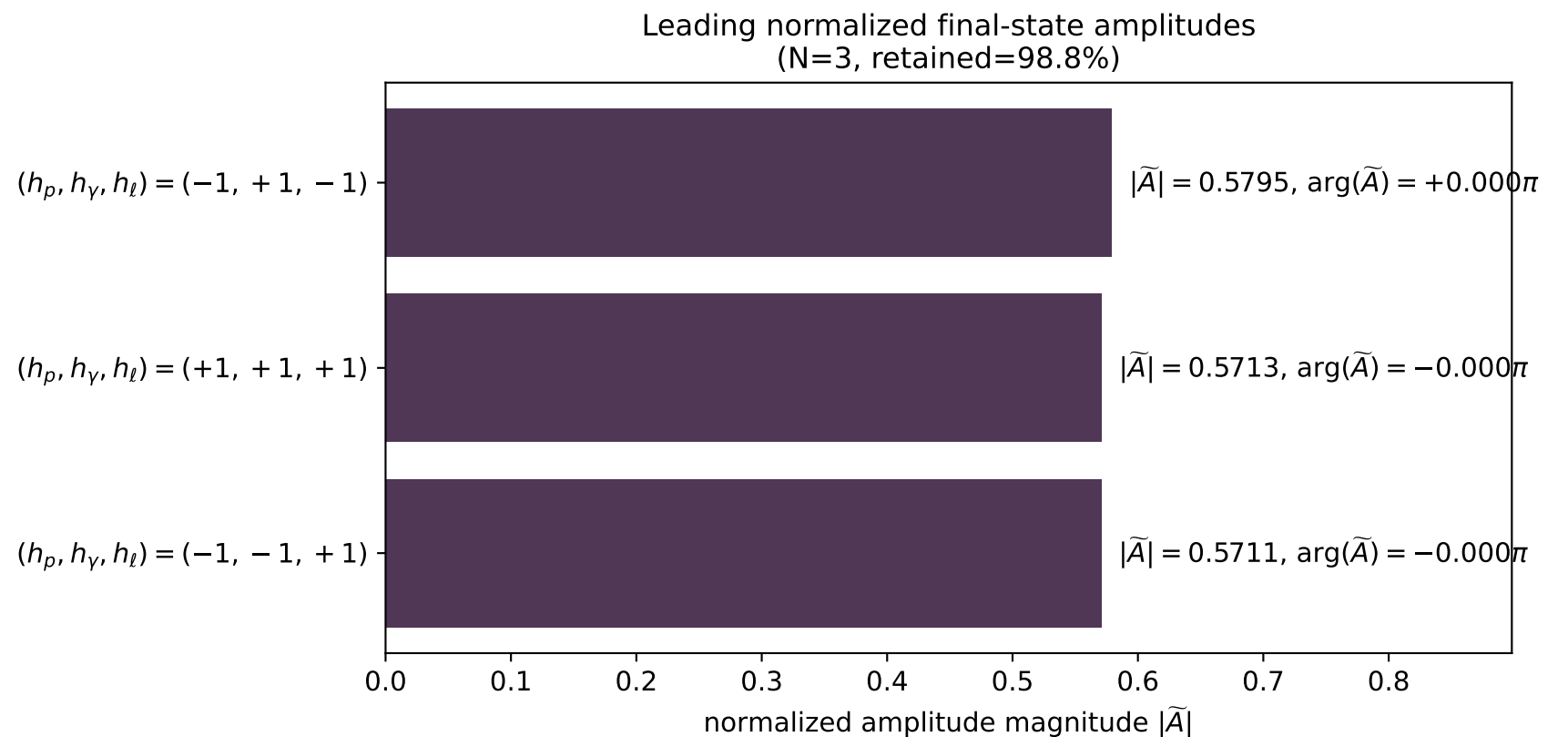
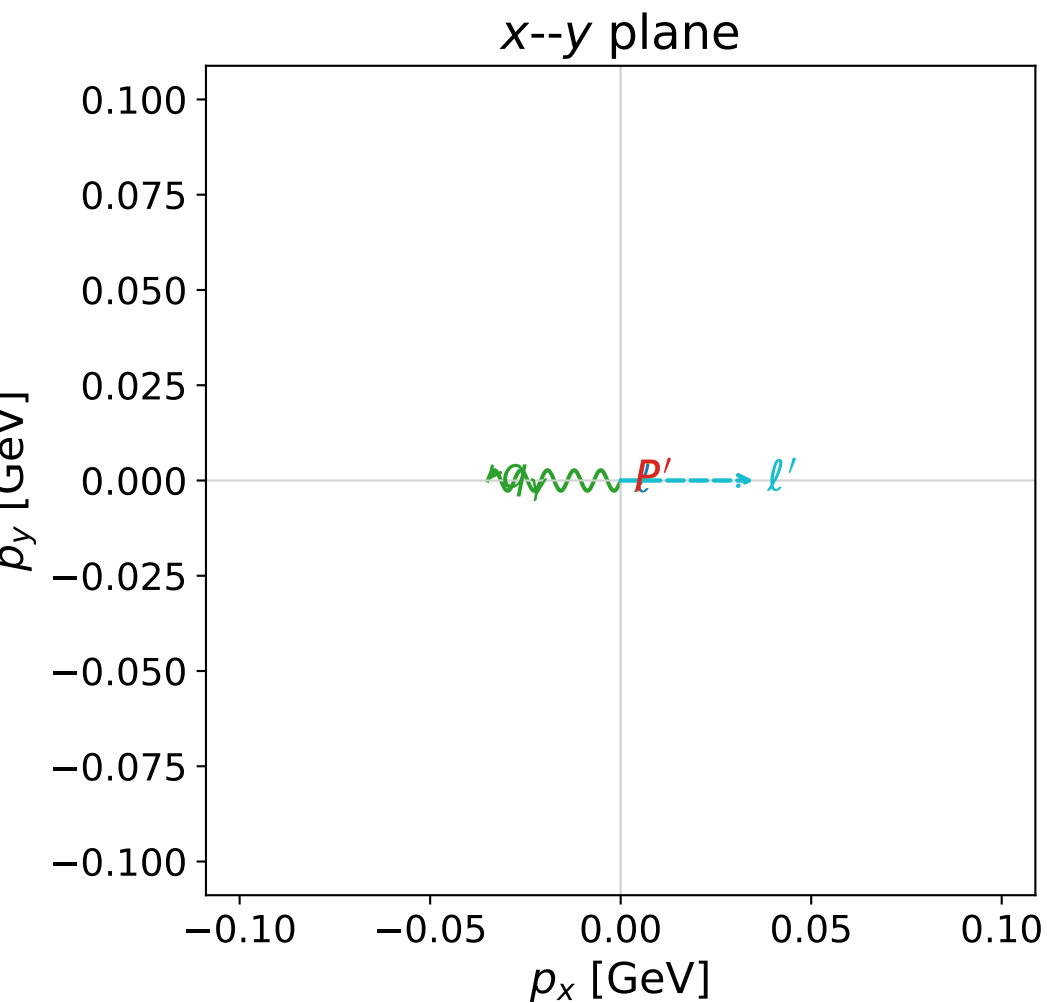


dw_mixing_angles_polarization_03_local_minimum_651
 $D_W=0.00554297$
 region: polarization_cluster_03_local_minimum_651
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0651
 lepton mixing angle: $\alpha_e=0.79607924$ rad
 incoming lepton: $0.699514|+\rangle + 0.714619|-\rangle$
 proton mixing angle: $\alpha_p=0.15671114$ rad
 incoming proton: $0.987746|+\rangle + 0.156071|-\rangle$

$s=1.06724$, $\sqrt{s}=1.03307$, $|\vec{P}|=0.0906976$, $|\vec{P}'|=0.0393062$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0.639302$, $\phi_{p'}=4.0112$, $\phi_\gamma=3.1416$
 $E_\gamma=0.0585098$, $\theta_{\ell'}=1.78649$, $\phi_{\ell'}=8.94111e-06$
 $Q^2=0.00509536$, $x_B=0.0429455$, $t=-0.0168884$, $W^2=0.993396$, $y=0.633133$

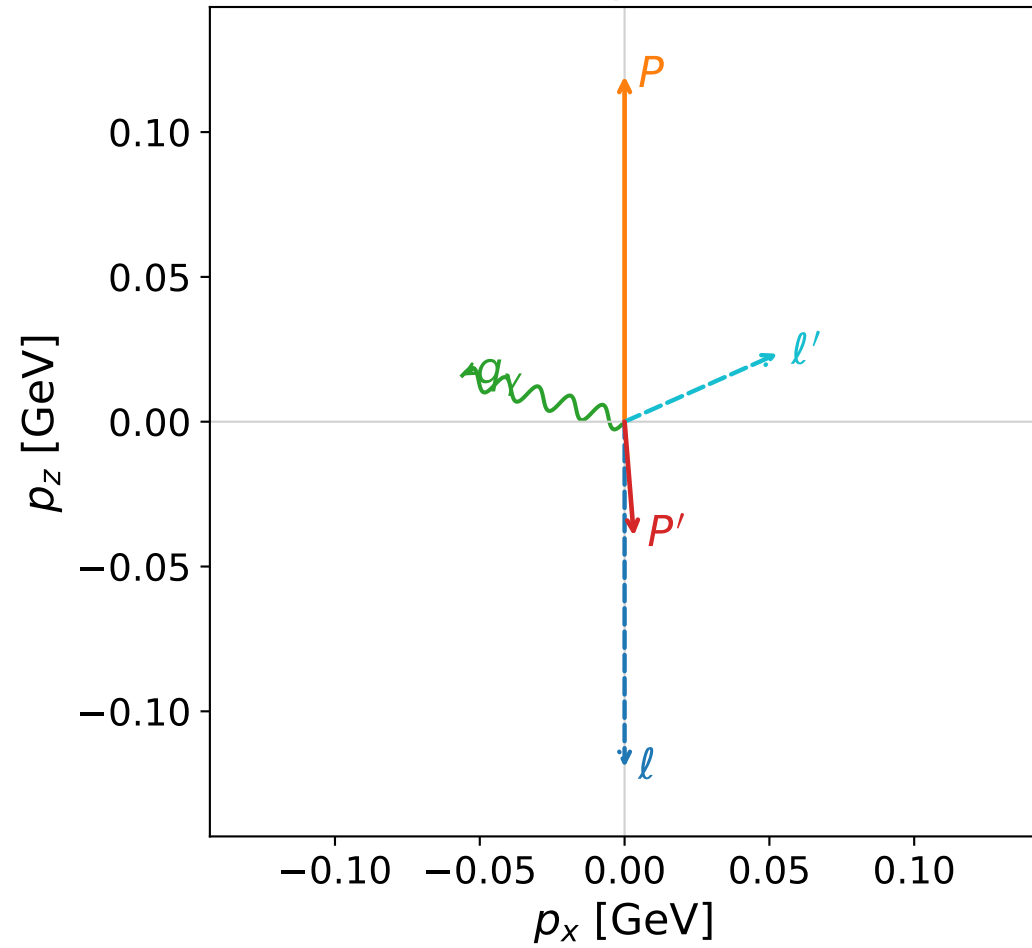
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.0907	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.0907
P	$E=$	0.9424	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.0907
ℓ'	$E=$	0.0357	$p_x=$	0.0349	$p_y=$	0.0000	$p_z=$	-0.0076
P'	$E=$	0.9388	$p_x=$	-0.0000	$p_y=$	-0.0000	$p_z=$	-0.0393
q_γ	$E=$	0.0585	$p_x=$	-0.0349	$p_y=$	-0.0000	$p_z=$	0.0470

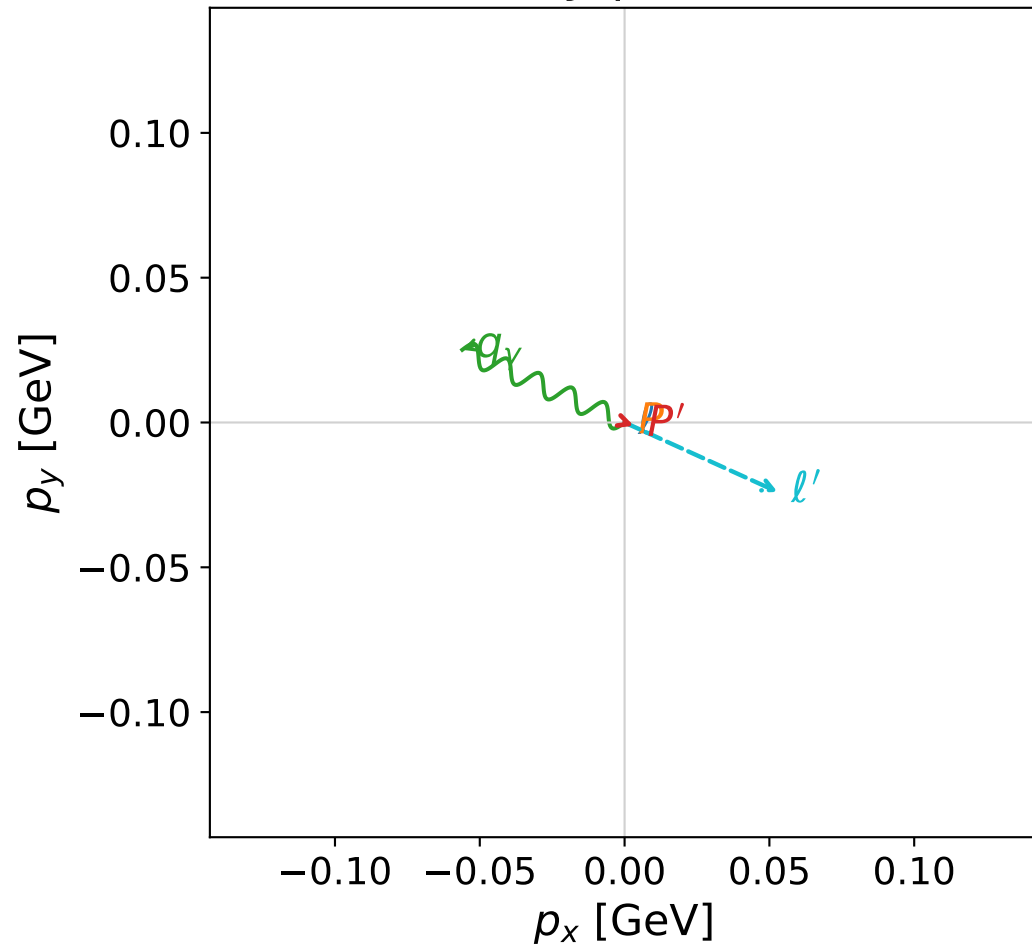


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

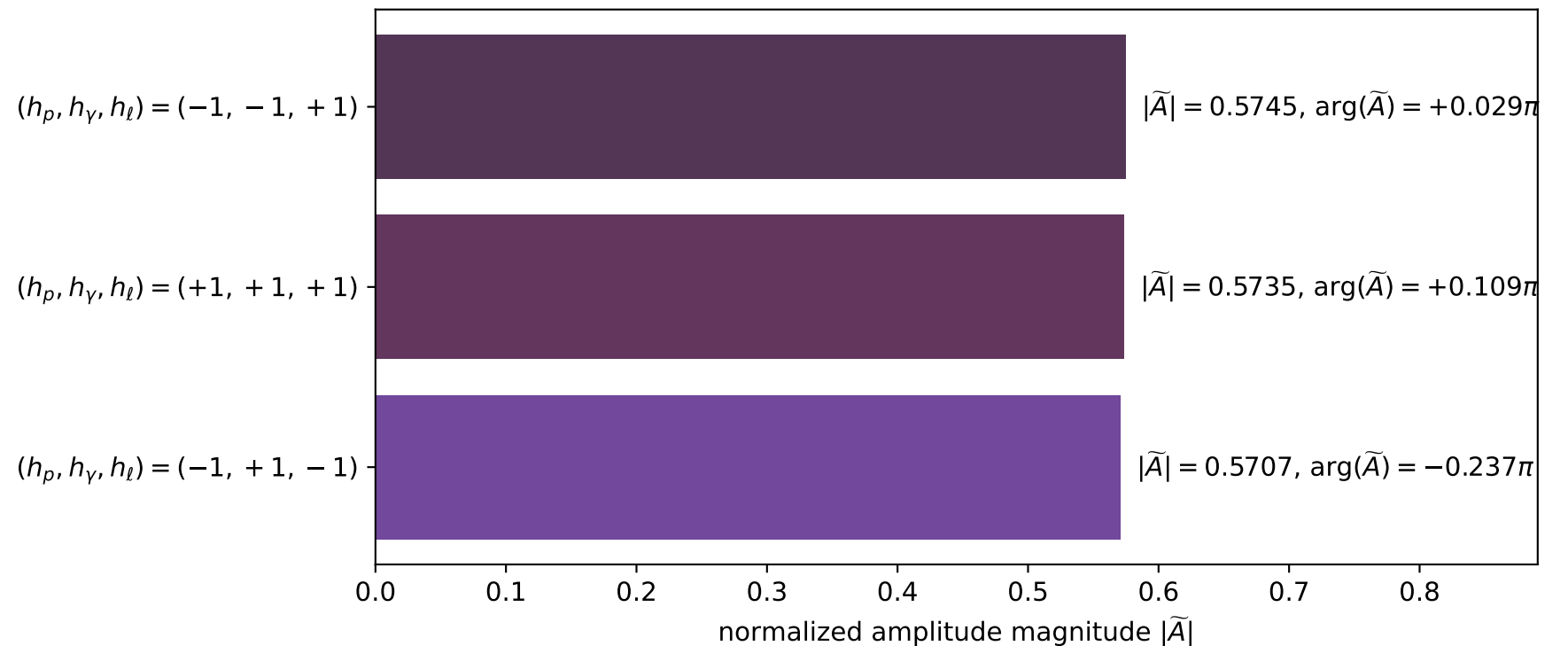


dw_mixing_angles_polarization_03_local_minimum_657
 $D_W=0.0057073$
 region: polarization_cluster_03_local_minimum_657
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0657
 lepton mixing angle: $\alpha_e=0.77071917$ rad
 incoming lepton: $0.71741|+\rangle + 0.696651|-\rangle$
 proton mixing angle: $\alpha_p=0.10578762$ rad
 incoming proton: $0.99441|+\rangle + 0.10559|-\rangle$

$s=1.13399$, $\sqrt{s}=1.06489$, $|\vec{P}|=0.11933$, $|\vec{P}'|=0.0395815$
 $\theta_{p'}=3.05612$, $\theta_\gamma=1.31674$, $\phi_{p'}=5.94507$, $\phi_\gamma=2.72054$
 $E_\gamma=0.0634418$, $\theta_{\ell'}=1.18619$, $\phi_{\ell'}=5.8573$
 $Q^2=0.0205499$, $x_B=0.14539$, $t=-0.0251732$, $W^2=1.00064$, $y=0.556145$

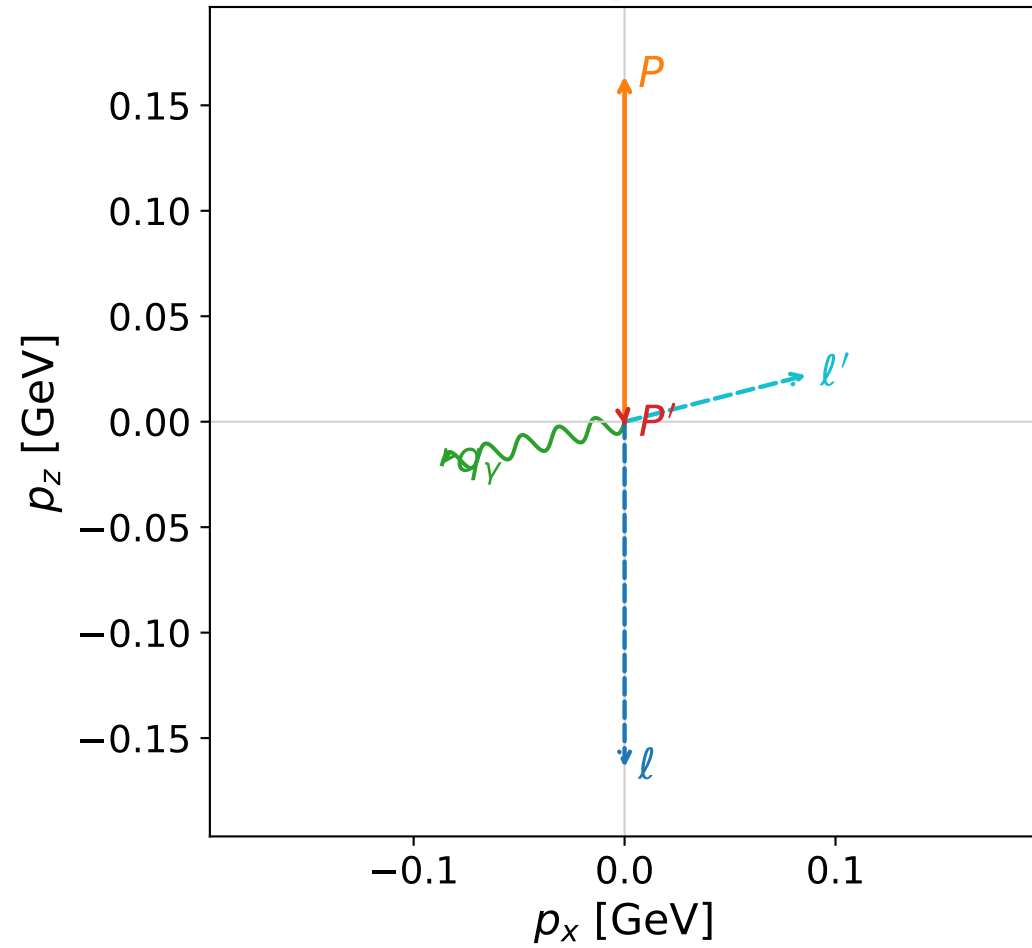
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1193$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1193$
 P $E= 0.9456$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1193$
 l' $E= 0.0626$ $p_x= 0.0529$ $p_y= -0.0240$ $p_z= 0.0235$
 P' $E= 0.9388$ $p_x= 0.0032$ $p_y= -0.0011$ $p_z= -0.0394$
 q_γ $E= 0.0634$ $p_x= -0.0560$ $p_y= 0.0251$ $p_z= 0.0159$

Leading normalized final-state amplitudes
(N=3, retained=98.5%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



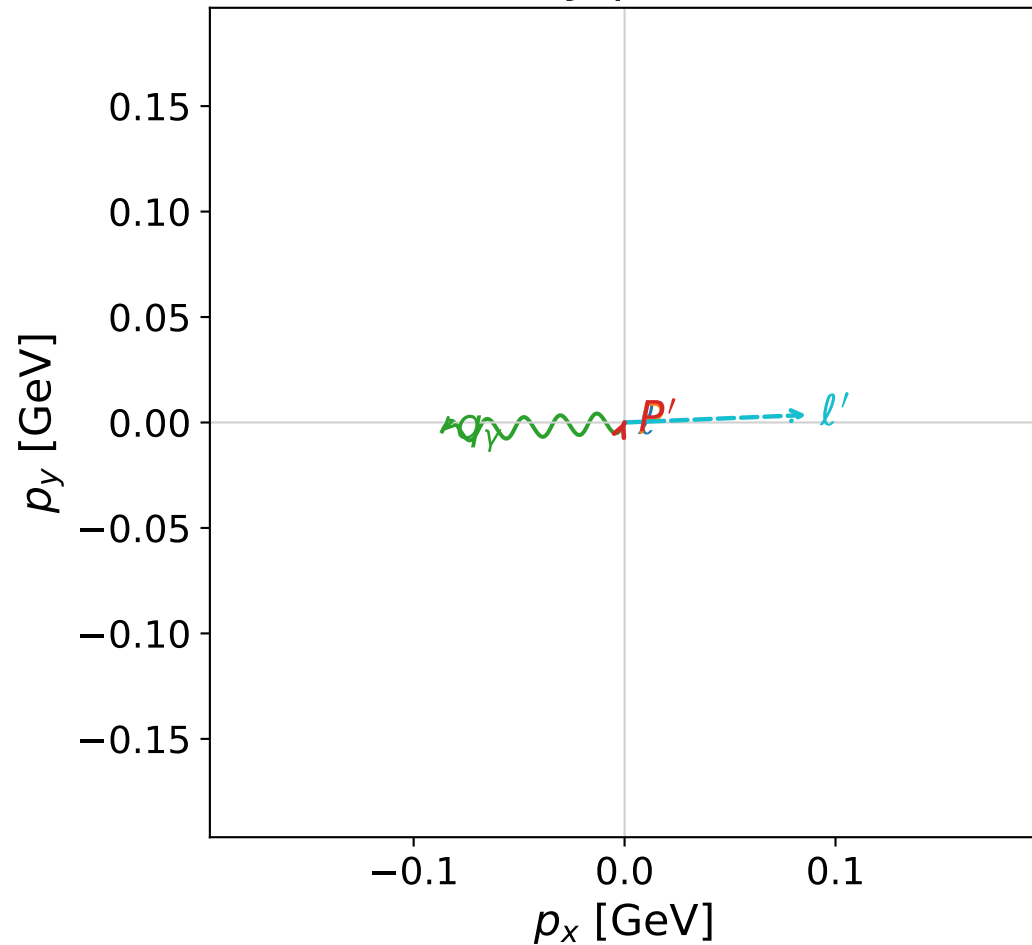
dw_mixing_angles_polarization_03_local_minimum_704
 $D_W=0.00743594$
 region: polarization_cluster_03_local_minimum_704
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0704
 lepton mixing angle: $\alpha_e=0.7651863$ rad
 incoming lepton: $0.721253|+\rangle + 0.692671|-\rangle$
 proton mixing angle: $\alpha_p=0.19293922$ rad
 incoming proton: $0.981445|+\rangle + 0.191744|-\rangle$

$s=1.24557$, $\sqrt{s}=1.11605$, $|\vec{P}|=0.163846$, $|\vec{P}'|=0.00200459$
 $\theta_{p'}=2.76927$, $\theta_\gamma=1.79959$, $\phi_{p'}=1.16918$, $\phi_\gamma=3.19046$
 $E_\gamma=0.0889545$, $\theta_{l'}=1.32079$, $\phi_{l'}=0.0412666$
 $Q^2=0.0364176$, $x_B=0.179153$, $t=-0.0272598$, $W^2=1.0467$, $y=0.555822$

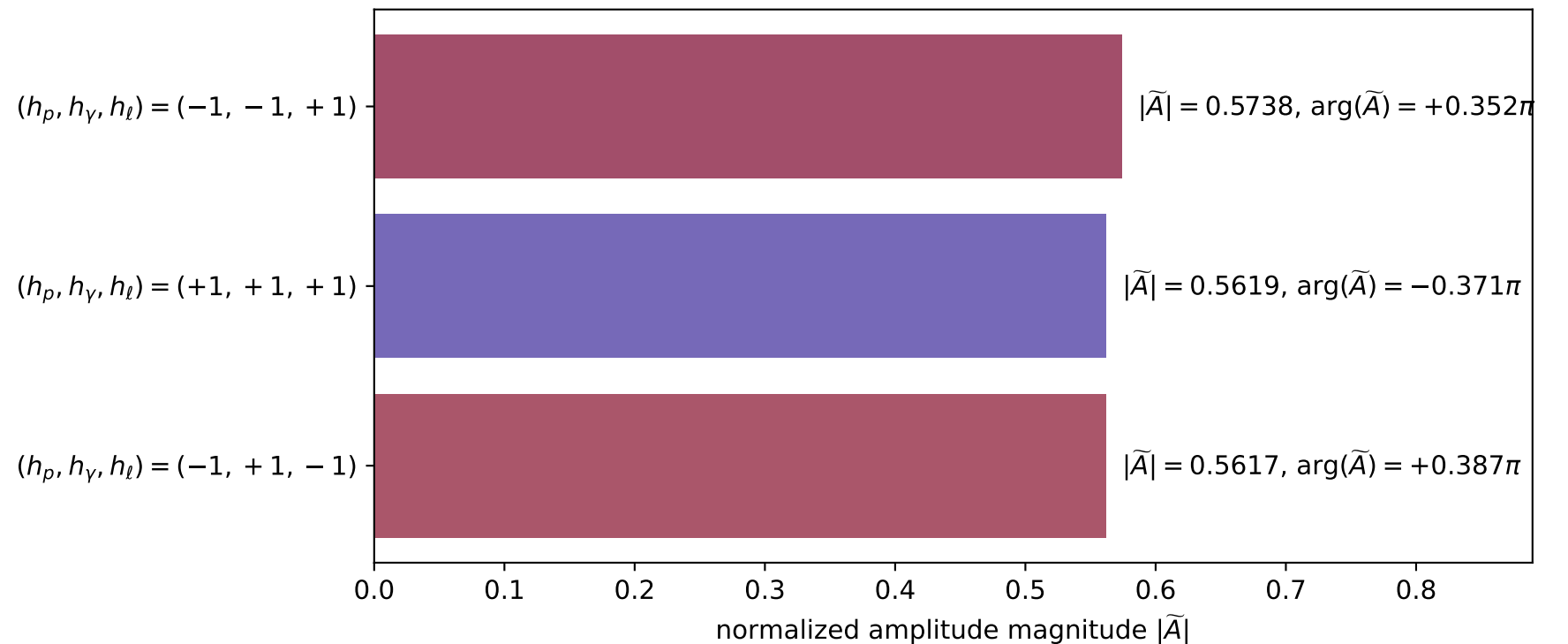
four-momenta $[E, p_x, p_y, p_z]$ GeV:

Particle	E	p_x	p_y	p_z
l	0.1638	0.0000	0.0000	-0.1638
P	0.9522	0.0000	0.0000	0.1638
l'	0.0891	0.0862	0.0036	0.0220
P'	0.9380	0.0003	0.0007	-0.0019
q_γ	0.0890	-0.0865	-0.0042	-0.0202

x--y plane

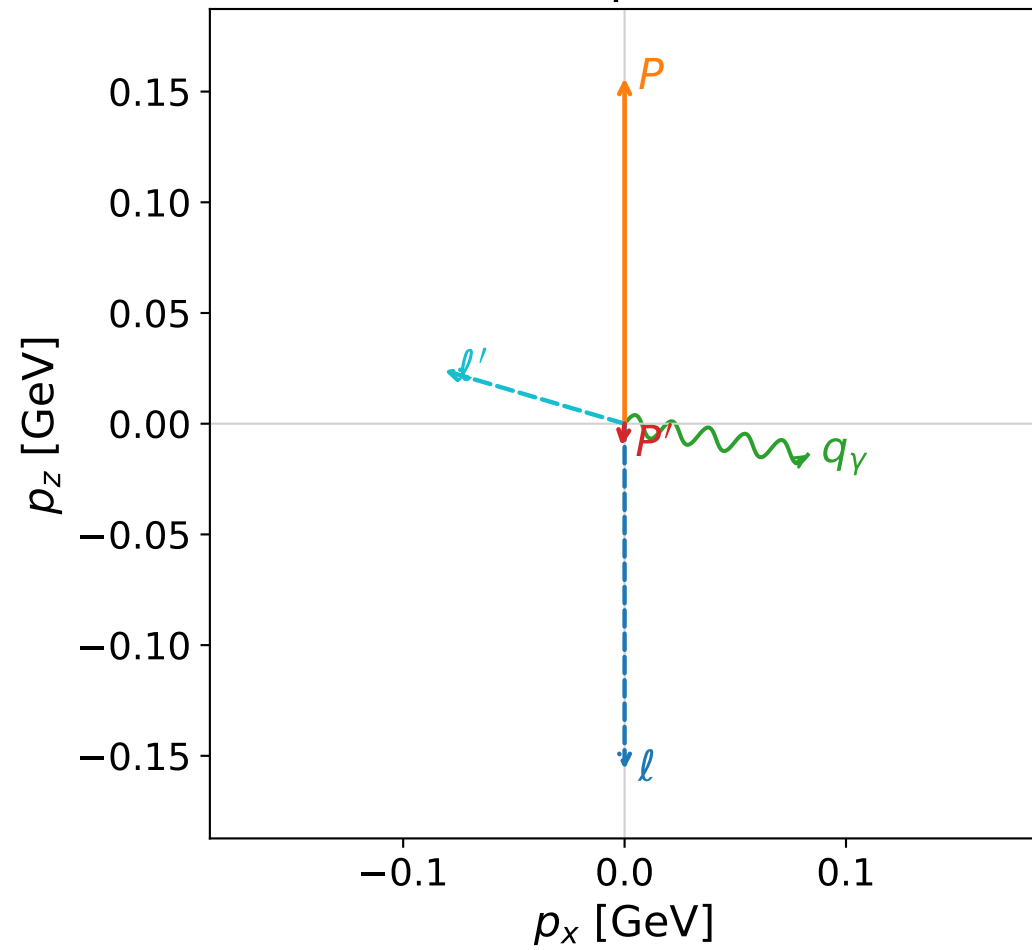


Leading normalized final-state amplitudes
(N=3, retained=96.1%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

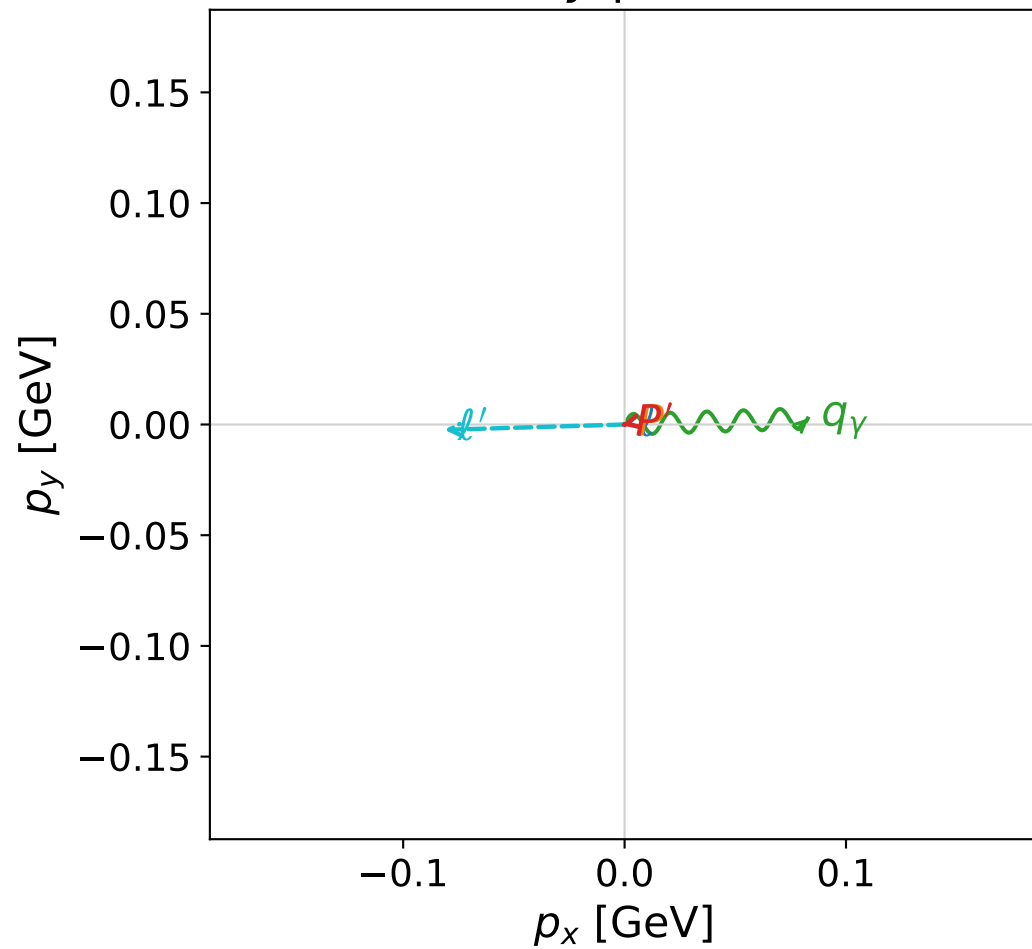


dw_mixing_angles_polarization_03_local_minimum_707
 $D_W=0.00749259$
 region: polarization_cluster_03_local_minimum_707
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0707
 lepton mixing angle: $\alpha_e=0.77624153$ rad
 incoming lepton: $0.713552|+\rangle + 0.700603|-\rangle$
 proton mixing angle: $\alpha_p=2.9784057$ rad
 incoming proton: $-0.986715|+\rangle + 0.162464|-\rangle$

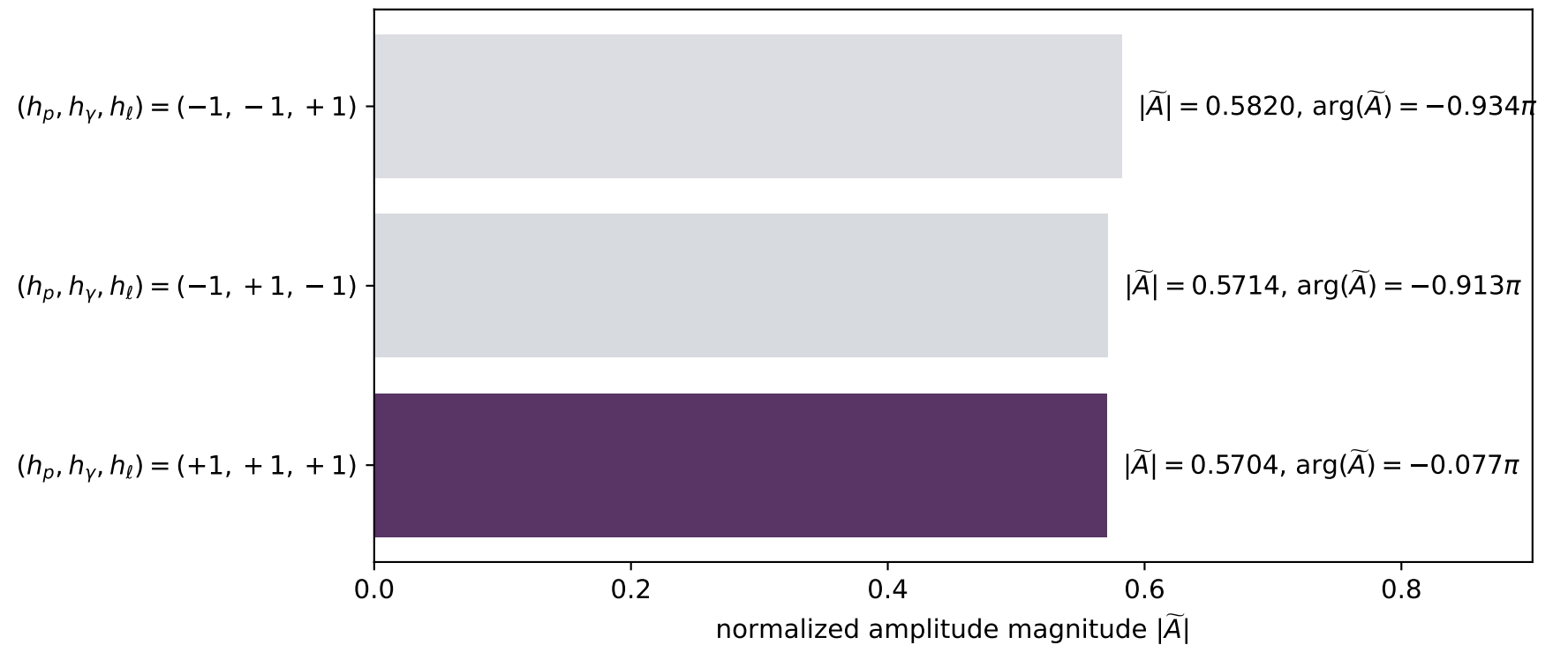
$s=1.22541$, $\sqrt{s}=1.10698$, $|\vec{P}|=0.156084$, $|\vec{P}'|=0.0100748$
 $\theta_{p'}=3.01025$, $\theta_\gamma=1.73909$, $\phi_{p'}=3.38298$, $\phi_\gamma=0.0329562$
 $E_\gamma=0.0839642$, $\theta_{\ell'}=1.28379$, $\phi_{\ell'}=3.1712$
 $Q^2=0.0340311$, $x_B=0.177718$, $t=-0.0274168$, $W^2=1.0373$, $y=0.554133$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1561$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1561$
 P $E= 0.9509$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1561$
 ℓ' $E= 0.0850$ $p_x= -0.0815$ $p_y= -0.0024$ $p_z= 0.0241$
 P' $E= 0.9381$ $p_x= -0.0013$ $p_y= -0.0003$ $p_z= -0.0100$
 q_γ $E= 0.0840$ $p_x= 0.0827$ $p_y= 0.0027$ $p_z= -0.0141$

x--y plane

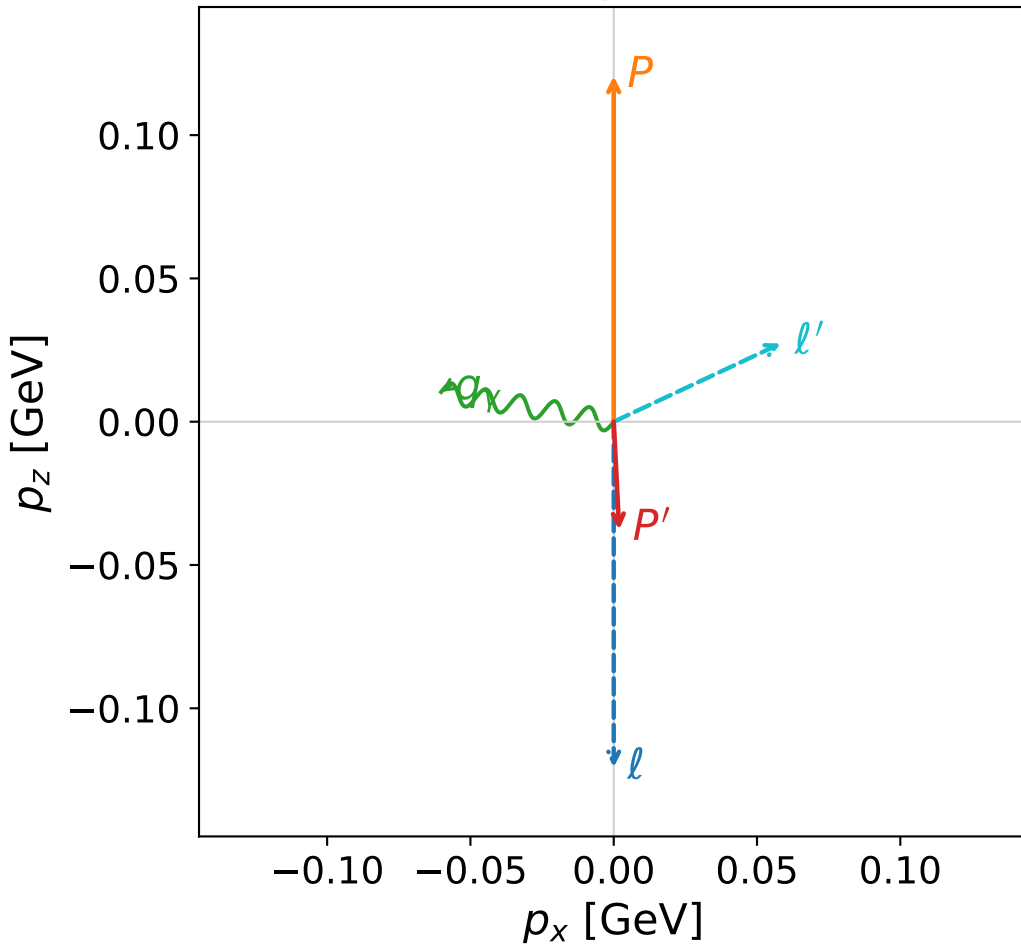


Leading normalized final-state amplitudes
 (N=3, retained=99.1%)

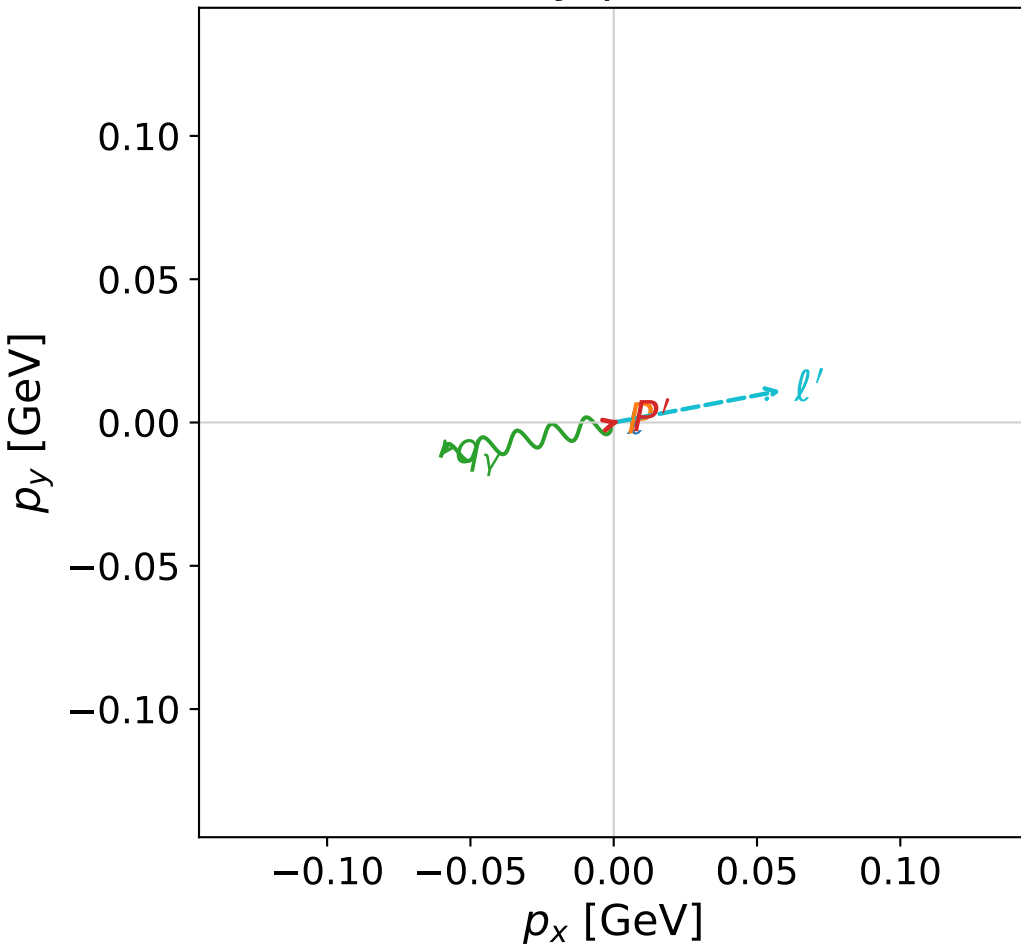


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_03_local_minimum_722

$D_W=0.00805759$

region: polarization_cluster_03_local_minimum_722

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0722

lepton mixing angle: $\alpha_e=0.76206041$ rad

incoming lepton: $0.723415|+\rangle + 0.690413|-\rangle$

proton mixing angle: $\alpha_p=0.11757213$ rad

incoming proton: $0.993096|+\rangle + 0.117301|-\rangle$

$s=1.13702$, $\sqrt{s}=1.06631$, $|\vec{P}|=0.120589$, $|\vec{P}'|=0.0377302$

$\theta_{p'}=3.08974$, $\theta_\gamma=1.40324$, $\phi_{p'}=0.260976$, $\phi_\gamma=3.33166$

$E_\gamma=0.0621959$, $\theta_{l'}=1.13973$, $\phi_{l'}=0.187734$

$Q^2=0.0223475$, $x_B=0.159463$, $t=-0.0250042$, $W^2=0.997639$, $y=0.544938$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E=0.1206$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1206$

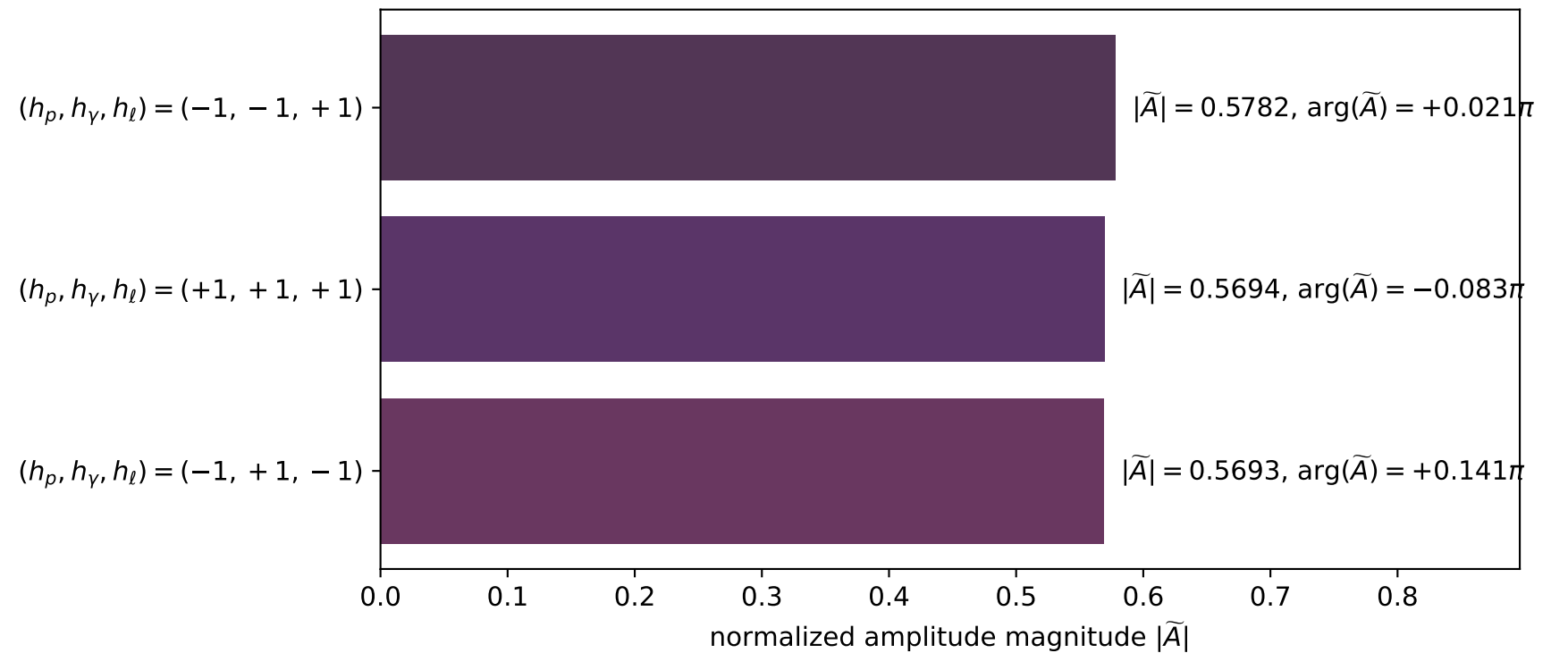
P $E=0.9457$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1206$

l' $E=0.0654$ $p_x=0.0583$ $p_y=0.0111$ $p_z=0.0273$

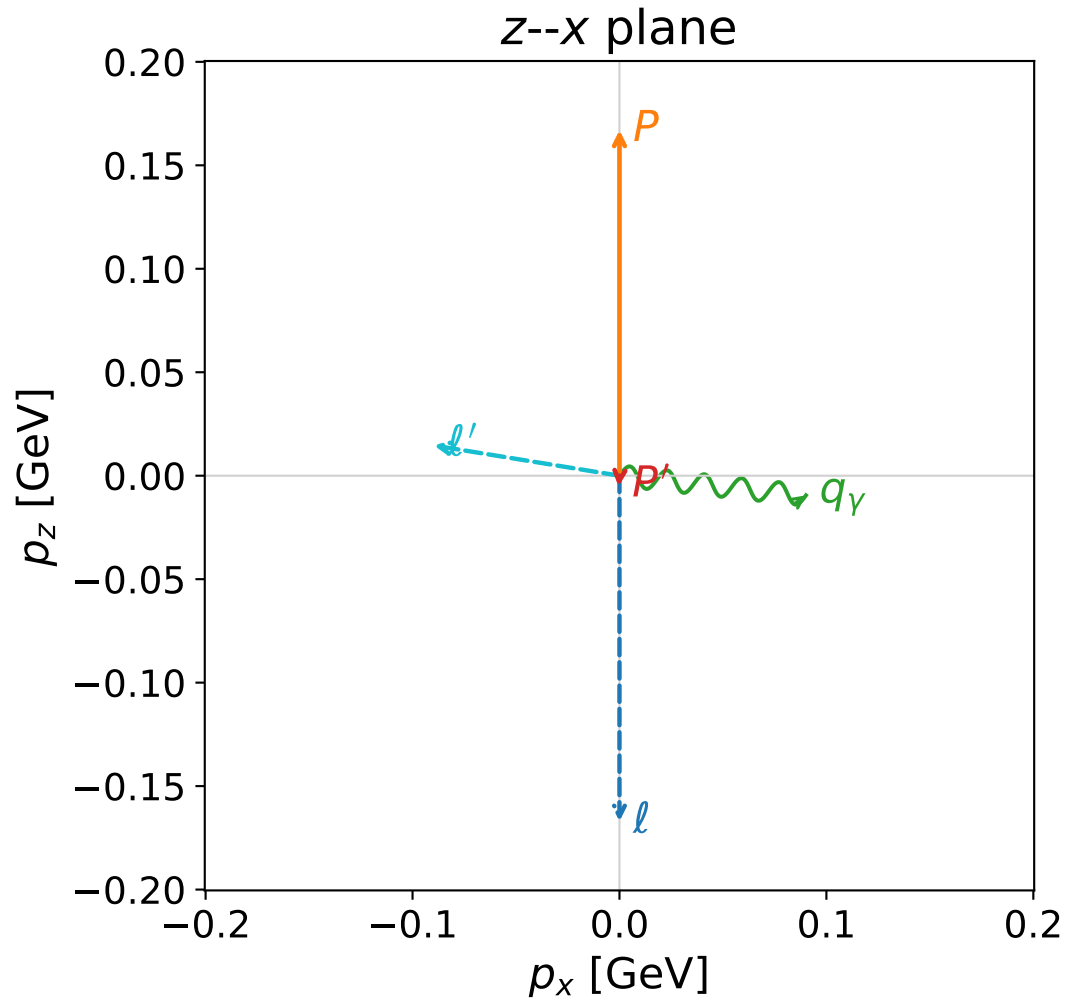
P' $E=0.9388$ $p_x=0.0019$ $p_y=0.0005$ $p_z=-0.0377$

q_γ $E=0.0622$ $p_x=-0.0602$ $p_y=-0.0116$ $p_z=0.0104$

Leading normalized final-state amplitudes
(N=3, retained=98.3%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

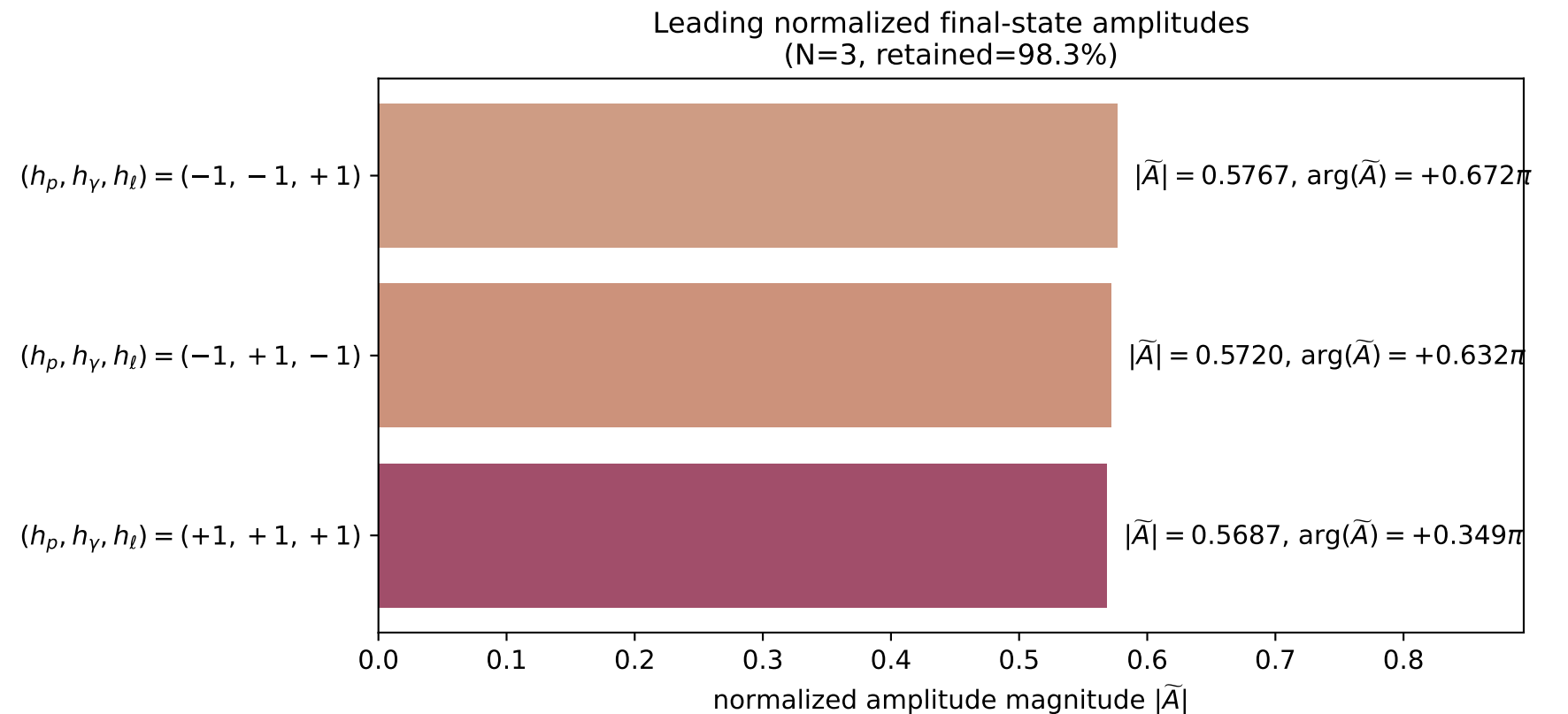
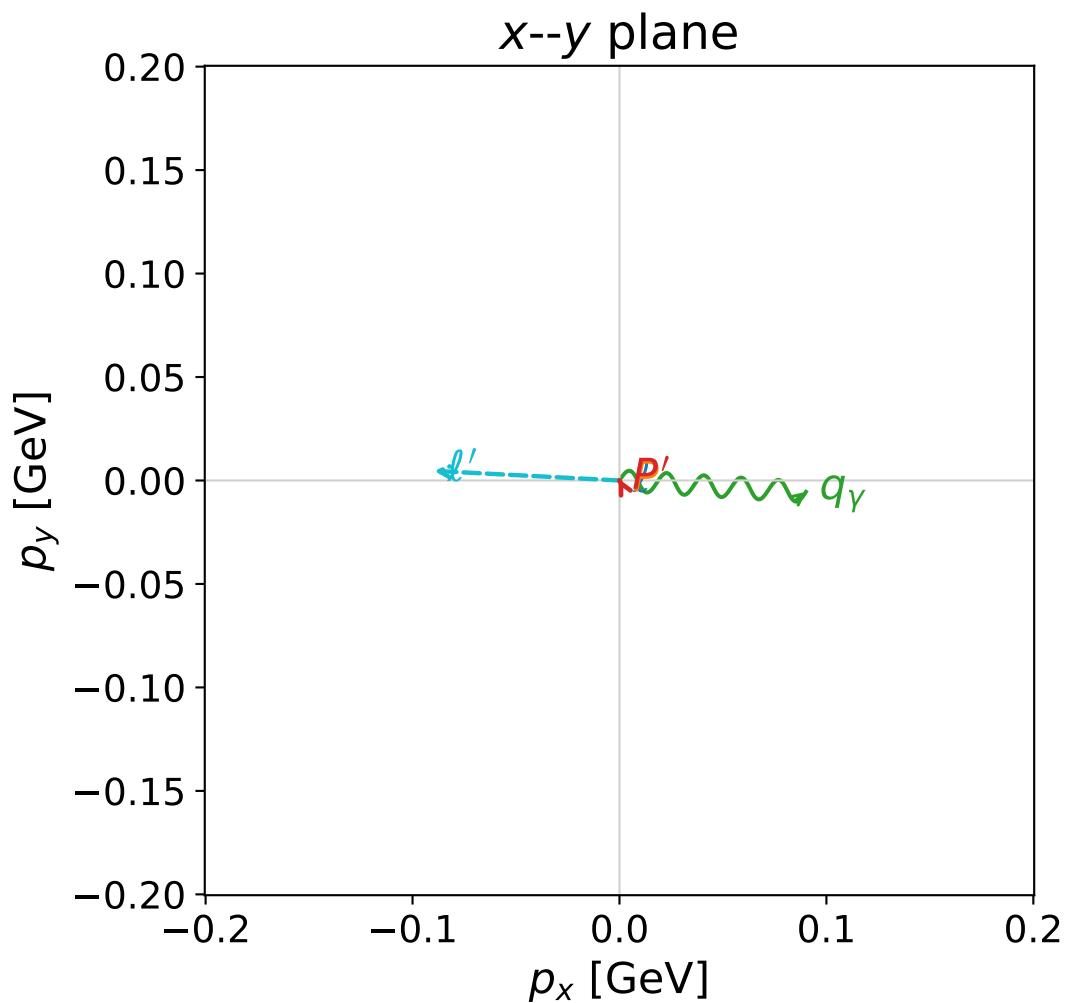


dw_mixing_angles_polarization_03_local_minimum_735
 $D_W=0.00848145$
 region: polarization_cluster_03_local_minimum_735
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0735
 lepton mixing angle: $\alpha_e=0.78322937$ rad
 incoming lepton: $0.708639|+\rangle + 0.705572|-\rangle$
 proton mixing angle: $\alpha_p=2.9506423$ rad
 incoming proton: $-0.981824|+\rangle + 0.189792|-\rangle$

$s=1.25386$, $\sqrt{s}=1.11976$, $|\vec{P}|=0.167008$, $|\vec{P}'|=0.0052075$
 $\theta_{p'}=2.95455$, $\theta_\gamma=1.67547$, $\phi_{p'}=2.04328$, $\phi_\gamma=6.22212$
 $E_\gamma=0.0907752$, $\theta_{\ell'}=1.4096$, $\phi_{\ell'}=3.08981$
 $Q^2=0.0352621$, $x_B=0.171549$, $t=-0.0294106$, $W^2=1.05013$, $y=0.549574$

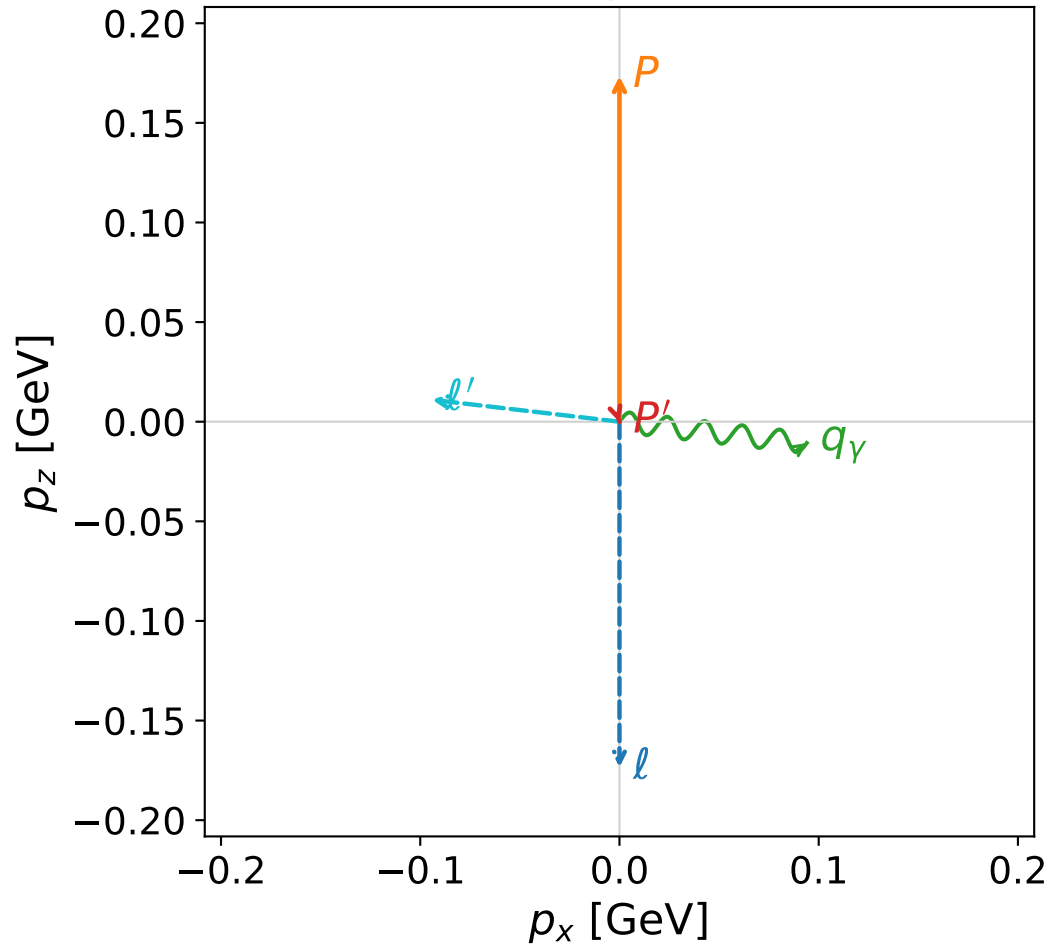
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.1670	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1670
P	$E=$	0.9528	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1670
ℓ'	$E=$	0.0910	$p_x=$	-0.0897	$p_y=$	0.0046	$p_z=$	0.0146
P'	$E=$	0.9380	$p_x=$	-0.0004	$p_y=$	0.0009	$p_z=$	-0.0051
q_γ	$E=$	0.0908	$p_x=$	0.0901	$p_y=$	-0.0055	$p_z=$	-0.0095



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



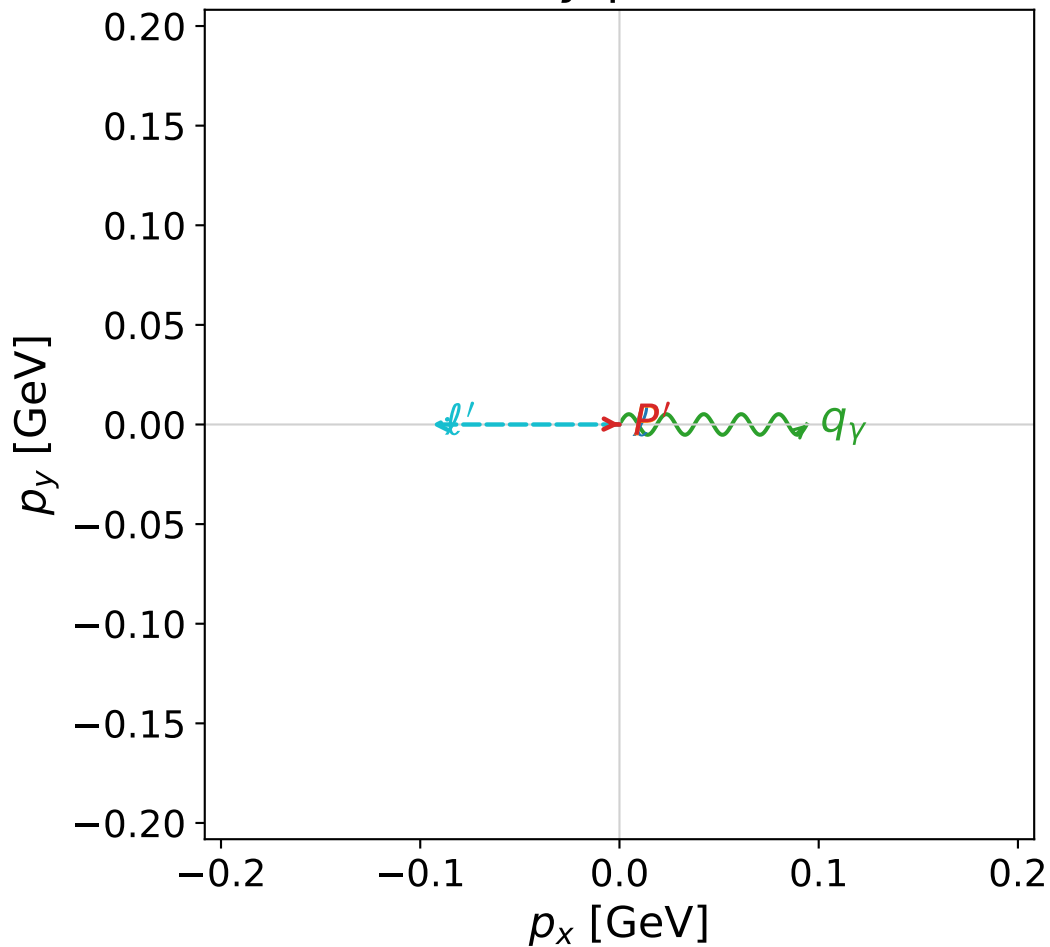
dw_mixing_angles_polarization_03_local_minimum_772
 $D_W=0.0095849$
 region: polarization_cluster_03_local_minimum_772
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0772
 lepton mixing angle: $\alpha_e=0.78701964$ rad
 incoming lepton: $0.705959|+\rangle + 0.708252|-\rangle$
 proton mixing angle: $\alpha_p=2.9424013$ rad
 incoming proton: $-0.980227|+\rangle + 0.197877|-\rangle$

$s=1.27096$, $\sqrt{s}=1.12737$, $|\vec{P}|=0.173463$, $|\vec{P}'|=0.000335174$
 $\theta_{p'}=2.84197$, $\theta_\gamma=1.68397$, $\phi_{p'}=6.25528$, $\phi_\gamma=0.000170876$
 $E_\gamma=0.0946156$, $\theta_{\ell'}=1.45439$, $\phi_{\ell'}=3.14173$
 $Q^2=0.0366894$, $x_B=0.171315$, $t=-0.0299475$, $W^2=1.05732$, $y=0.547572$

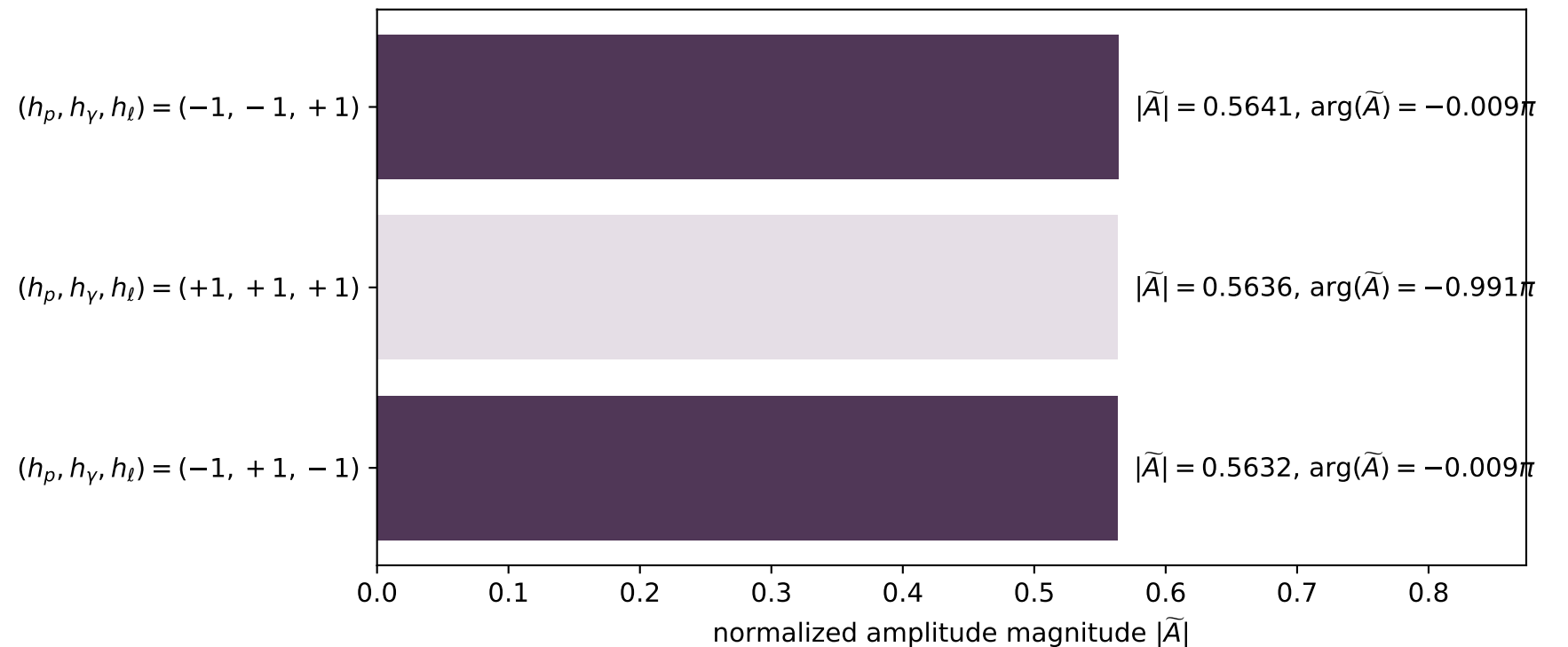
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.1735	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1735
P	$E=$	0.9539	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1735
ℓ'	$E=$	0.0948	$p_x=$	-0.0941	$p_y=$	-0.0000	$p_z=$	0.0110
P'	$E=$	0.9380	$p_x=$	0.0001	$p_y=$	-0.0000	$p_z=$	-0.0003
q_γ	$E=$	0.0946	$p_x=$	0.0940	$p_y=$	0.0000	$p_z=$	-0.0107

x--y plane

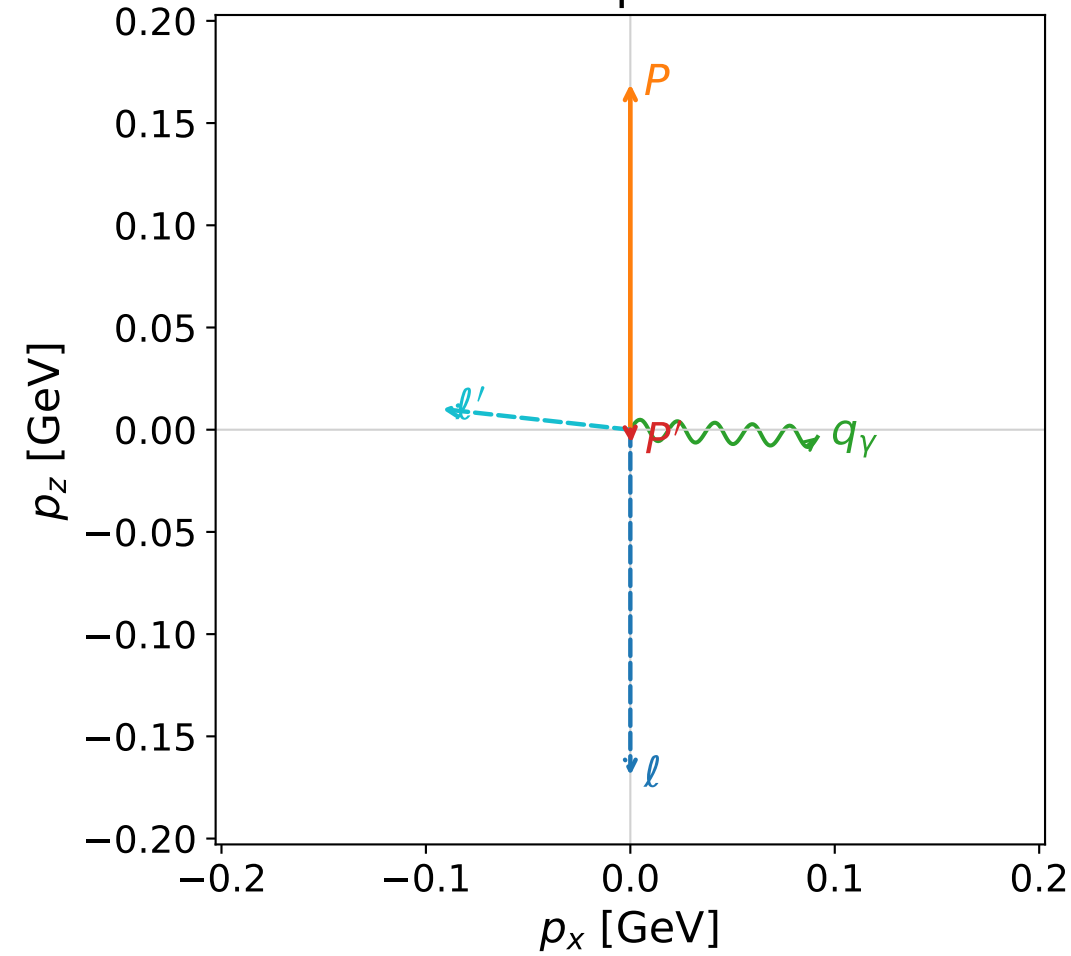


Leading normalized final-state amplitudes
 (N=3, retained=95.3%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



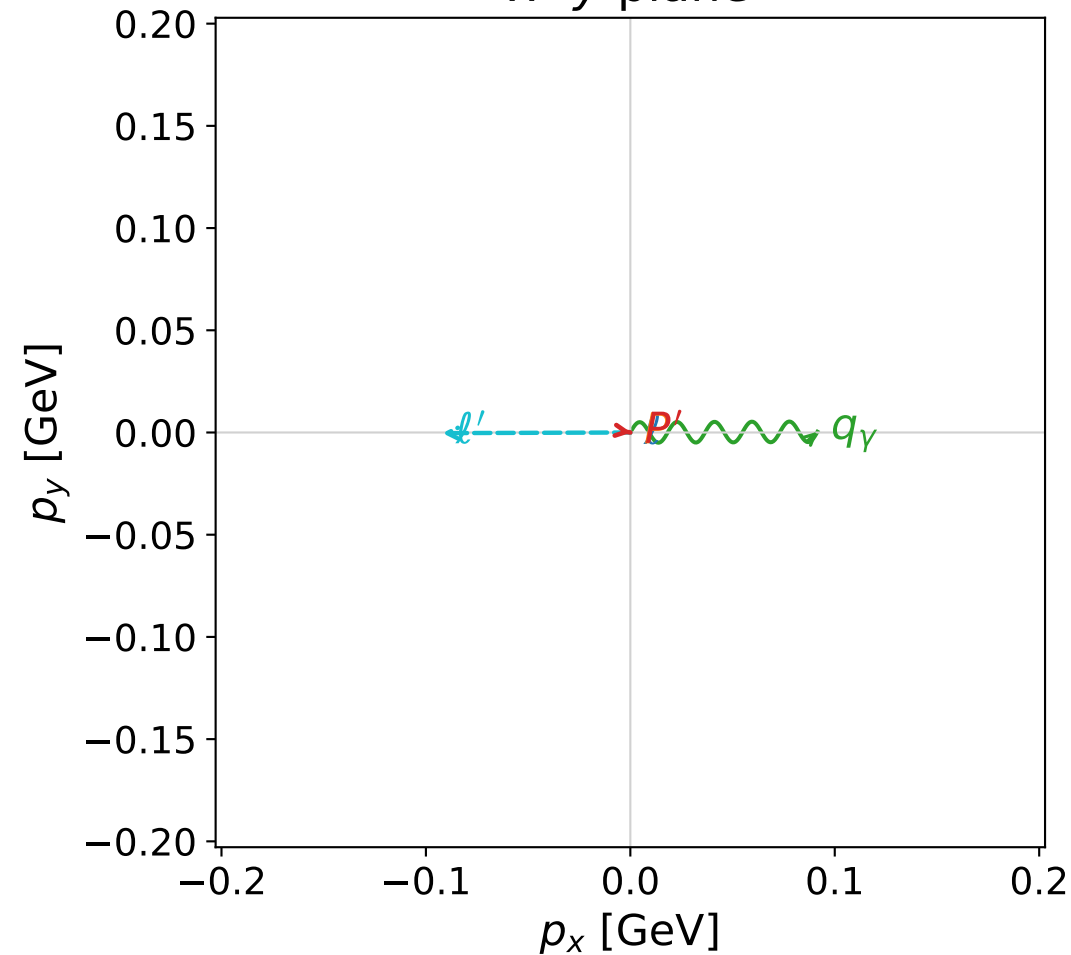
dw_mixing_angles_polarization_03_local_minimum_800
 $D_W=0.0103956$
 region: polarization_cluster_03_local_minimum_800
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0800
 lepton mixing angle: $\alpha_e=0.78689655$ rad
 incoming lepton: $0.706046|+\rangle + 0.708166|-\rangle$
 proton mixing angle: $\alpha_p=2.9368835$ rad
 incoming proton: $-0.97912|+\rangle + 0.203282|-\rangle$

$s=1.25926$, $\sqrt{s}=1.12217$, $|\vec{P}|=0.169055$, $|\vec{P}'|=0.00653172$
 $\theta_{p'}=3.07246$, $\theta_\gamma=1.61004$, $\phi_{p'}=6.13637$, $\phi_\gamma=0.00404817$
 $E_\gamma=0.0916075$, $\theta_{\ell'}=1.46132$, $\phi_{\ell'}=3.1449$
 $Q^2=0.0347062$, $x_B=0.16812$, $t=-0.0305978$, $W^2=1.05158$, $y=0.54409$

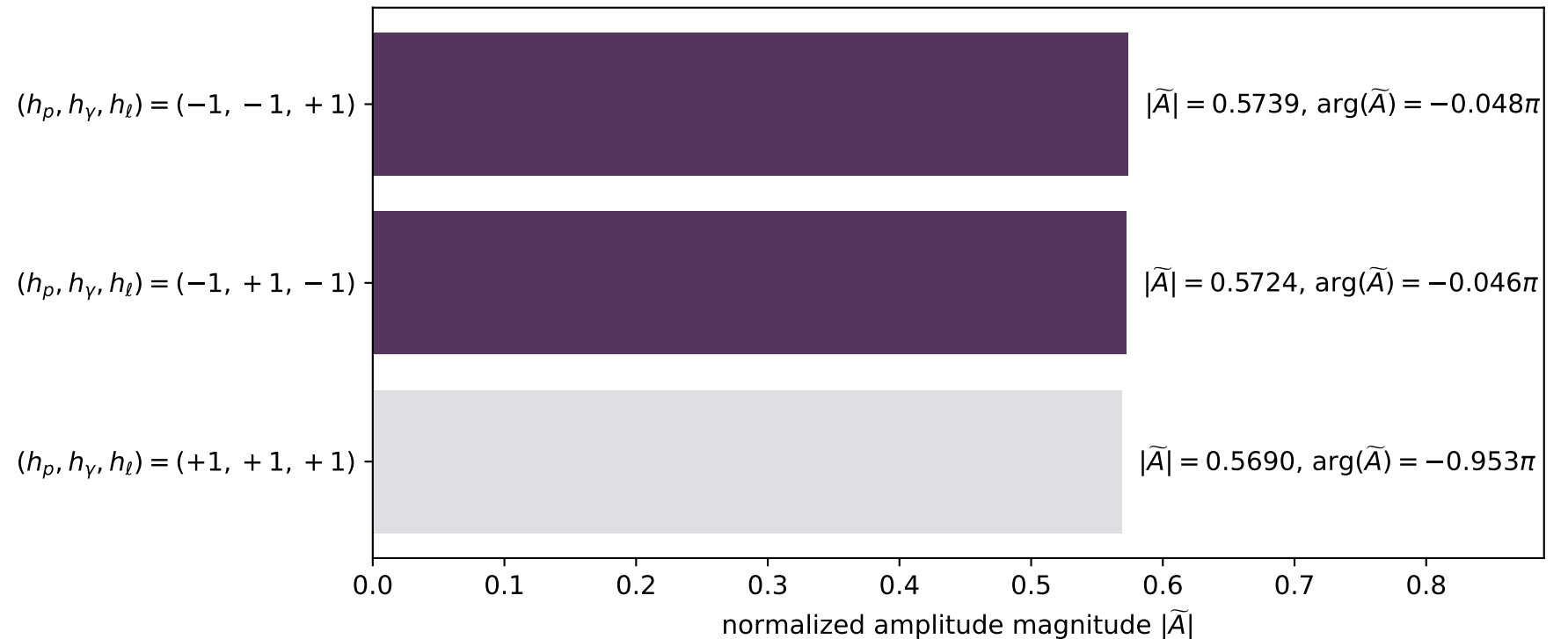
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.1691	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1691
P	$E=$	0.9531	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1691
ℓ'	$E=$	0.0925	$p_x=$	-0.0920	$p_y=$	-0.0003	$p_z=$	0.0101
P'	$E=$	0.9380	$p_x=$	0.0004	$p_y=$	-0.0001	$p_z=$	-0.0065
q_γ	$E=$	0.0916	$p_x=$	0.0915	$p_y=$	0.0004	$p_z=$	-0.0036

x--y plane

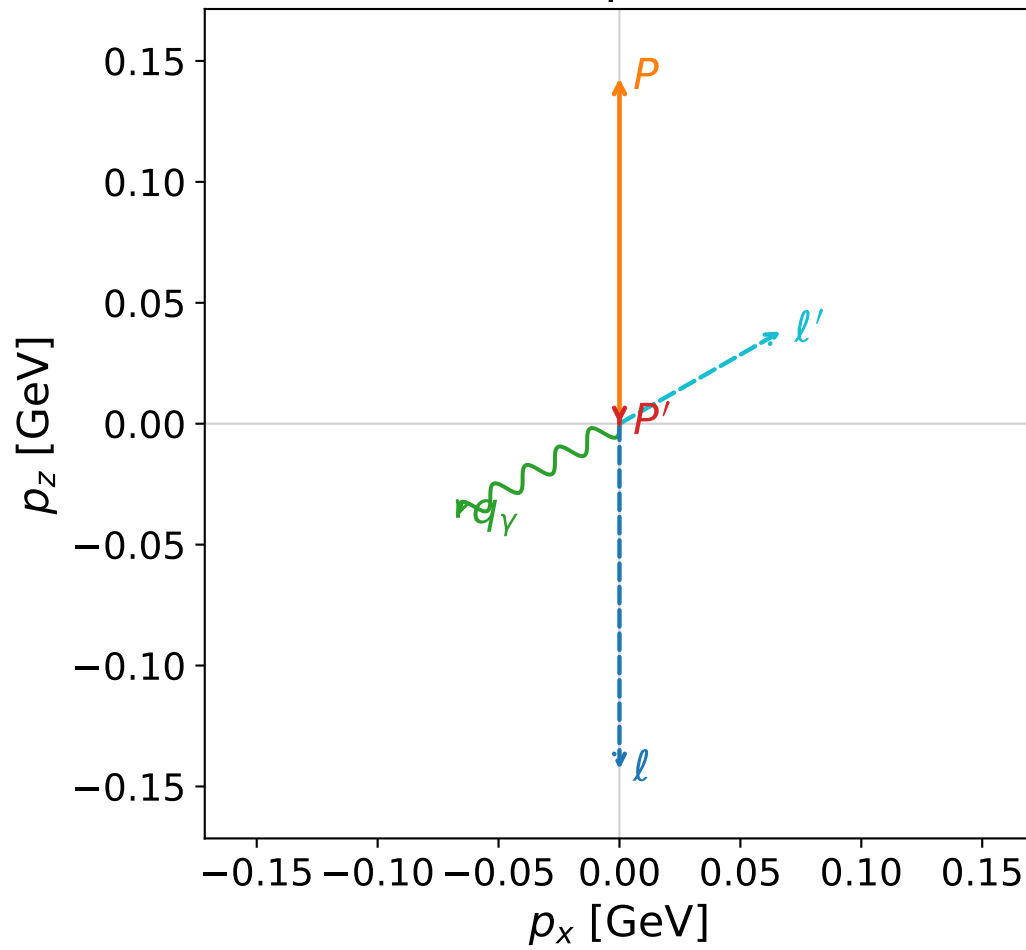


Leading normalized final-state amplitudes
(N=3, retained=98.1%)

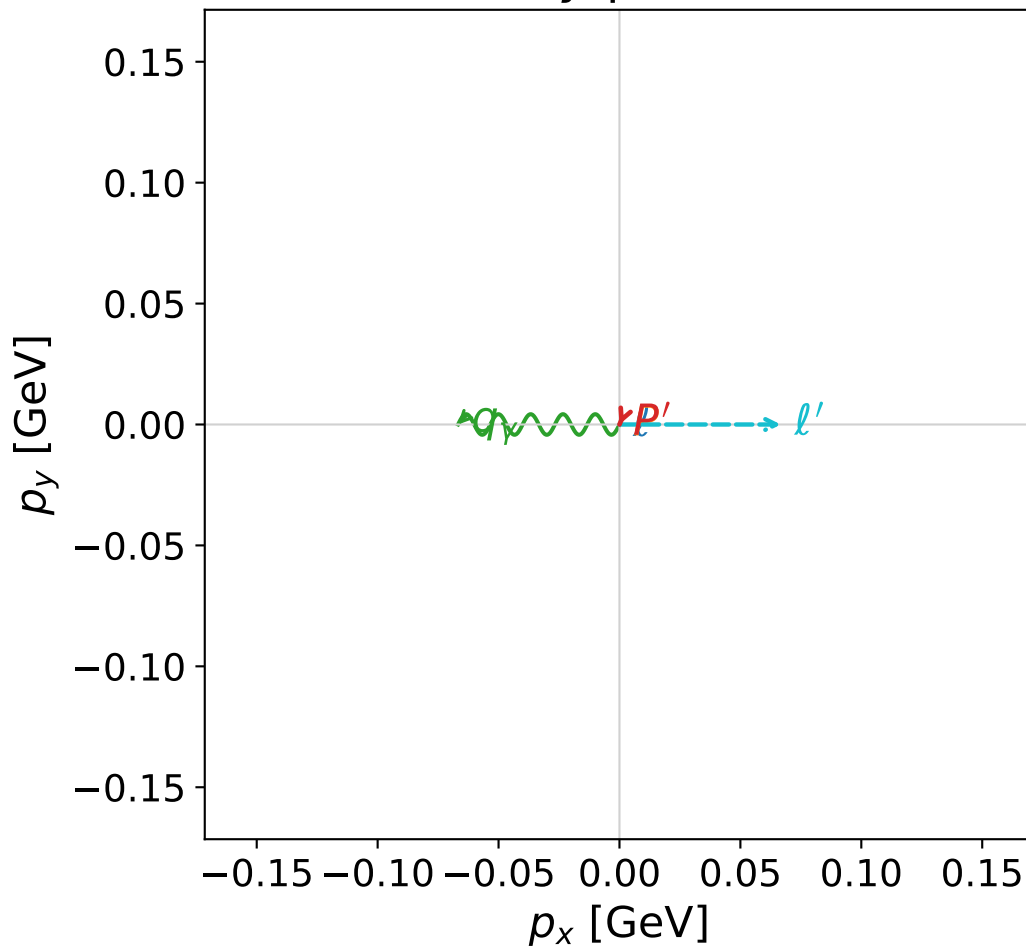


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

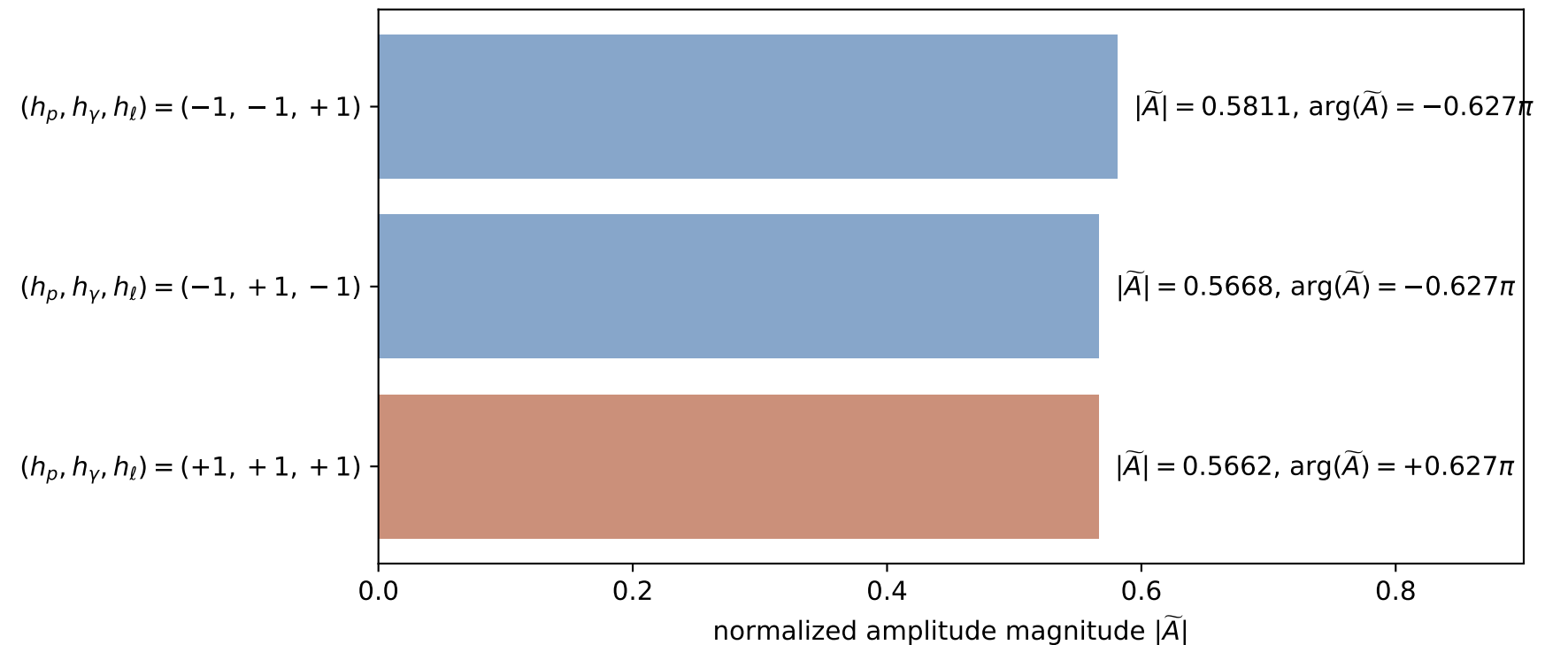


dw_mixing_angles_polarization_03_local_minimum_813
 $D_W=0.0107951$
 region: polarization_cluster_03_local_minimum_813
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0813
 lepton mixing angle: $\alpha_e=0.75167736$ rad
 incoming lepton: $0.730544|+\rangle + 0.682865|-\rangle$
 proton mixing angle: $\alpha_p=0.12679047$ rad
 incoming proton: $0.991973|+\rangle + 0.126451|-\rangle$

$s=1.19188$, $\sqrt{s}=1.09173$, $|\vec{P}|=0.142907$, $|\vec{P}'|=3.11807e-05$
 $\theta_{p'}=3.13792$, $\theta_\gamma=2.08925$, $\phi_{p'}=4.3132$, $\phi_\gamma=3.14151$
 $E_\gamma=0.076857$, $\theta_{l'}=1.05199$, $\phi_{l'}=6.28311$
 $Q^2=0.0328655$, $x_B=0.185632$, $t=-0.0203141$, $W^2=1.02403$, $y=0.567398$

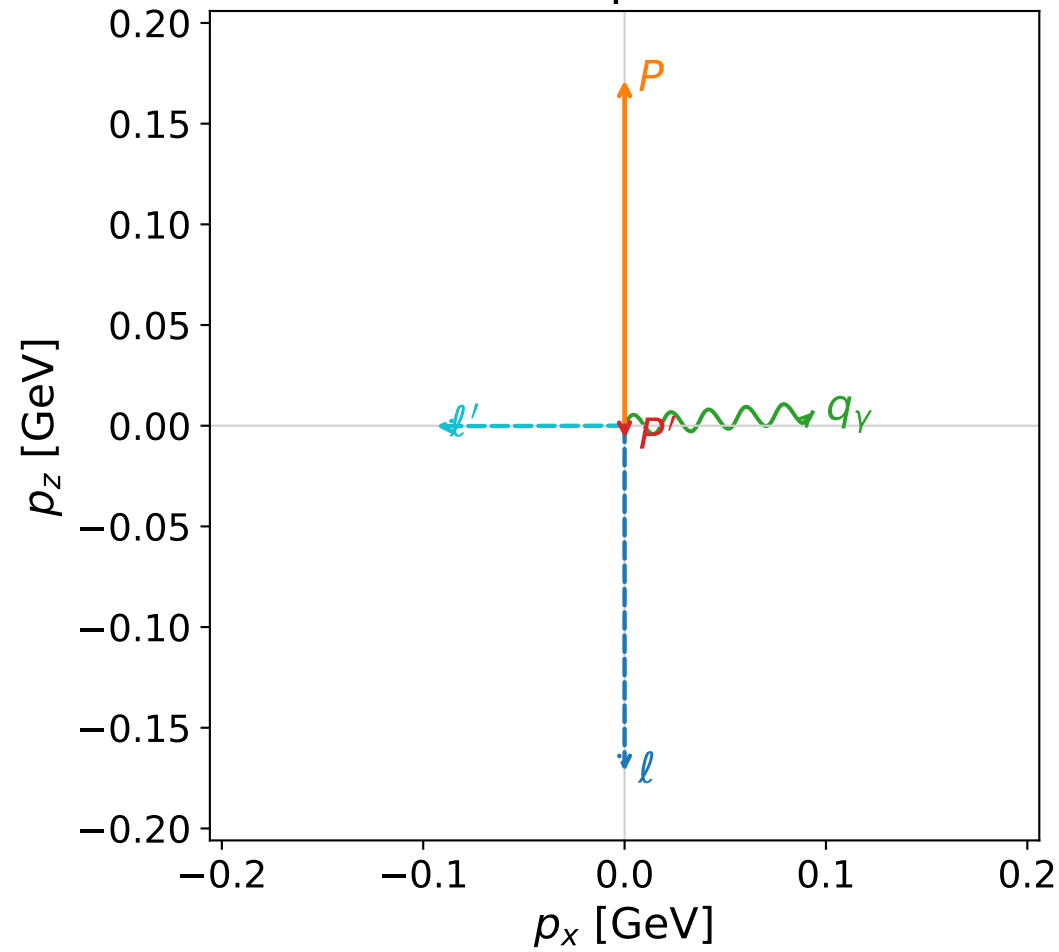
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1429$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1429$
 P $E= 0.9488$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1429$
 l' $E= 0.0769$ $p_x= 0.0668$ $p_y= -0.0000$ $p_z= 0.0381$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_γ $E= 0.0769$ $p_x= -0.0668$ $p_y= 0.0000$ $p_z= -0.0381$

Leading normalized final-state amplitudes
 (N=3, retained=98.0%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

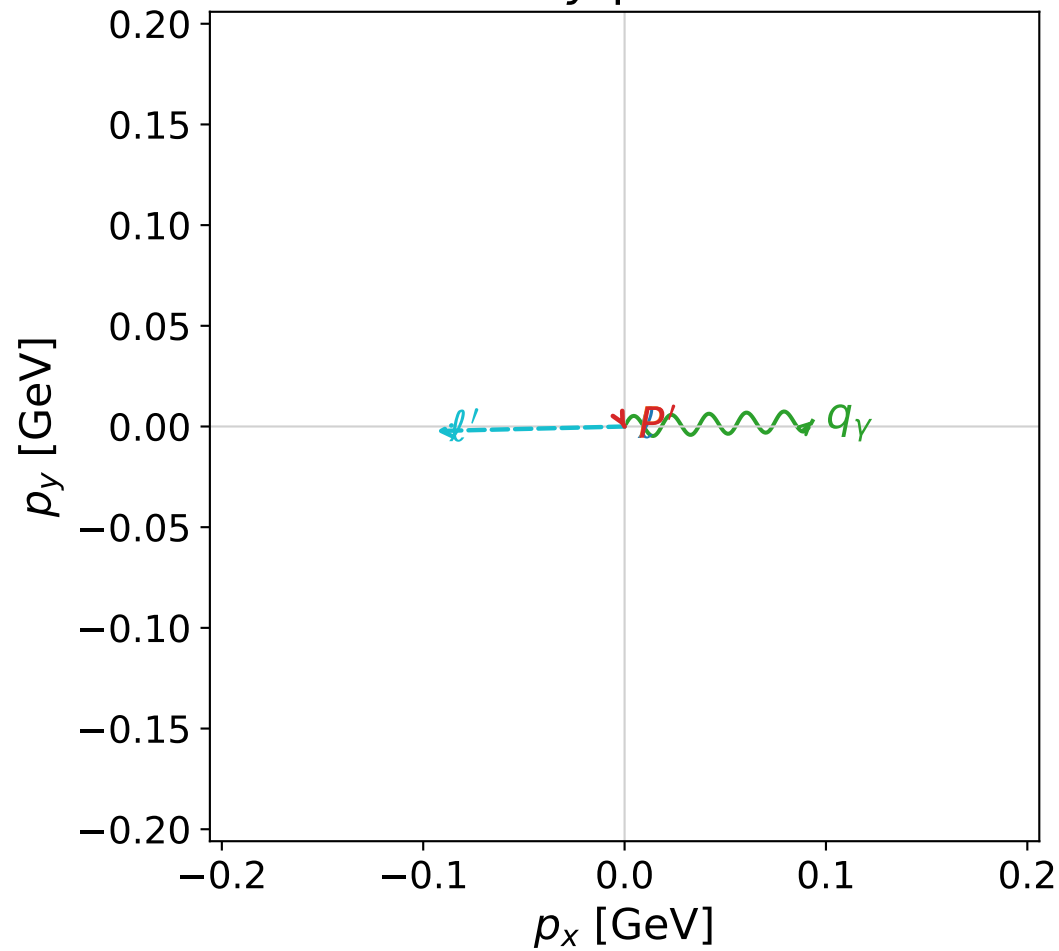


dw_mixing_angles_polarization_03_local_minimum_815
 $D_W=0.010864$
 region: polarization_cluster_03_local_minimum_815
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0815
 lepton mixing angle: $\alpha_e=0.79883948$ rad
 incoming lepton: $0.697539|+\rangle +0.716547|-\rangle$
 proton mixing angle: $\alpha_p=2.9379666$ rad
 incoming proton: $-0.97934|+\rangle +0.202222|-\rangle$

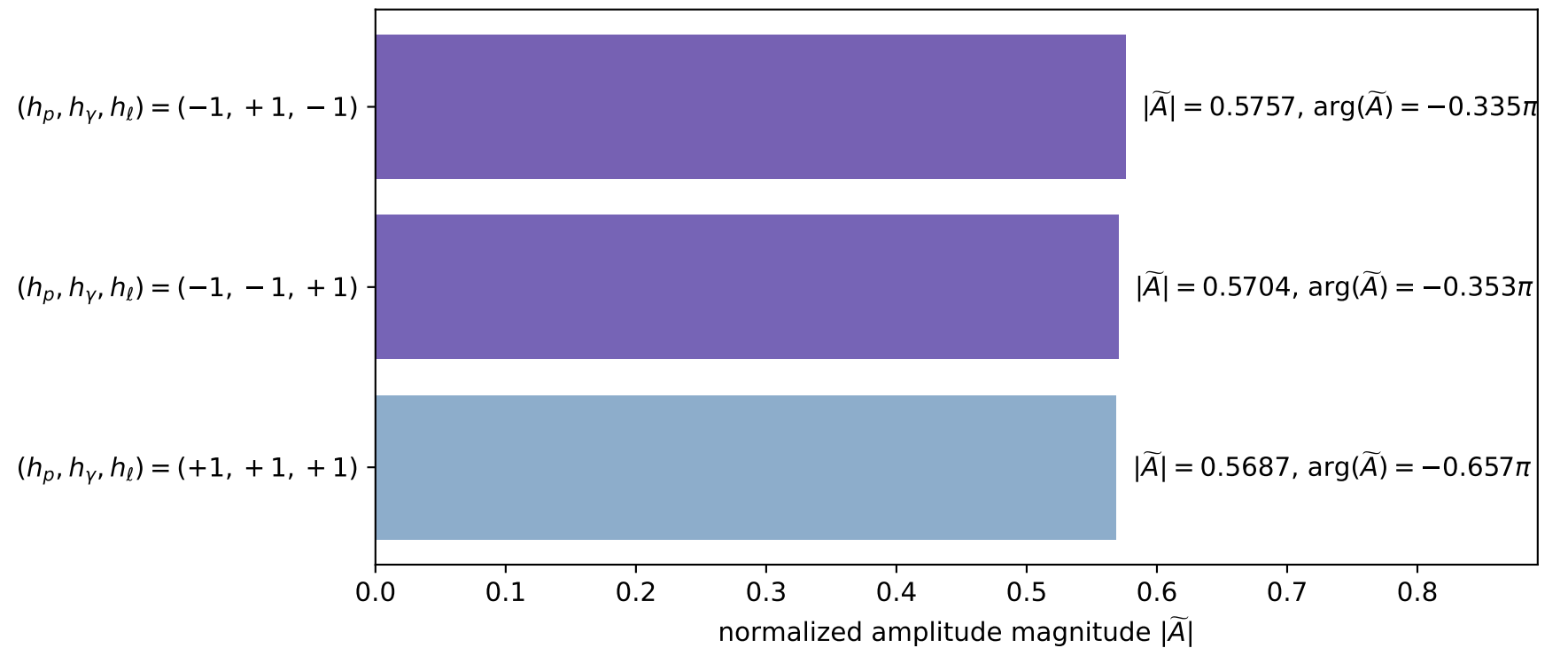
$s=1.26605$, $\sqrt{s}=1.12519$, $|\vec{P}|=0.171619$, $|\vec{P}'|=0.00636176$
 $\theta_{p'}=3.05711$, $\theta_\gamma=1.50047$, $\phi_{p'}=5.20611$, $\phi_\gamma=0.0292564$
 $E_\gamma=0.0935787$, $\theta_{\ell'}=1.57333$, $\phi_{\ell'}=3.16572$
 $Q^2=0.0320423$, $x_B=0.154318$, $t=-0.0314278$, $W^2=1.05544$, $y=0.53763$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1716$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1716$
 P $E= 0.9536$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1716$
 ℓ' $E= 0.0936$ $p_x= -0.0936$ $p_y= -0.0023$ $p_z= -0.0002$
 P' $E= 0.9380$ $p_x= 0.0003$ $p_y= -0.0005$ $p_z= -0.0063$
 q_γ $E= 0.0936$ $p_x= 0.0933$ $p_y= 0.0027$ $p_z= 0.0066$

x--y plane

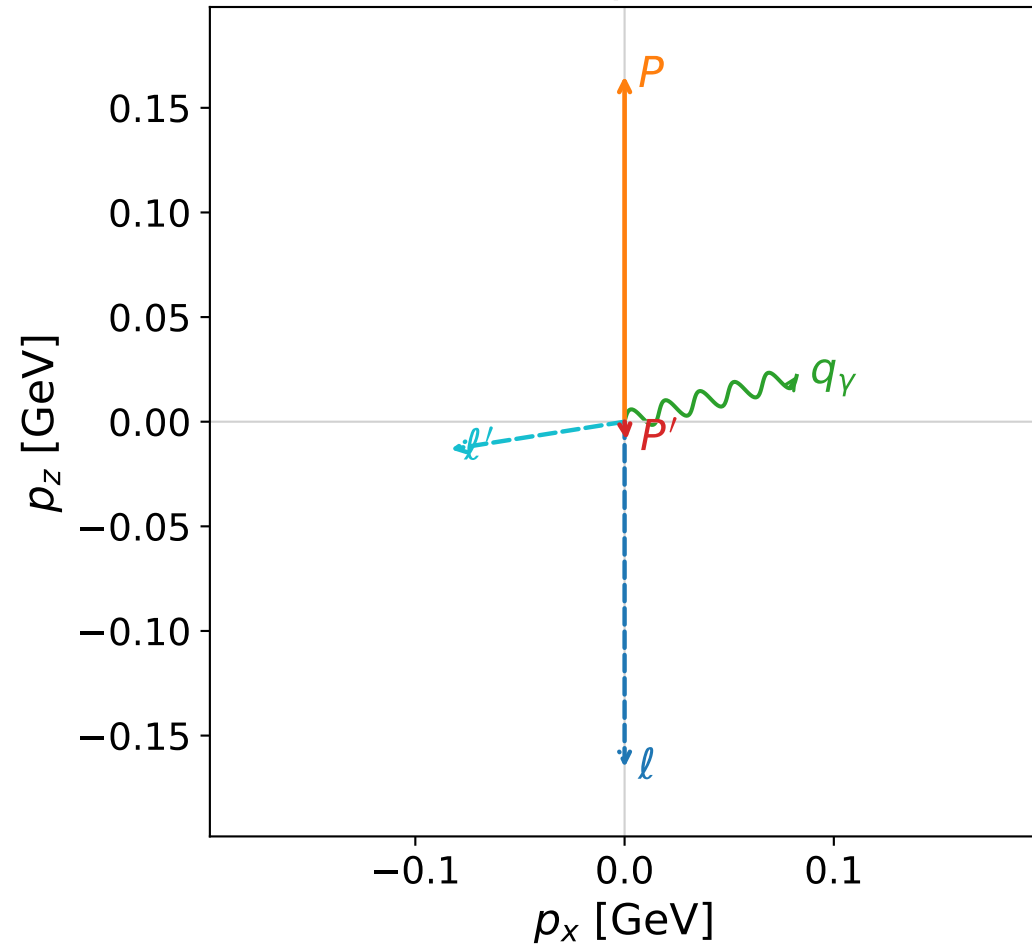


Leading normalized final-state amplitudes
 (N=3, retained=98.0%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

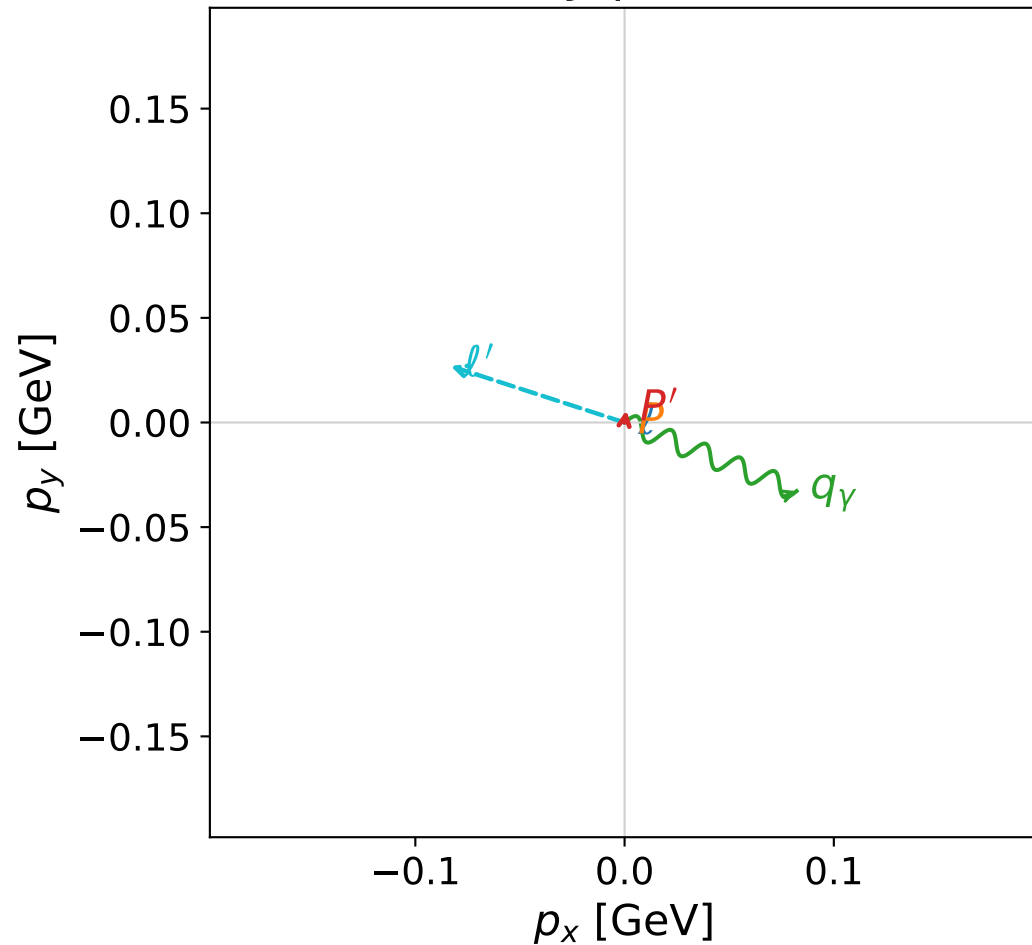


dw_mixing_angles_polarization_03_local_minimum_839
 $D_W=0.0116253$
 region: polarization_cluster_03_local_minimum_839
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0839
 lepton mixing angle: $\alpha_e=0.81111552$ rad
 incoming lepton: $0.68869|+\rangle + 0.725056|-\rangle$
 proton mixing angle: $\alpha_p=2.9294071$ rad
 incoming proton: $-0.977573|+\rangle + 0.210597|-\rangle$

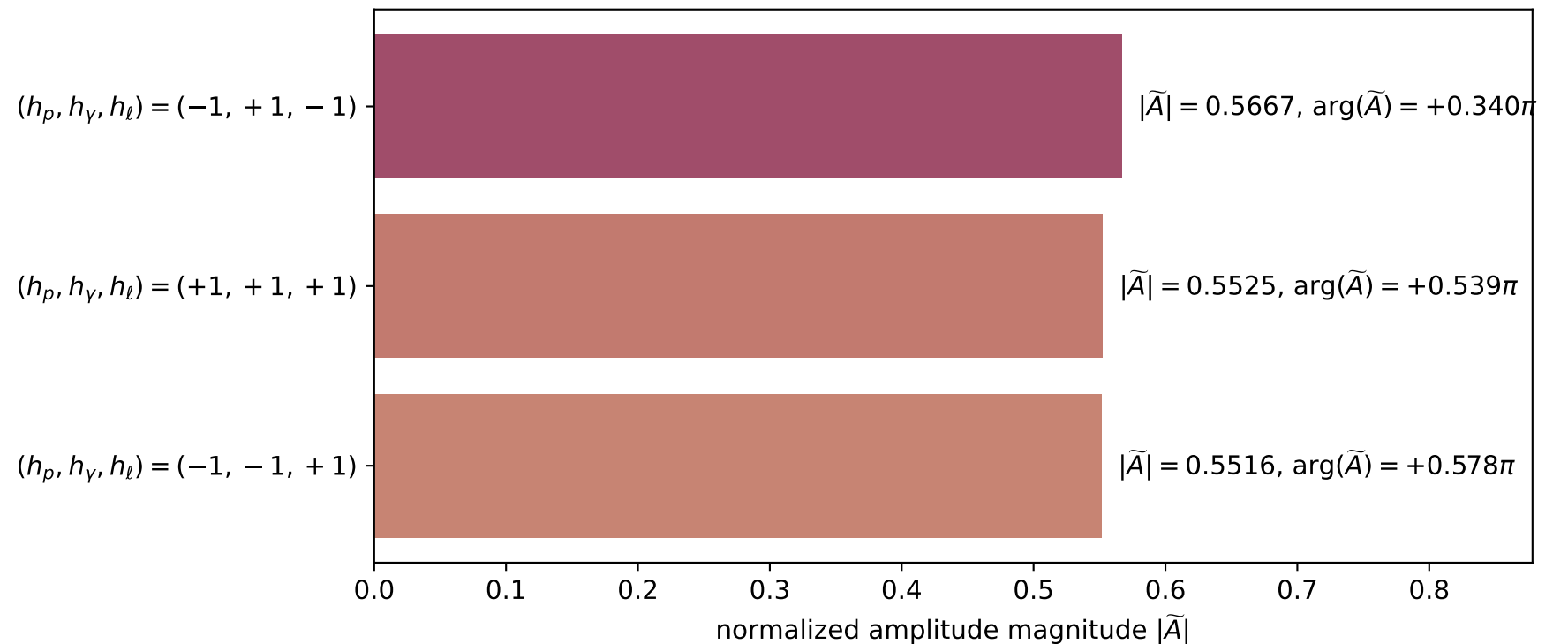
$s=1.24899$, $\sqrt{s}=1.11758$, $|\vec{P}|=0.165155$, $|\vec{P}'|=0.0105191$
 $\theta_{p'}=2.54466$, $\theta_\gamma=1.32836$, $\phi_{p'}=1.42117$, $\phi_\gamma=5.9038$
 $E_\gamma=0.0911783$, $\theta_{l'}=1.72064$, $\phi_{l'}=2.82816$
 $Q^2=0.024825$, $x_B=0.126333$, $t=-0.0300539$, $W^2=1.05152$, $y=0.532319$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E=0.1652$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1652$
 P $E=0.9524$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1652$
 l' $E=0.0883$ $p_x=-0.0831$ $p_y=0.0269$ $p_z=-0.0132$
 P' $E=0.9381$ $p_x=0.0009$ $p_y=0.0058$ $p_z=-0.0087$
 q_γ $E=0.0912$ $p_x=0.0822$ $p_y=-0.0328$ $p_z=0.0219$

x--y plane

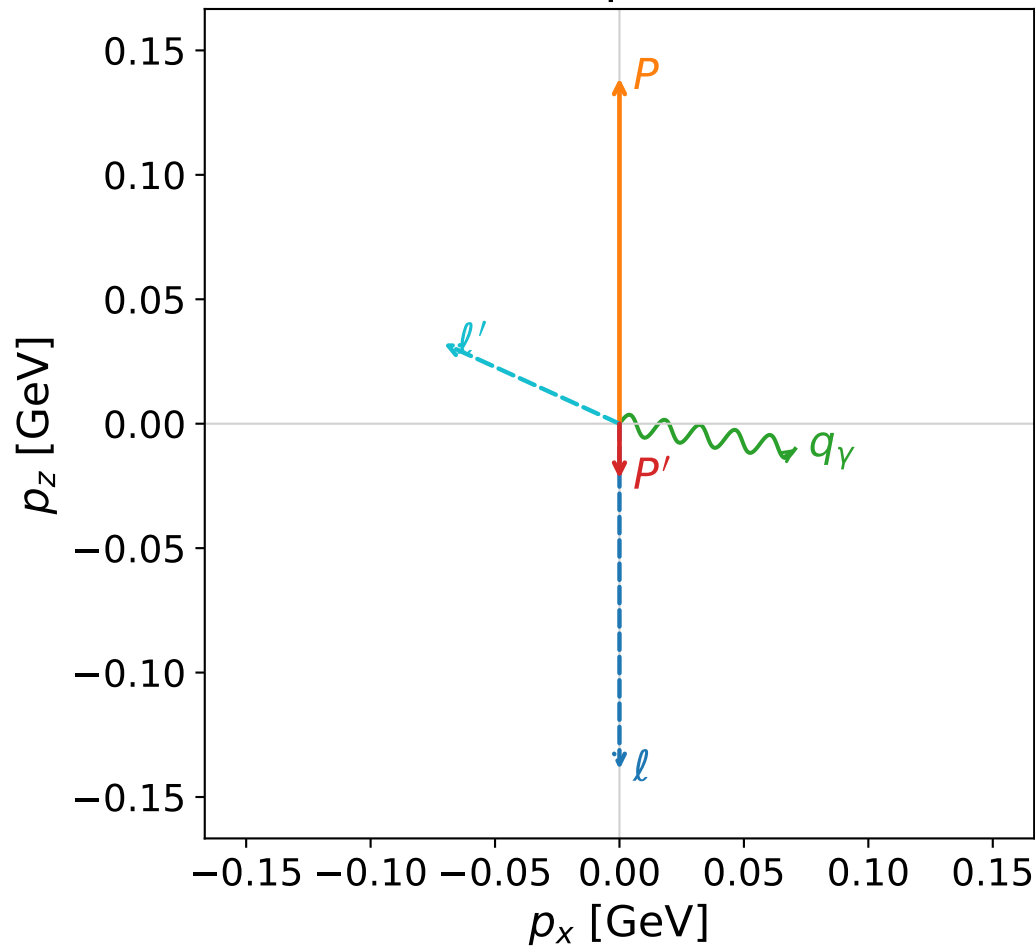


Leading normalized final-state amplitudes
 (N=3, retained=93.1%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



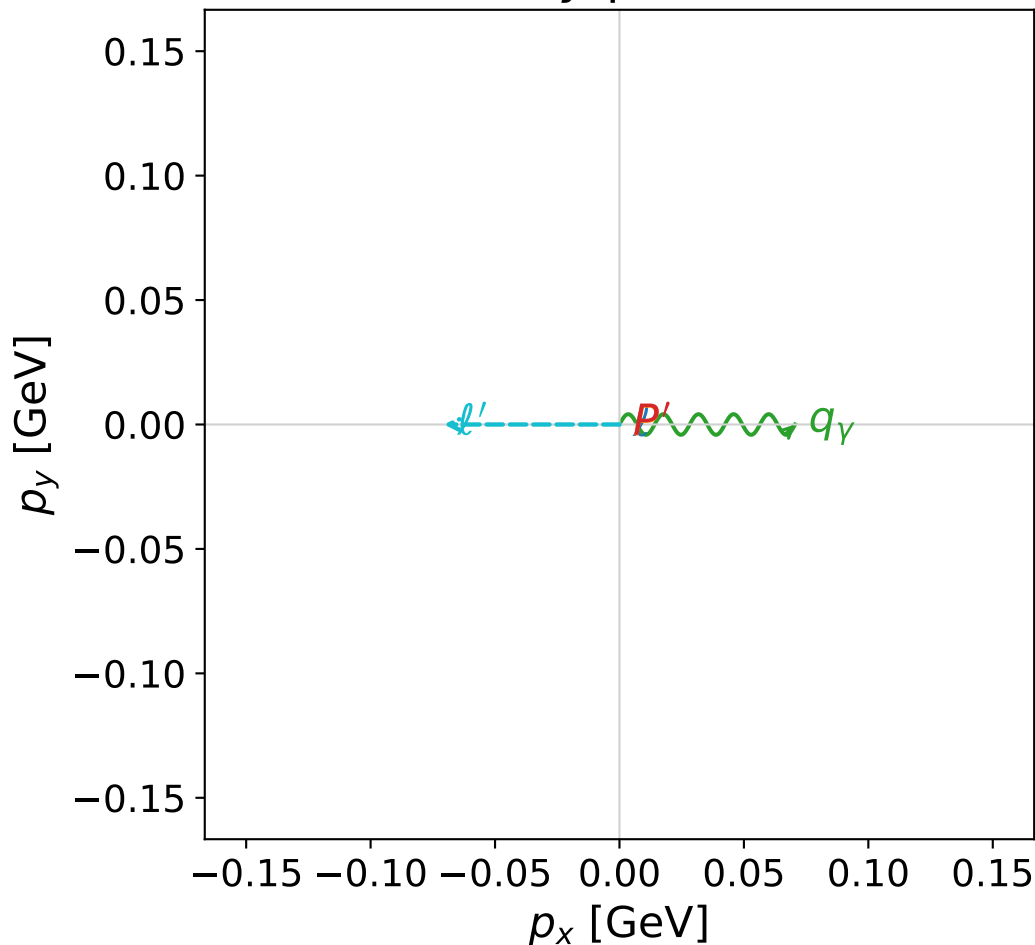
dw_mixing_angles_polarization_03_local_minimum_870
 $D_W=0.0123808$
 region: polarization_cluster_03_local_minimum_870
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0870
 lepton mixing angle: $\alpha_e=0.76538427$ rad
 incoming lepton: $0.721116|+\rangle + 0.692814|-\rangle$
 proton mixing angle: $\alpha_p=2.99225$ rad
 incoming proton: $-0.988869|+\rangle + 0.148788|-\rangle$

$s=1.18176$, $\sqrt{s}=1.08709$, $|\vec{P}|=0.138864$, $|\vec{P}'|=0.0220325$
 $\theta_{p'}=3.14159$, $\theta_\gamma=1.71361$, $\phi_{p'}=5.23833$, $\phi_\gamma=1.72592e-05$
 $E_\gamma=0.0712811$, $\theta_{\ell'}=1.14291$, $\phi_{\ell'}=3.14161$
 $Q^2=0.0304738$, $x_B=0.186058$, $t=-0.0257884$, $W^2=1.01316$, $y=0.542491$

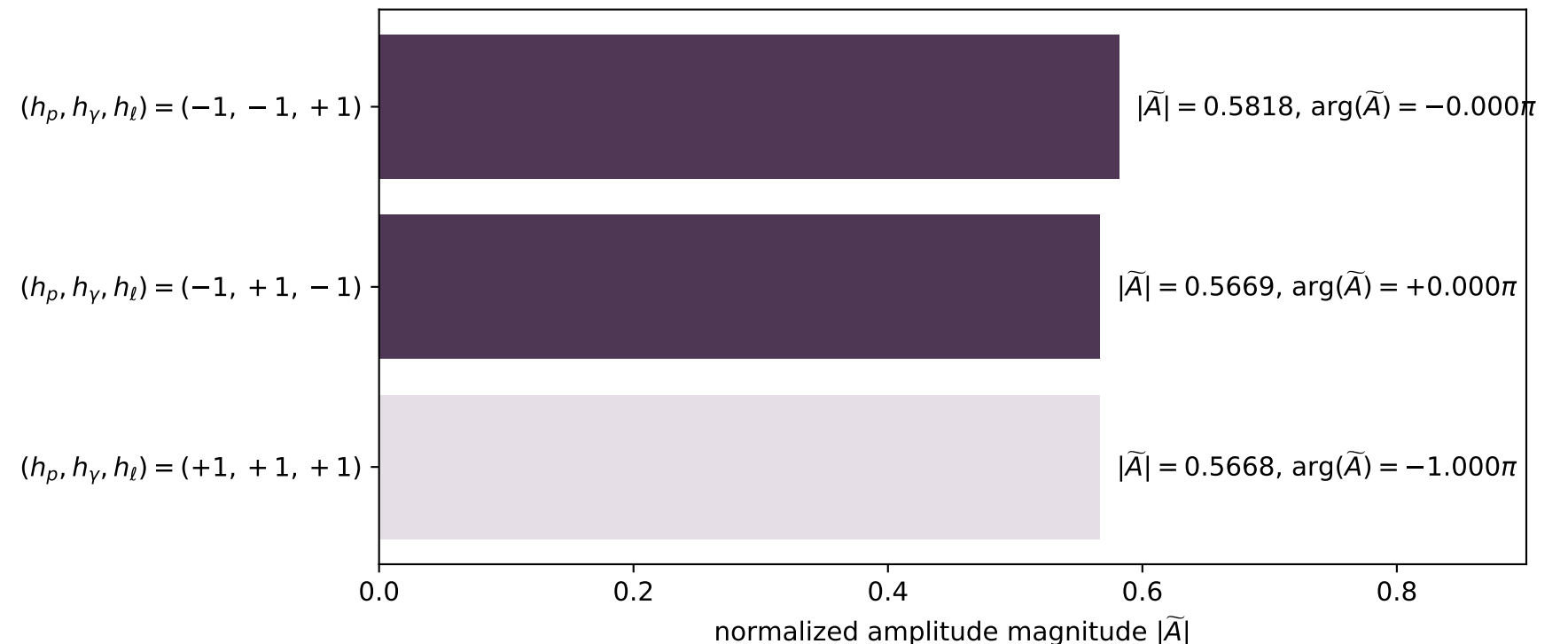
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	0.1389	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.1389
P	$E=$	0.9482	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.1389
ℓ'	$E=$	0.0775	$p_x=$	-0.0706	$p_y=$	-0.0000	$p_z=$	0.0322
P'	$E=$	0.9383	$p_x=$	0.0000	$p_y=$	-0.0000	$p_z=$	-0.0220
q_γ	$E=$	0.0713	$p_x=$	0.0706	$p_y=$	0.0000	$p_z=$	-0.0101

x--y plane

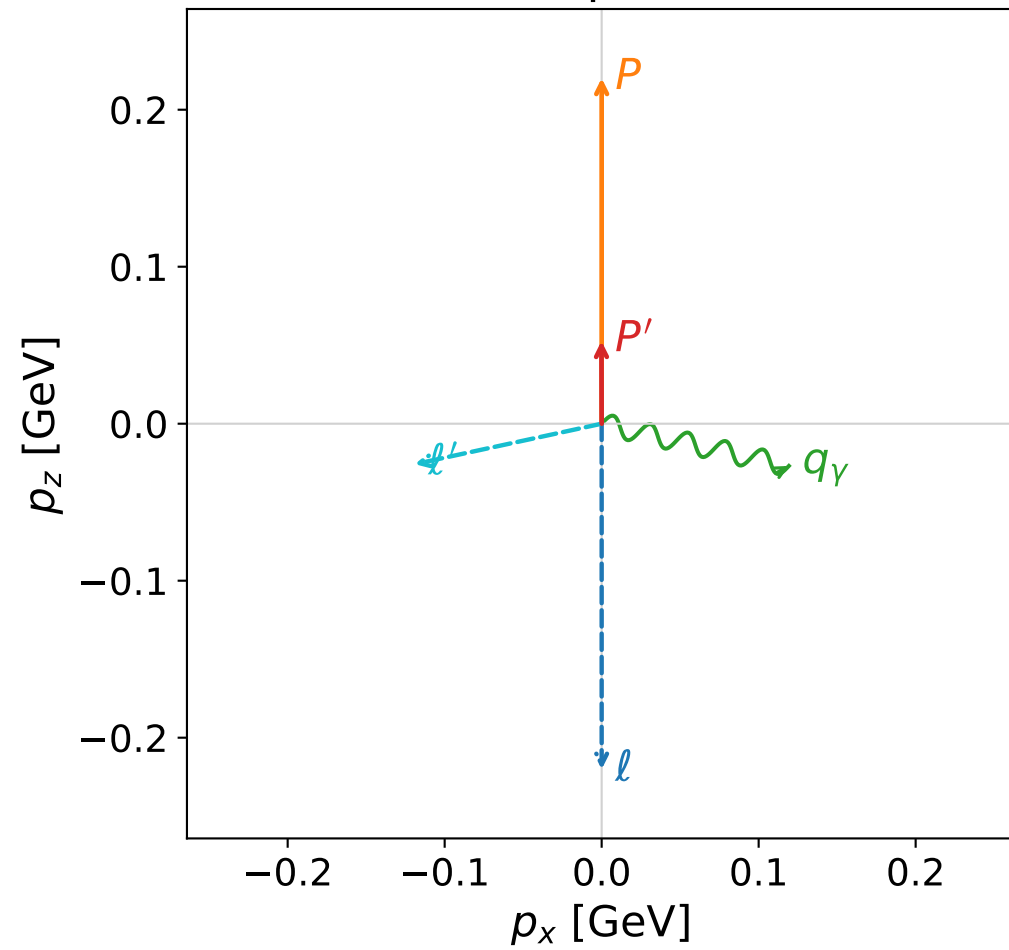


Leading normalized final-state amplitudes
(N=3, retained=98.1%)

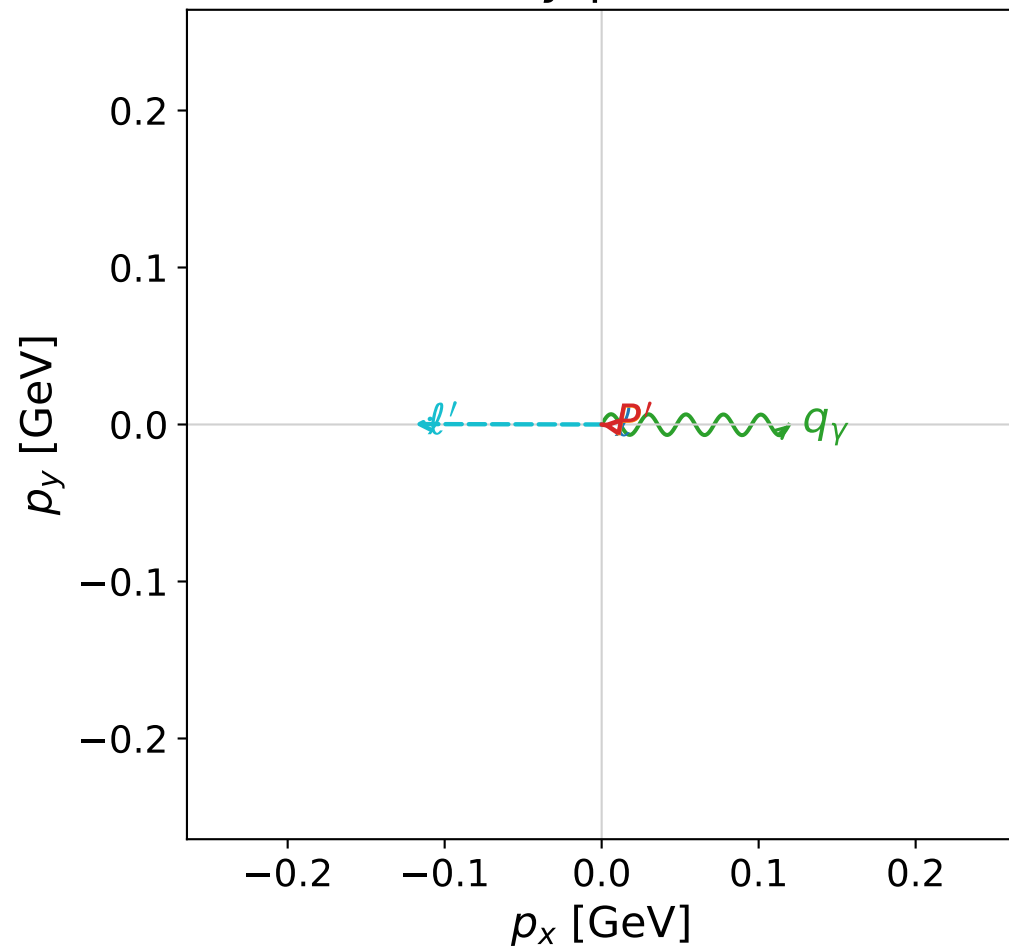


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_03_local_minimum_885

$D_W=0.0130708$

region: polarization_cluster_03_local_minimum_885

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0885

lepton mixing angle: $\alpha_e=0.81698053$ rad

incoming lepton: $0.684426|+\rangle + 0.729083|-\rangle$

proton mixing angle: $\alpha_p=2.9786241$ rad

incoming proton: $-0.98675|+\rangle + 0.162248|-\rangle$

$s=1.40095$, $\sqrt{s}=1.18362$, $|\vec{P}|=0.220131$, $|\vec{P}'|=0.0526706$

$\theta_{p'}=0.00278182$, $\theta_\gamma=1.79256$, $\phi_{p'}=3.00775$, $\phi_\gamma=6.28075$

$E_\gamma=0.122259$, $\theta_{\ell'}=1.78392$, $\phi_{\ell'}=3.13932$

$Q^2=0.0423086$, $x_B=0.153908$, $t=-0.0274667$, $W^2=1.11243$, $y=0.527529$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.2201$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2201$

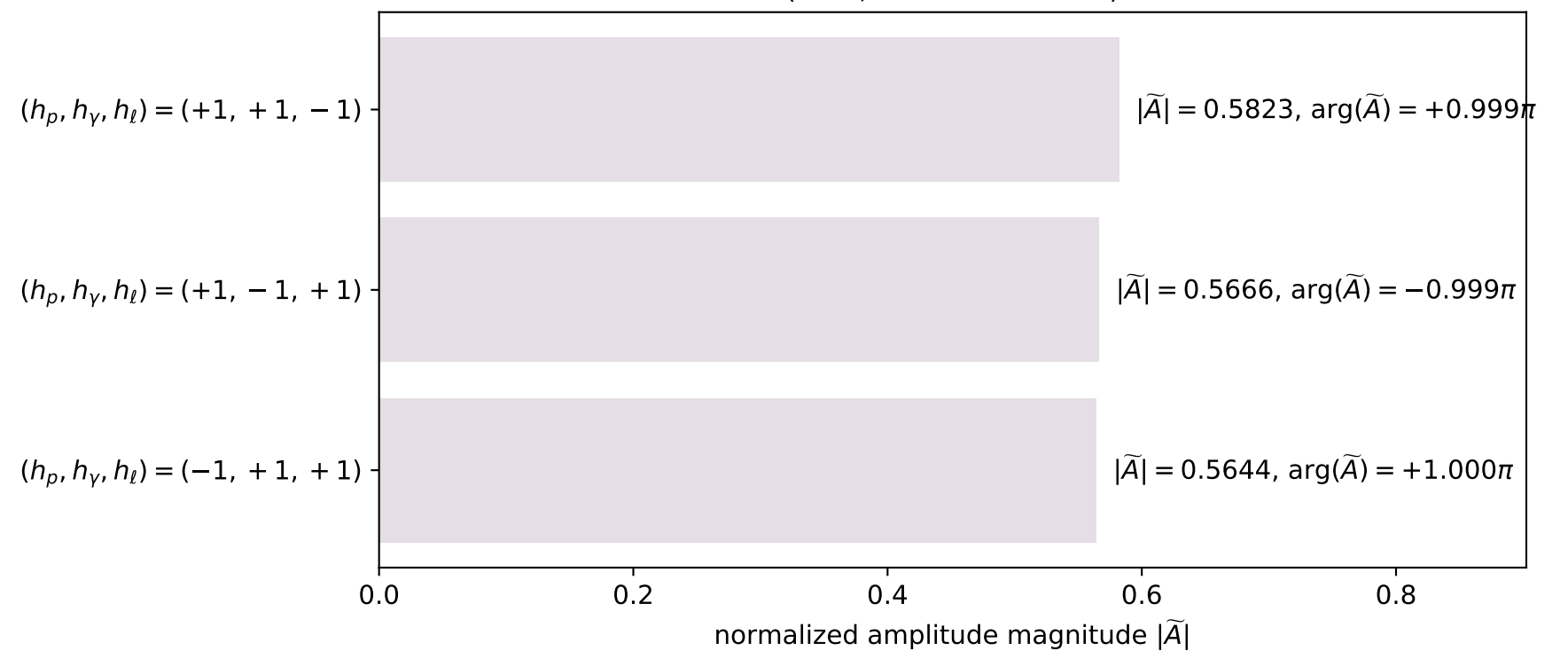
P $E= 0.9635$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2201$

ℓ' $E= 0.1219$ $p_x= -0.1191$ $p_y= 0.0003$ $p_z= -0.0258$

P' $E= 0.9395$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 0.0527$

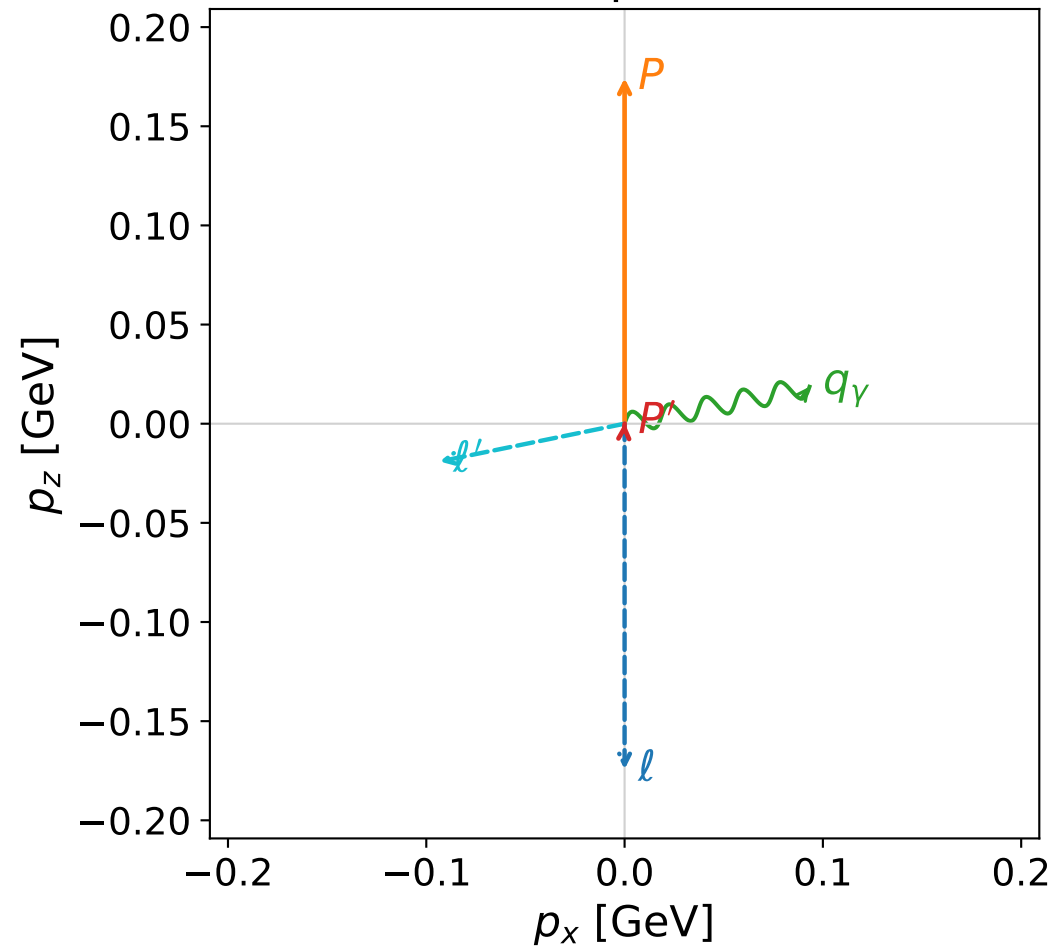
q_γ $E= 0.1223$ $p_x= 0.1193$ $p_y= -0.0003$ $p_z= -0.0269$

Leading normalized final-state amplitudes
(N=3, retained=97.9%)

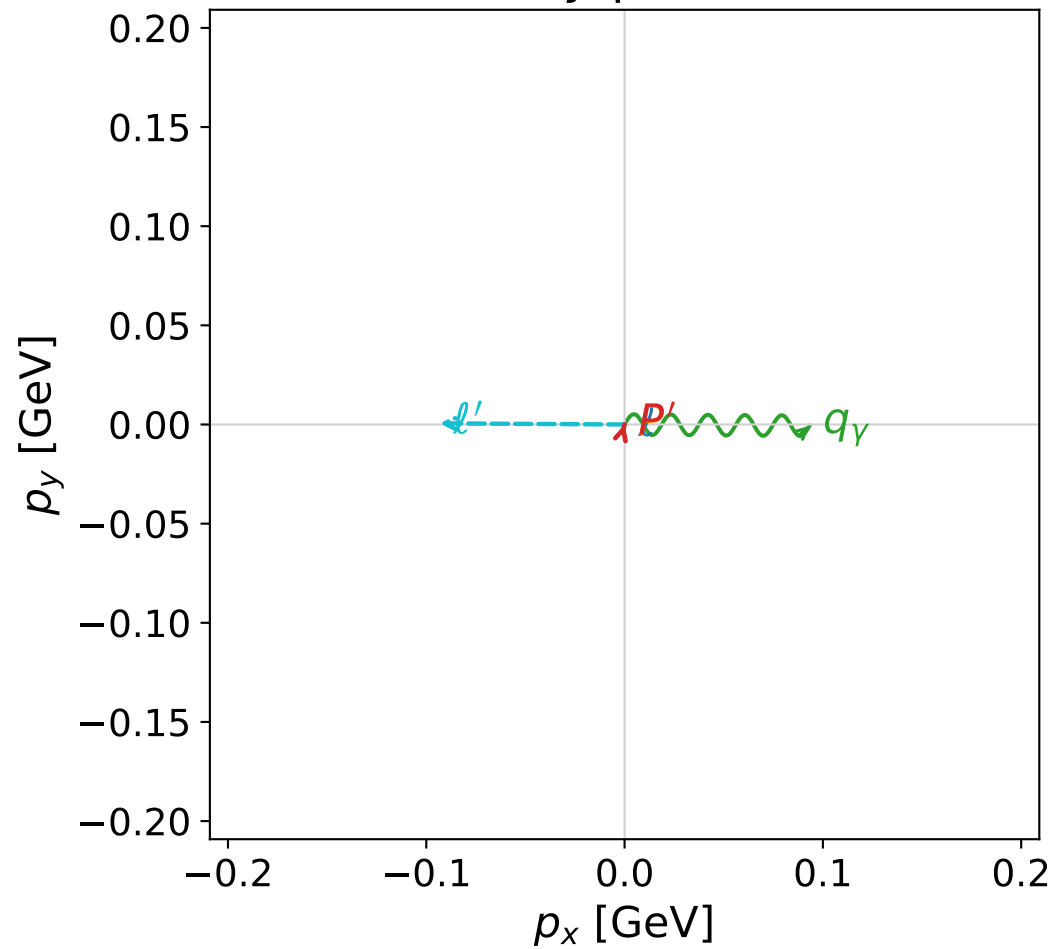


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane

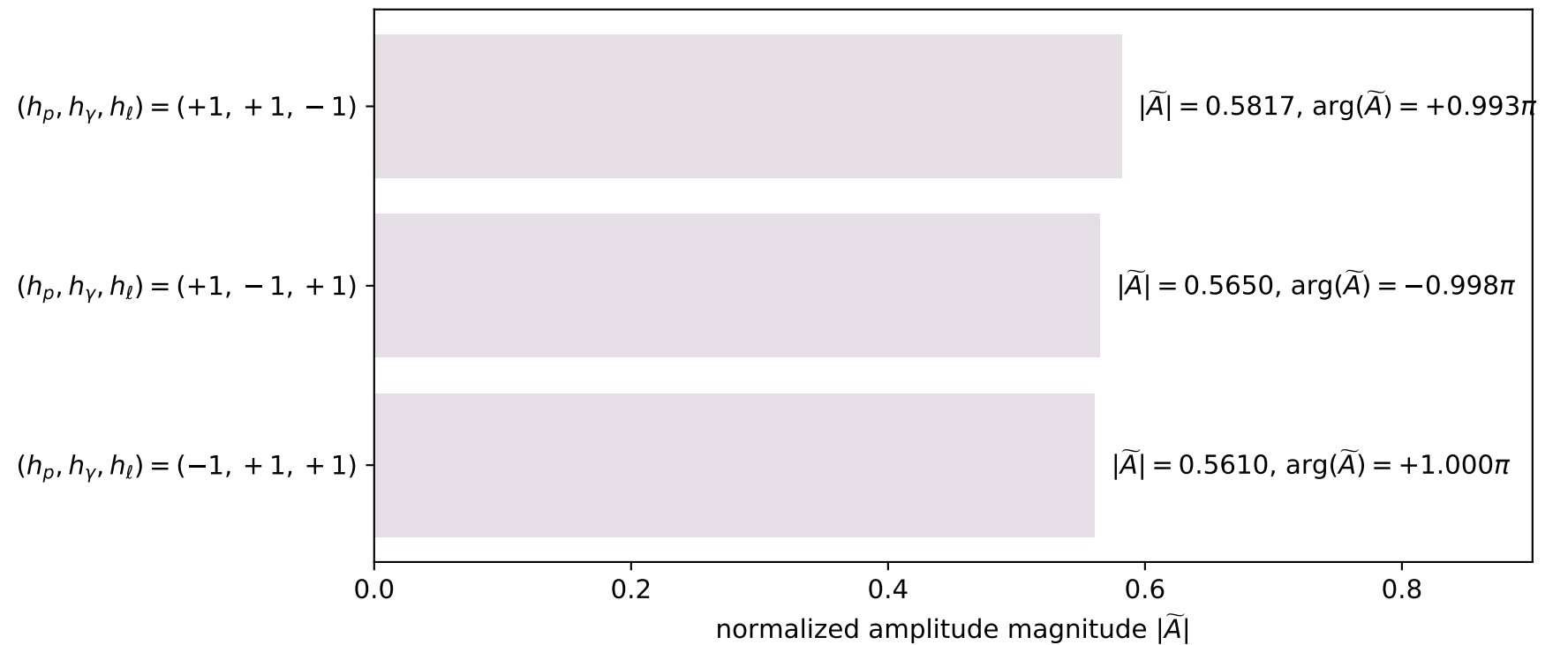


dw_mixing_angles_polarization_03_local_minimum_888
 $D_W=0.0132357$
 region: polarization_cluster_03_local_minimum_888
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0888
 lepton mixing angle: $\alpha_e=0.81471988$ rad
 incoming lepton: $0.686072|+\rangle + 0.727533|-\rangle$
 proton mixing angle: $\alpha_p=2.9588101$ rad
 incoming proton: $-0.983342|+\rangle + 0.181767|-\rangle$

$s=1.27313$, $\sqrt{s}=1.12833$, $|\vec{P}|=0.174277$, $|\vec{P}'|=0.000416078$
 $\theta_{p'}=0.200961$, $\theta_Y=1.37346$, $\phi_{p'}=1.30137$, $\phi_Y=6.27608$
 $E_Y=0.0951137$, $\theta_{\ell'}=1.77228$, $\phi_{\ell'}=3.13535$
 $Q^2=0.0265461$, $x_B=0.129518$, $t=-0.029973$, $W^2=1.05826$, $y=0.521148$

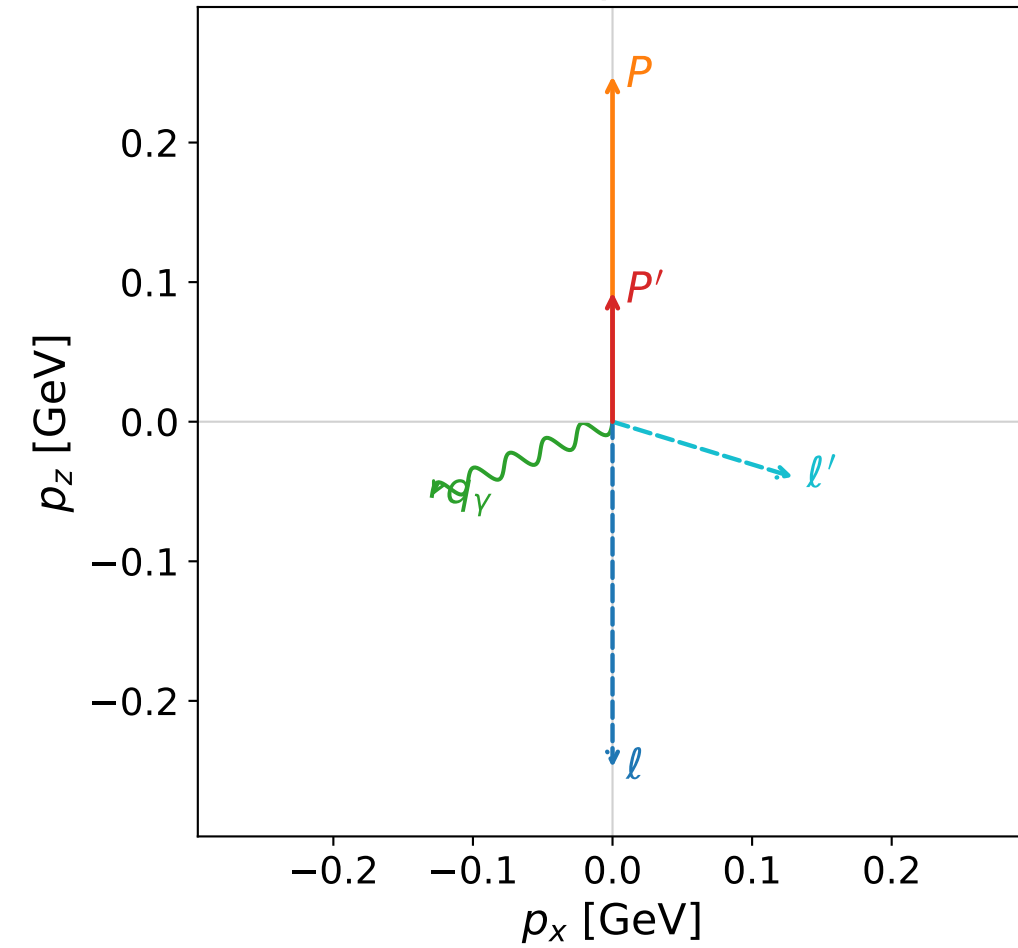
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1743$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1743$
 P $E= 0.9541$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1743$
 ℓ' $E= 0.0952$ $p_x= -0.0933$ $p_y= 0.0006$ $p_z= -0.0191$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0001$ $p_z= 0.0004$
 q_Y $E= 0.0951$ $p_x= 0.0933$ $p_y= -0.0007$ $p_z= 0.0186$

Leading normalized final-state amplitudes
 (N=3, retained=97.2%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

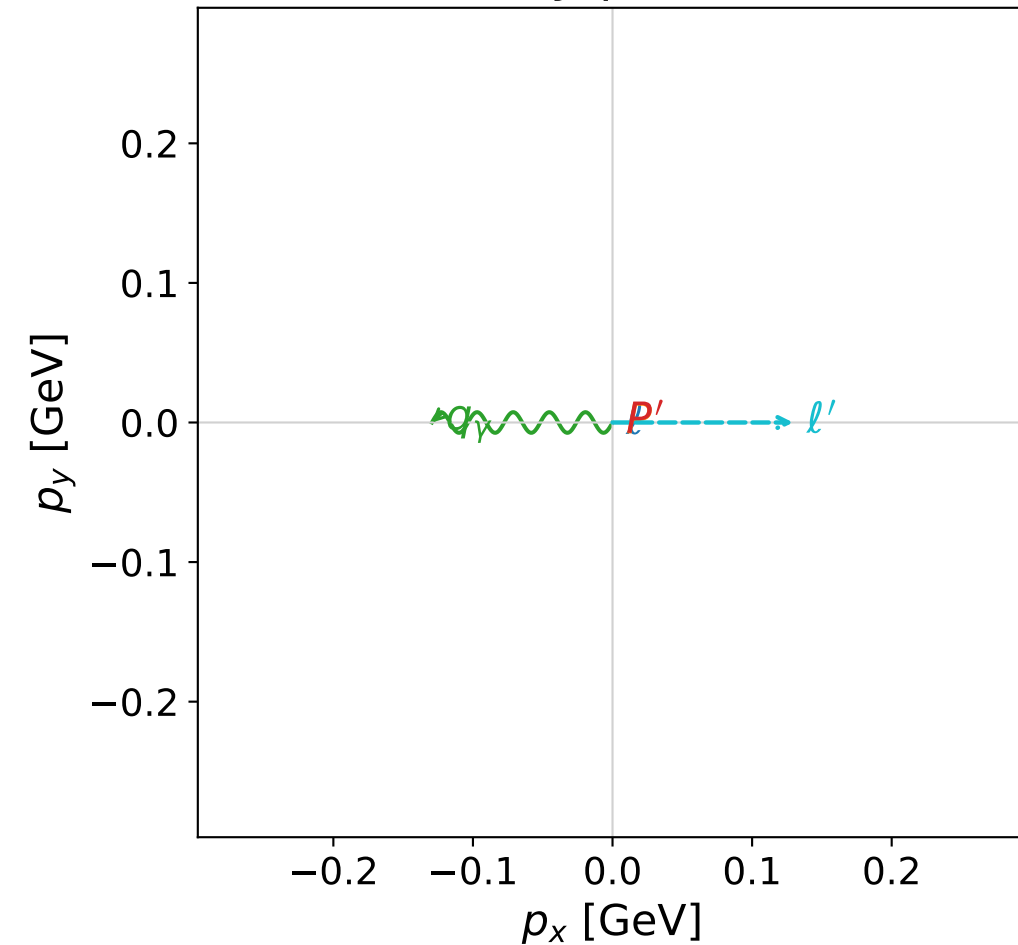


dw_mixing_angles_polarization_03_local_minimum_910
 $D_W=0.0140542$
 region: polarization_cluster_03_local_minimum_910
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0910
 lepton mixing angle: $\alpha_e=0.81021801$ rad
 incoming lepton: $0.689341|+\rangle + 0.724437|-\rangle$
 proton mixing angle: $\alpha_p=0.13099092$ rad
 incoming proton: $0.991433|+\rangle + 0.130617|-\rangle$

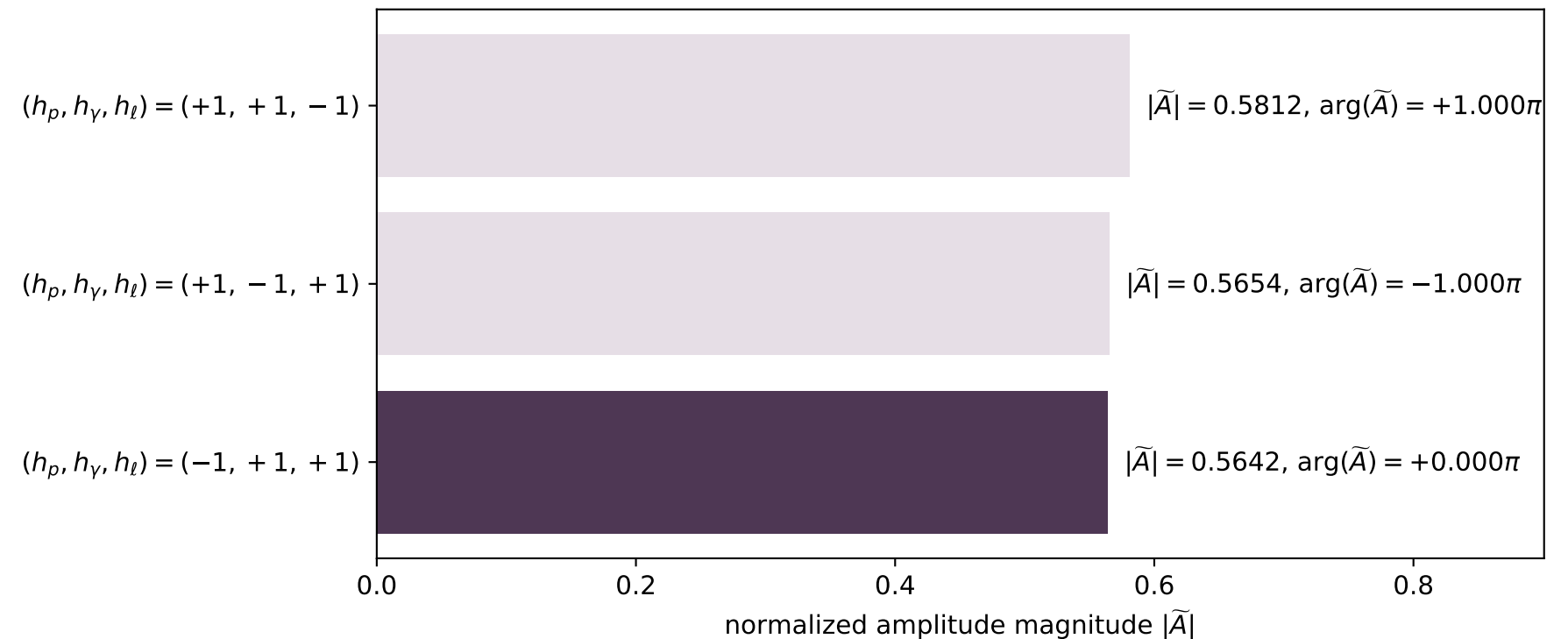
$s=1.48284$, $\sqrt{s}=1.21772$, $|\vec{P}|=0.247591$, $|\vec{P}'|=0.0926797$
 $\theta_{p'}=0$, $\theta_\gamma=1.96038$, $\phi_{p'}=3.1836$, $\phi_\gamma=3.14158$
 $E_\gamma=0.139858$, $\theta_{l'}=1.86755$, $\phi_{l'}=6.28318$
 $Q^2=0.047404$, $x_B=0.147721$, $t=-0.0232381$, $W^2=1.15334$, $y=0.53218$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.2476$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2476$
 P $E= 0.9701$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2476$
 l' $E= 0.1353$ $p_x= 0.1294$ $p_y= -0.0000$ $p_z= -0.0396$
 P' $E= 0.9426$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0927$
 q_γ $E= 0.1399$ $p_x= -0.1294$ $p_y= 0.0000$ $p_z= -0.0531$

x--y plane

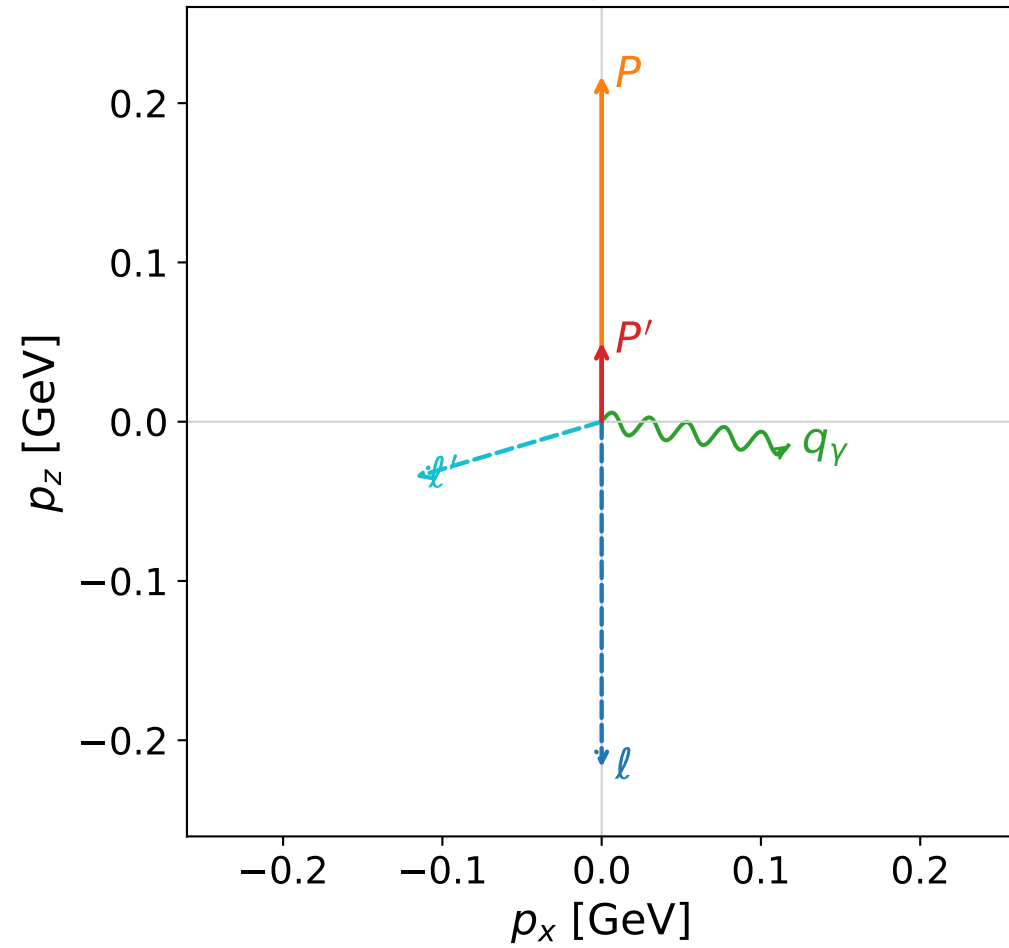


Leading normalized final-state amplitudes
 (N=3, retained=97.6%)

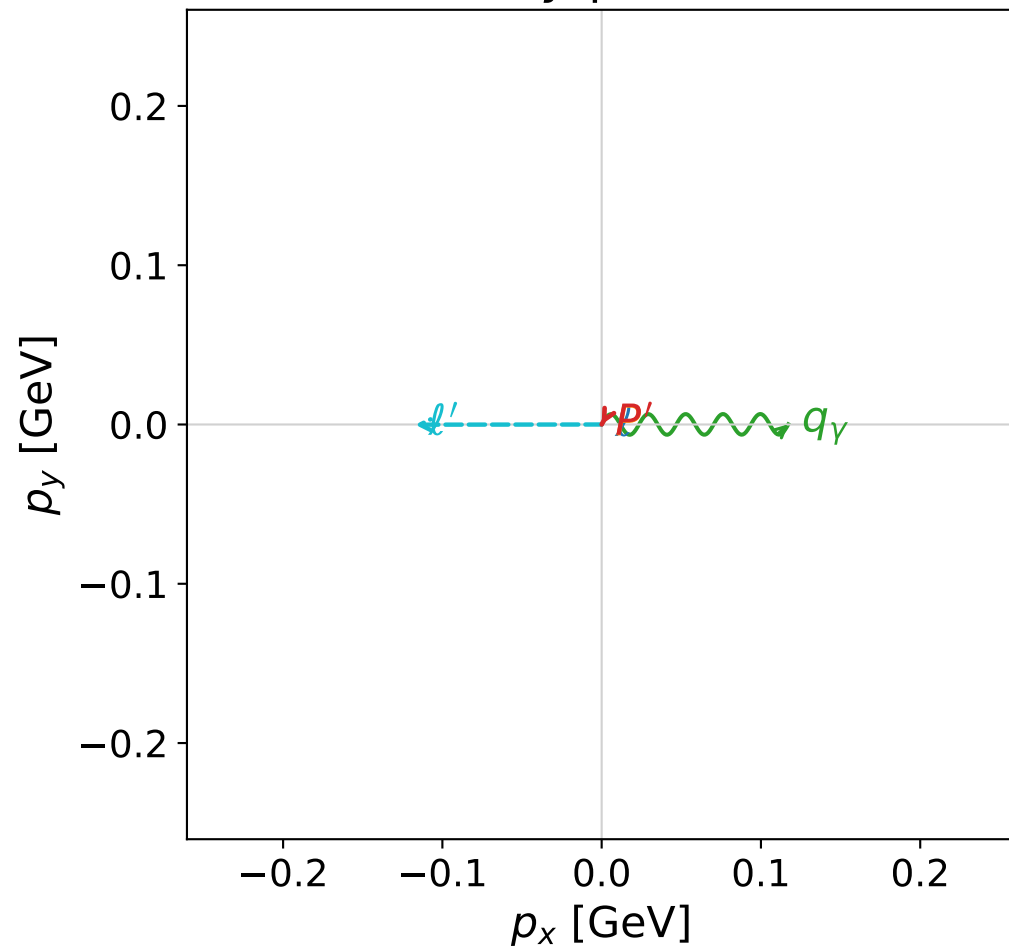


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



dw_mixing_angles_polarization_03_local_minimum_950

D_W=0.01601

region: polarization_cluster_03_local_minimum_950

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0950

lepton mixing angle: $\alpha_e=0.81877057$ rad

incoming lepton: $0.68312|+\rangle + 0.730307|-\rangle$

proton mixing angle: $\alpha_p=2.9968052$ rad

incoming proton: $-0.989537|+\rangle + 0.144282|-\rangle$

$s=1.39179$, $\sqrt{s}=1.17974$, $|\vec{P}|=0.216973$, $|\vec{P}'|=0.0497768$

$\theta_{p'}=0.000110249$, $\theta_Y=1.69747$, $\phi_{p'}=4.21159$, $\phi_Y=0.00102048$

$E_Y=0.118149$, $\theta_{\ell'}=1.85983$, $\phi_{\ell'}=3.14257$

$Q^2=0.0379363$, $x_B=0.145137$, $t=-0.0274049$, $W^2=1.10329$, $y=0.510567$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.2170$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2170$

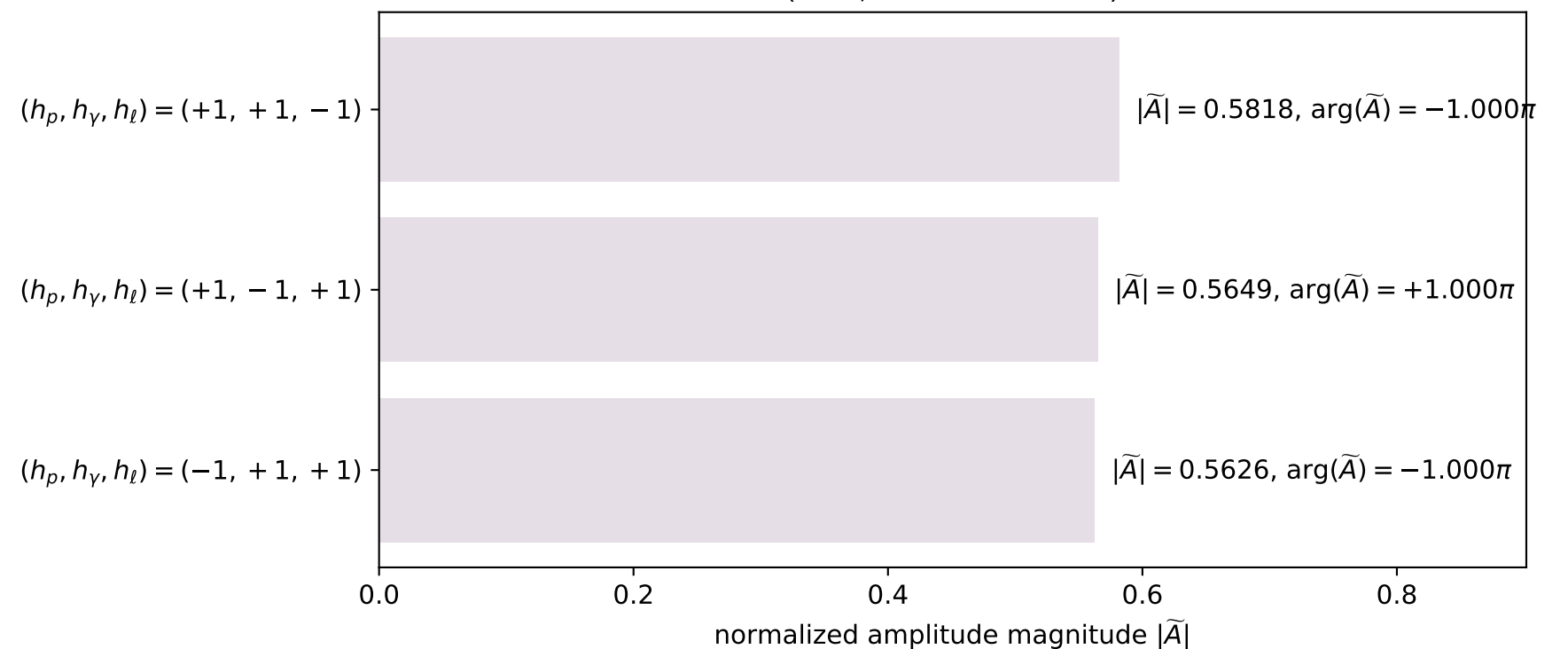
P $E= 0.9628$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2170$

ℓ' $E= 0.1223$ $p_x= -0.1172$ $p_y= -0.0001$ $p_z= -0.0349$

P' $E= 0.9393$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0498$

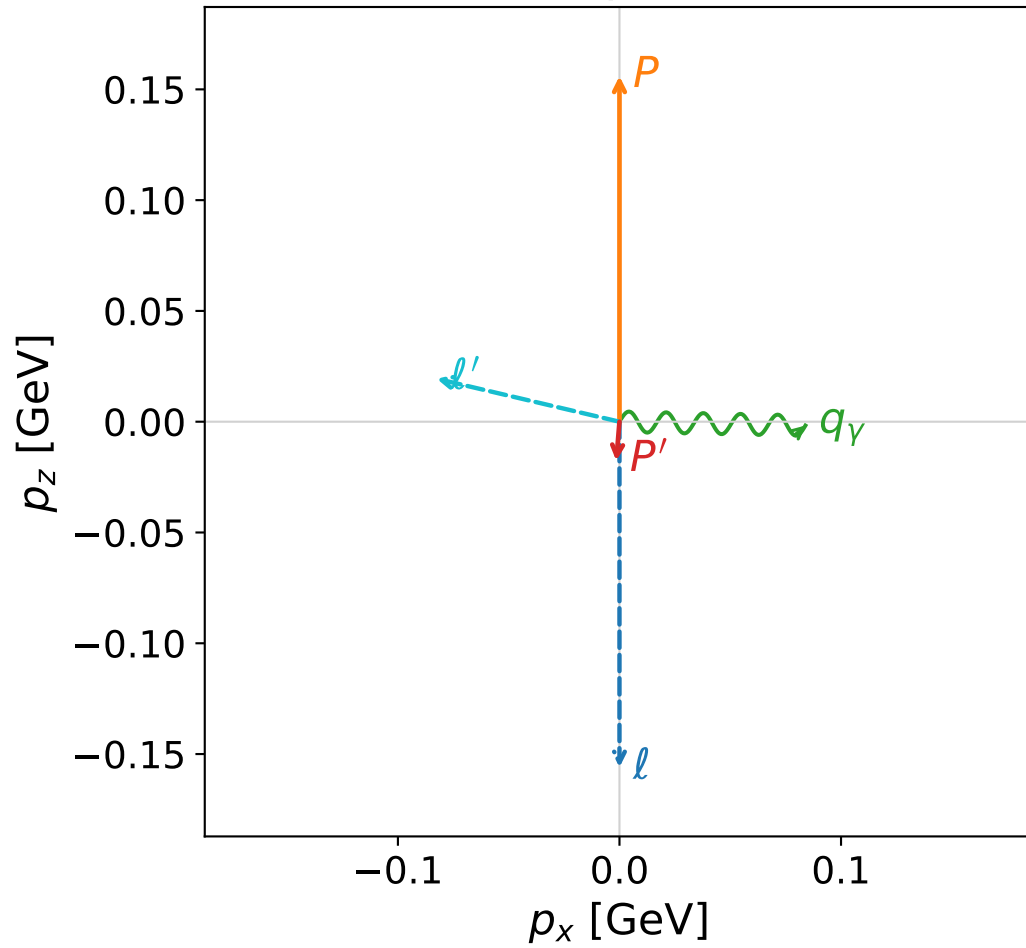
q_Y $E= 0.1181$ $p_x= 0.1172$ $p_y= 0.0001$ $p_z= -0.0149$

Leading normalized final-state amplitudes
(N=3, retained=97.4%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

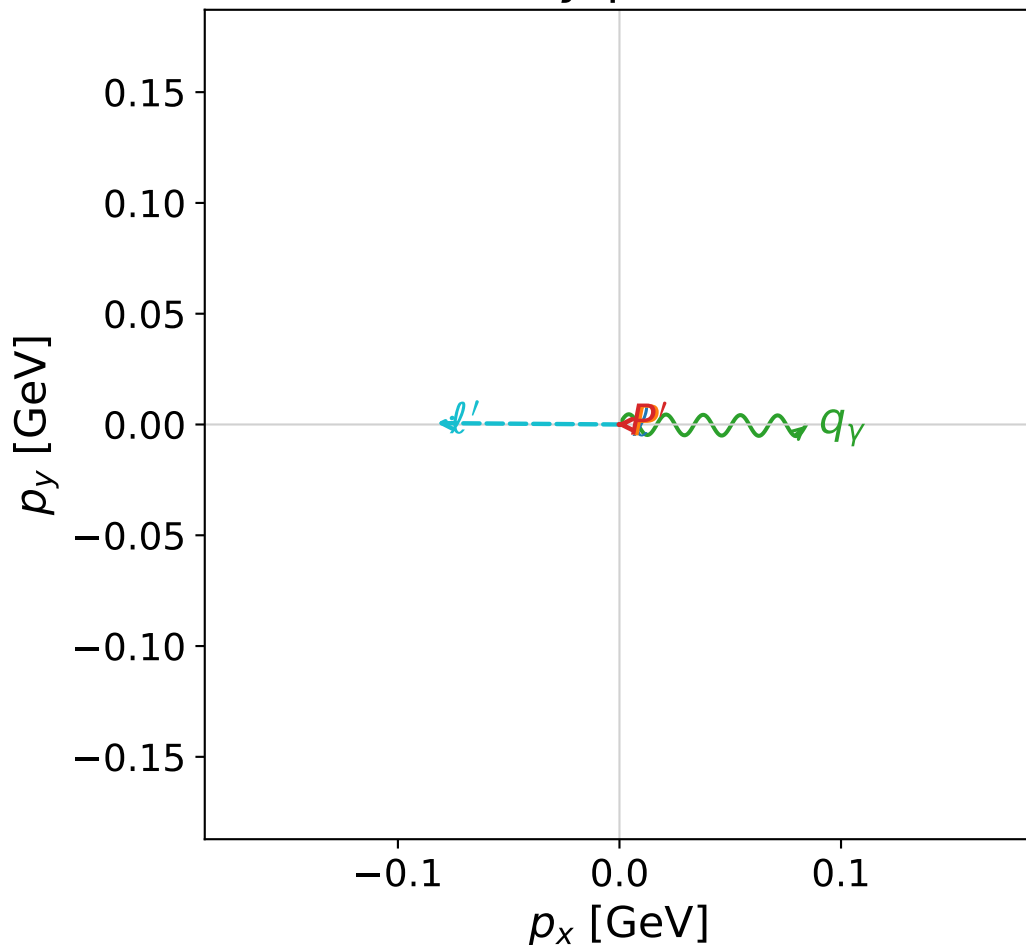


dw_mixing_angles_polarization_03_local_minimum_991
 $D_W=0.0179842$
 region: polarization_cluster_03_local_minimum_991
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0991
 lepton mixing angle: $\alpha_e=0.78916052$ rad
 incoming lepton: $0.704441|+\rangle + 0.709762|-\rangle$
 proton mixing angle: $\alpha_p=3.112534$ rad
 incoming proton: $-0.999578|+\rangle + 0.0290546|-\rangle$

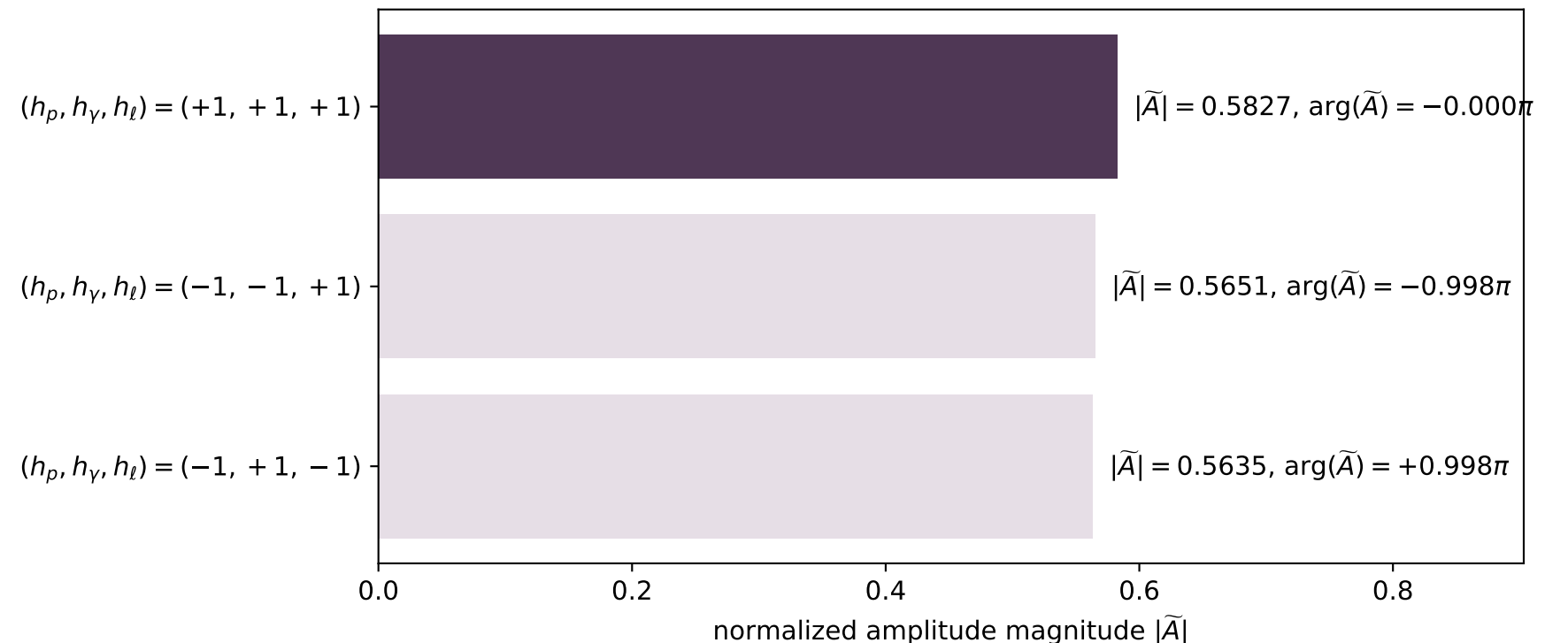
$s=1.22514$, $\sqrt{s}=1.10686$, $|\vec{P}|=0.155981$, $|\vec{P}'|=0.0177058$
 $\theta_{p'}=3.06042$, $\theta_\gamma=1.59143$, $\phi_{p'}=3.14178$, $\phi_\gamma=6.27569$
 $E_\gamma=0.0839506$, $\theta_{\ell'}=1.34006$, $\phi_{\ell'}=3.13397$
 $Q^2=0.0324823$, $x_B=0.170796$, $t=-0.0299871$, $W^2=1.03754$, $y=0.550773$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1560$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1560$
 P $E= 0.9509$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1560$
 ℓ' $E= 0.0847$ $p_x= -0.0825$ $p_y= 0.0006$ $p_z= 0.0194$
 P' $E= 0.9382$ $p_x= -0.0014$ $p_y= -0.0000$ $p_z= -0.0176$
 q_γ $E= 0.0840$ $p_x= 0.0839$ $p_y= -0.0006$ $p_z= -0.0017$

x--y plane

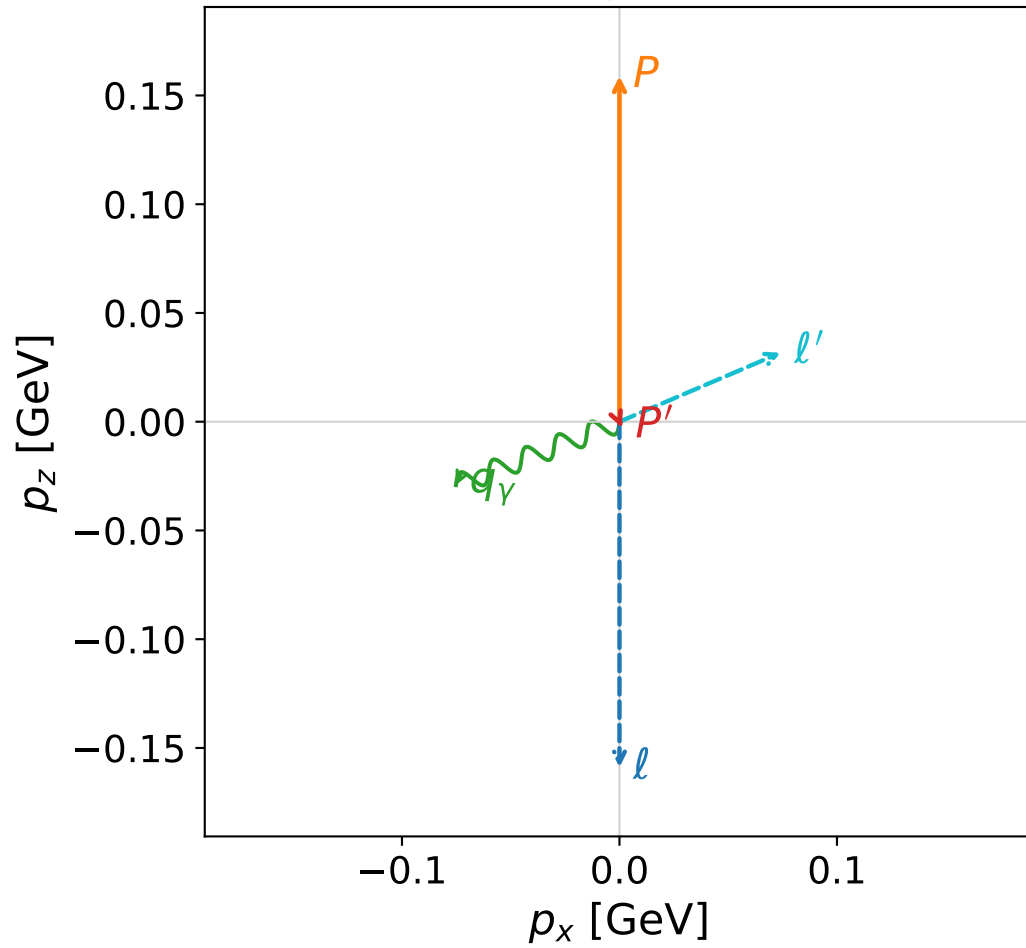


Leading normalized final-state amplitudes
 (N=3, retained=97.6%)



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane

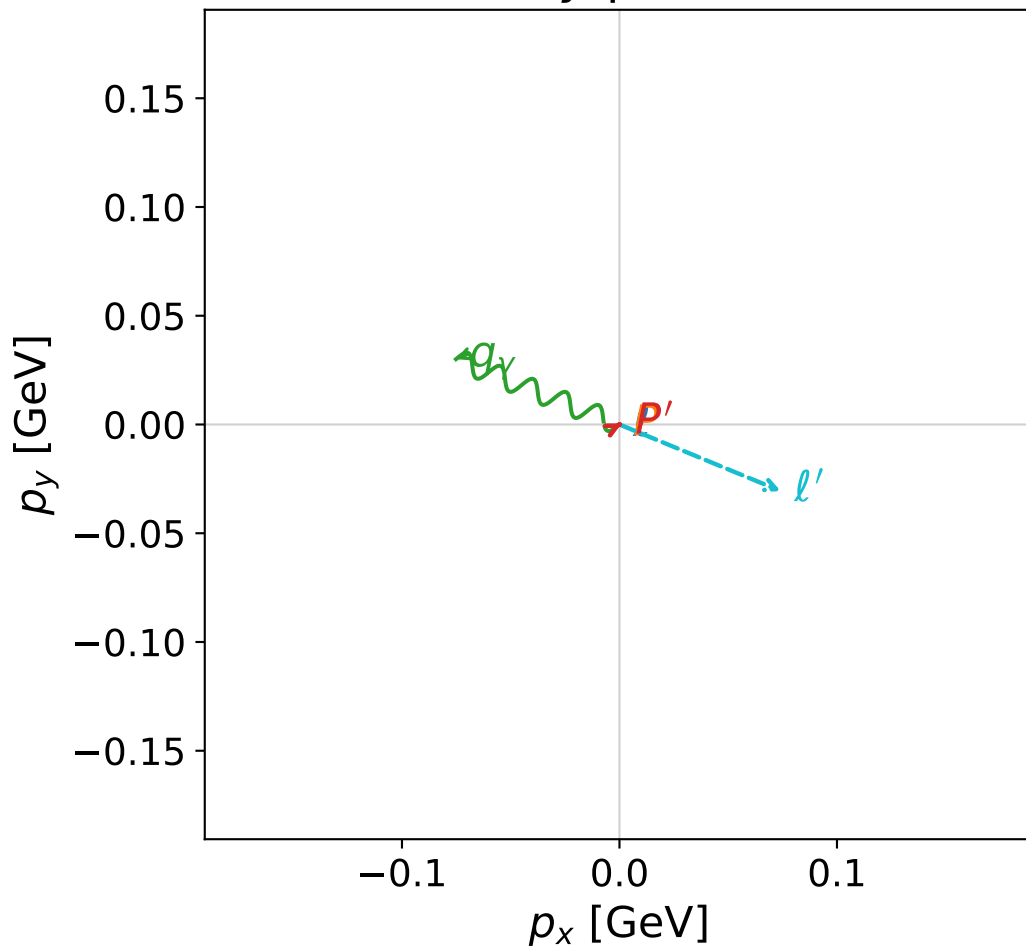


dw_mixing_angles_polarization_03_local_minimum_994
 $D_W=0.0180986$
 region: polarization_cluster_03_local_minimum_994
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0994
 lepton mixing angle: $\alpha_e=0.77750627$ rad
 incoming lepton: $0.712665|+\rangle + 0.701504|-\rangle$
 proton mixing angle: $\alpha_p=0.028423775$ rad
 incoming proton: $0.999596|+\rangle + 0.0284199|-\rangle$

$s=1.23267$, $\sqrt{s}=1.11026$, $|\vec{P}|=0.158894$, $|\vec{P}'|=0.00281842$
 $\theta_{p'}=2.70197$, $\theta_Y=1.91653$, $\phi_{p'}=0.467859$, $\phi_Y=2.76154$
 $E_Y=0.0860442$, $\theta_{\ell'}=1.19413$, $\phi_{\ell'}=5.89191$
 $Q^2=0.0374729$, $x_B=0.188427$, $t=-0.0258873$, $W^2=1.04124$, $y=0.563652$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1589$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1589$
 P $E= 0.9514$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1589$
 ℓ' $E= 0.0862$ $p_x= 0.0741$ $p_y= -0.0306$ $p_z= 0.0317$
 P' $E= 0.9380$ $p_x= 0.0011$ $p_y= 0.0005$ $p_z= -0.0026$
 q_Y $E= 0.0860$ $p_x= -0.0752$ $p_y= 0.0300$ $p_z= -0.0292$

x--y plane



Leading normalized final-state amplitudes
(N=3, retained=96.0%)

